arXiv:2010.07836v2 [math.GT] 10 Nov 2020

INSTANTON FLOER HOMOLOGY, SUTURES, AND HEEGAARD DIAGRAMS

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Abstract. In this paper, we introduce a method to extract information about the instanton Floer homology of rational homology spheres and knots in them from Heegaard diagrams, based on sutured manifolds and bypass maps. As an application, we prove that for a (1,1)-knot K in a lens space Y, we have $\dim_{\mathbb{C}} KHI(Y,K) \leqslant \mathrm{rk}_{\mathbb{Z}} \widehat{HFK}(Y,K)$. If $Y = S^3$ and K is also a Heegaard Floer L-space knot, for example, K is a torus knot, then $\dim_{\mathbb{C}} KHI(Y,K) = \mathrm{rk}_{\mathbb{Z}} \widehat{HFK}(Y,K)$. Also, we prove that all simple knots in lens spaces are instanton Floer simple knots and admit instanton L-space surgeries. Another application is that for a rationally null-homologous knot K in a general 3-manifold Y, we prove the dimension inequality $\dim_{\mathbb{C}} I^{\sharp}(Y) \leqslant \dim_{\mathbb{C}} KHI(Y,K)$.

In the second part of this paper, for a rationally null-homologous knot $K \subset Y$, we construct a decomposition of $I^{\sharp}(Y)$ as a candidate for the counterpart of the torsion spin^c decompositions in $\widetilde{HM}(Y)$ and $\widehat{HF}(Y)$. Moreover, we construct a spectral sequence from $KHI(Y,K)$ to $I^{\sharp}(Y)$ respecting this decomposition. As an application, we derive the relation $\tau_I(\bar{K}) = -\tau_I(K)$, where $K \subset S^3$ is a knot, $\bar{K}$ is its mirror, and τ_I is the instanton tau invariant defined by the first author. Also, we use this decomposition to derive a dimension formula of framed instanton Floer homology for large surgeries on a null-homologous knot in a 3-manifold.

Contents

1. Introduction 2
2. Instanton Floer homology and sutured manifolds 9
2.1. Instanton Floer homology 9
2.2. Balanced sutured manifolds 10
2.3. Bypass attachments 15
3. Instanton Floer homology and Heegaard diagrams 18
3.1. Sutured manifold with tangles and Heegaard diagrams 18
3.2. A dimension inequality for tangles 20
3.3. The instanton knot homology of (1,1)-knots 32
3.4. Large surgeries on simple knots 38
4. Instanton Floer homology and the decomposition 40
4.1. Basic setups 40
4.2. Bypasses on knot complements 41
4.3. Decomposing framed instanton Floer homology 47
4.4. Commutative diagrams for bypass maps 53
4.5. Spectral sequences and tau invariants 62
4.6. The stabilization of Integral surgeries 68
4.7. The splitting of exact triangles 70
5. Some remarks for future directions 75
References 80

1. Introduction

The instanton homology of closed 3-manifolds and knots in 3-manifolds was introduced by Floer [Flo88, Flo90], which became a powerful tool in the study of 3-dimensional topology. Some related constructions were made by Kronheimer and Mrowka [KM10b, KM11b, KM11a], Li [Li19], and Daemi and Scaduto [DS19]. Apart from instanton Floer homology, there are three Floer homologies of closed 3-manifolds, knots, and balanced sutured manifolds: Heegaard Floer homology by Ozsváth and Szabó [OS04b, OS04a], Rasmussen [Ras03], and Juhász [Juh06], monopole Floer homology by Kronheimer and Mrowka [KM07, KM10b], and embedded contact homology (ECH) by Hutchings [Hut10], Colin, Ghiggini, Honda, and Hutchings [CGHH10]. For closed 3-manifolds, all these three Floer homologies are isomorphic by work of Kutluhan, Lee, and Taubes [KLT10a, KLT10b, KLT10c, KLT11, KLT12], or Taubes [Tau10a, Tau10b, Tau10c, Tau10d, Tau10e] combined with Colin, Ghiggini, and Honda [CGH11]. However, instanton Floer homology remains isolated from the rest. The following conjecture is still open.

Conjecture 1.1 ([KM10b])**.** For a balanced sutured manifold (M,γ), we have

$$SHI(M,\gamma) \cong SFH(M,\gamma) \otimes \mathbb{C}.$$

In particular, for a knot K in a closed 3-manifold Y, there are isomorphisms

$$I^\sharp(Y) \cong \widehat{HF}(Y) \otimes \mathbb{C}, \text{ and } KHI(Y,K) \cong \widehat{HFK}(Y,K) \otimes \mathbb{C}.$$

Here SHI is sutured instanton Floer homology [KM10b], SFH is sutured (Heegaard) Floer homology [Juh06], $I^\sharp$ is framed instanton Floer homology [KM11a], $\widehat{HF}$ is the hat version of Heegaard Floer homology [OS04b], KHI is instanton knot homology [KM10b], and $\widehat{HFK}$ is (Heegaard) knot Floer homology [OS04a].

Instanton Floer homology serves as a topological invariant of closed 3-manifolds or knots. Meanwhile, closed 3-manifolds and knots can be determined by Heegaard diagrams. Thus, in principle, Heegaard diagrams also determine instanton Floer homology. However, unlike Heegaard Floer homology, which is built directly upon Heegaard diagrams, the construction of instanton Floer homology is based on a set of partial differential equations on $SO(3)$-bundles over suitably chosen 3-manifolds. This makes the relation between instanton Floer homology and Heegaard diagrams implicit. In the first part of the paper, we introduce a method to extract some information about instanton Floer homology from Heegaard diagrams.

Theorem 1.2. *Suppose Y is a rational homology sphere, and $K \subset Y$ is a knot. Suppose $(\Sigma,\alpha,\beta,z,w)$ is a double-pointed Heegaard diagram of (Y,K). Then there is a balanced sutured handlebody (H,γ) induced by $(\Sigma,\alpha,\beta,z,w)$ so that*

$$\dim_{\mathbb{C}} I^\sharp(-Y) \leqslant \dim_{\mathbb{C}} KHI(-Y,K) \leqslant \dim_{\mathbb{C}} SHI(-H,-\gamma),$$

The proof of Theorem 1.2 makes use of sutured instanton Floer homology of balanced sutured manifolds with tangles (*c.f.* Xie and Zhang [XZ19]). The technique also applies to a generalization of the first inequality, which also generalizes a result for null-homologous knots by Wang [Wan20].

Proposition 1.3. *Suppose Y is a closed 3-manifold and $K \subset Y$ is a knot so that*

$$[K] = 0 \in H_1(Y;\mathbb{Q}).$$

Then we have

$$\dim_{\mathbb{C}} I^\sharp(-Y) \leqslant \dim_{\mathbb{C}} KHI(-Y,K).$$

In Theorem 1.2, we bound the dimensions of $I^\sharp(-Y)$ and $KHI(-Y,K)$ by that of another instanton Floer homology $SHI(-H,-\gamma)$, which is still difficult to compute. However, in the special case when H is a handlebody, an upper bound of $\dim_{\mathbb{C}} SHI(H,\gamma)$ can be obtained via bypass exact triangles (for bypass exact triangle, *c.f.* Baldwin and Sivek [BS16a], and for the algorithm to obtain an upper bound, *c.f.* Ghosh and Li [GL19]). In particular, we apply this theorem to (1,1)-knots in lens spaces, whose Heegaard diagrams are known to be simple, and obtain the following theorem.

Theorem 1.4. *Suppose Y is a lens space, and $K \subset Y$ is a (1,1)-knot. Then we have*

$$\dim_{\mathbb{C}} KHI(Y,K) \leqslant \mathrm{rk}_{\mathbb{Z}} \widehat{HFK}(Y,K).$$

Previously, Hedden, Herald, and Kirk [HHK14] studied explicit sets of generators of the instanton chain complexes for some special families of knots, which would also offer upper bounds of $\dim_{\mathbb{C}} KHI$. However, their bounds might not always be sharp. Theorem 1.4 thus offers a new way to obtain upper bounds of $\dim_{\mathbb{C}} KHI$, and the following corollary indicates that these bounds are sharp in many occasions.

Corollary 1.5. *Suppose $K \subset S^3$ is a (1,1)-knot that is also a Heegaard Floer L-space knot. Then we have*

$$\dim_{\mathbb{C}} KHI(S^3,K) = \mathrm{rk}_{\mathbb{Z}} \widehat{HFK}(S^3,K).$$

Proof. Suppose the Alexander polynomial of K is $\Delta_K(t) = \sum_{i\in\mathbb{Z}} c_i t^i$. From Ozsváth and Szabó [OS05], we have

$$\mathrm{rk}_{\mathbb{Z}} \widehat{HFK}(S^3,K) = \sum_{i\in\mathbb{Z}} |c_i|.$$

Then from Kronheimer and Mrowka [KM10a], or Lim [Lim09], we have

$$\dim_{\mathbb{C}} KHI(Y,K) \geqslant \sum_{i\in\mathbb{Z}} |c_i|.$$

Hence Theorem 1.4 applies and we conclude the desired equality. □

There is a spectral sequence from reduced Khovanov homology to instanton knot homology (*c.f.* Kronheimer and Mrowka [KM11a]). Corollary 1.5 would provide many examples whose related spectral sequences have some non-trivial intermediate pages. In particular, for torus knots, previously there were only partial computations of KHI from the related specatral sequences (*c.f.* [KM14, LZ20]), while Corollary 1.5 applies to torus knots directly since torus knots admit lens spaces surgeries (*c.f.* Moser [Mos71]).

Corollary 1.6. *For a torus knot $K = T_{(p,q)}$, suppose its Alexander polynomial is*

$$\Delta_K(t) = t^{-\frac{(p-1)(q-1)}{2}} \frac{(t^{pq}-1)(t-1)}{(t^p-1)(t^q-1)} = \sum_{i=-\frac{(p-1)(q-1)}{2}}^{\frac{(p-1)(q-1)}{2}} c_i t^i.$$

Then we have

$$\dim_{\mathbb{C}} KHI(S^3,K,i) = |c_i|,$$

where i denotes the Alexander grading of $KHI(S^3,K)$.

In general lens spaces, we may obtain lower bounds from graded Euler characteristics and show more examples of Conjecture 1.1. For example, the Heegaard Floer homology of constrained knots

in lens spaces, introduced by the second author [Ye20], is determined by their graded Euler characteristics. However, in the manifold other than S^3, it is not known if graded Euler characteristics of $\widehat{HFK}$ and KHI are equal. We will deal with this in a forthcoming paper [LY].

A **(Heegaard) Floer simple knot** is a knot K in a rational homology sphere Y so that

$$\mathrm{rk}_{\mathbb{Z}}\widehat{HFK}(Y,K) = \mathrm{rk}_{\mathbb{Z}}\widehat{HF}(Y) = |H_1(Y)|, \tag{1.1}$$

where $|H_1(Y)|$ is the order of the first homology group of Y with $\mathbb{Z}$ coefficient. A **(Heegaard Floer) L-space** is a rational homology sphere Y so that the latter equality in (1.1) holds. Examples of Floer simple knots include simple knots in lens spaces (*c.f.* Rasmussen [Ras07]). Rasmussen and Rasmussen [RR17] proved that for Floer simple knots, there is an interval in $\mathbb{Q} \cup \{\infty\}$ so that a Dehn surgery gives an L-space if and only if the surgery slope is in the interval. In the interior of the interval, the dual knots are also Floer simple knots.

A similar result may hold for instanton Floer homology. A knot K in a rational homology sphere Y is called a **instanton Floer simple knot** if

$$\dim_{\mathbb{C}} KHI(Y,K) = \dim_{\mathbb{C}} I^{\sharp}(Y) = |H_1(Y)|. \tag{1.2}$$

A **instanton Floer L-space** is a rational homology sphere Y so that the latter equality in (1.2) holds.

Proposition 1.7. *Simple knots in lens spaces are instanton Floer simple knots.*

Proof. Suppose K is a simple knot in $Y = L(p,q)$. Combining Theorem 1.2, Theorem 1.4 and results in [Ras07], we have

$$\dim_{\mathbb{C}} I^{\sharp}(Y) \leqslant \dim_{\mathbb{C}} KHI(Y,K) \leqslant \mathrm{rk}_{\mathbb{Z}}\widehat{HFK}(Y,K) = p.$$

By Scaduto [Sca15, Corollary 1.4], we have

$$\dim_{\mathbb{C}} I^{\sharp}(Y) \geqslant |H_1(Y)| = p.$$

Hence we conclude the desired equality. □

Inspired by the proof of Theorem 1.4, we prove the following theorem by induction on Heegaard diagrams.

Theorem 1.8. *Suppose K is a simple knot in a lens space Y. Fixing a framing on $\partial Y(K)$ by picking an arbitrary longitude of K. Then there exists an integer $N \geqslant 0$ so that for any $r \in \mathbb{Q}$ with $|r| \geqslant N$, the manifold obtained by a surgery of slope r along K is an instanton L-space, and the dual knot is also an instanton Floer simple knot.*

Remark 1.9. In general, framed instanton Floer homology is very difficult to compute. Only some special families of 3-manifolds were studied (*c.f.* [SS18, LPCS20, BS20, ABDS20]). Theorem 1.8 provides many new examples whose framed instanton Floer homology can be computed.

In the second part, we derive some properties of framed instanton Floer homology in analogy to monopole or Heegaard Floer homology.

Theorem 1.10. *Suppose $\widehat{Y}$ is a closed 3-manifold, and $\widehat{K} \subset \widehat{Y}$ is a rationally null-homologous knot. Let $\widehat{Y}(\widehat{K}) = \widehat{Y} \backslash \mathrm{int}(N(\widehat{K}))$ be the knot complement and let S be a Seifert surface of $\widehat{K}$. Suppose futher that $\lambda = \partial S \subset \partial\widehat{Y}(\widehat{K})$ is connected and $|\hat{\mu} \cdot \lambda| = q$, where $\hat{\mu}$ is the meridian of $\widehat{K}$ on*

$\partial\widehat{Y}(\widehat{K})$ and the dot $\cdot$ denotes the algebraic intersection number. Then there is a decomposition up to isomorphism associated to $\widehat{K}$:

$$I^\sharp(\widehat{Y}) \cong \bigoplus_{i=0}^{q-1} I^\sharp(\widehat{Y}, i).$$

When $H_1(\widehat{Y}) = \mathbb{Z}_q$ and $\widehat{K}$ represents a generator of $H_1(\widehat{Y})$, we can regard the decomposition in Theorem 1.10 as an analog of the torsion spin^c decompositions

$$\widetilde{HM}(\widehat{Y}) = \bigoplus_{\mathfrak{s}\in \mathrm{Spin}^c(\widehat{Y})} \widetilde{HM}(\widehat{Y}, \mathfrak{s}), \text{ and } \widehat{HF}(\widehat{Y}) = \bigoplus_{\mathfrak{s}\in \mathrm{Spin}^c(\widehat{Y})} \widehat{HF}(\widehat{Y}, \mathfrak{s}).$$

Here $\widetilde{HM}$ is the tilde version of monopole Floer homology [Blo09]. An evidence is the following.

Proposition 1.11. *Under the hypothesis and the statement of Theorem 1.10, there is a spectral sequence $\{E_r, d_r\}$ from $E_1 = KHI(\widehat{Y}, \widehat{K})$ to $I^\sharp(\widehat{Y})$. Moreover, this spectral sequence respects the decomposition in Theorem 1.10, i.e., there is a decomposition*

$$(E_r, d_r) = \bigoplus_{i=0}^{q-1} (E_{r,i}, d_{r,i})$$

so that $E_{r,i}$ converges to $I^\sharp(\widehat{Y}, i)$.

To provide more evidence, we will prove the following result in a forthcoming paper [LY].

Proposition 1.12. *Under the hypothesis and the statement of Theorem 1.10, there is a well-defined $\mathbb{Z}_2$ grading on $I^\sharp(\widehat{Y}, i)$. Suppose $\mu \subset \partial\widehat{Y}(\widehat{K})$ is a simple closed curve so that $|\mu \cdot \lambda| = 1$ and suppose Y is the manifold obtained by Dehn filling along μ. For any integer $i \in [0, q-1]$, we have*

$$\chi(I^\sharp(\widehat{Y}, i)) = \chi(I^\sharp(Y)).$$

In particular, if $\widehat{Y}$ is an instanton L-space and $Y = S^3$, then for any integer $i \in [0, q-1]$, we have

$$I^\sharp(\widehat{Y}, i) \cong \mathbb{C}.$$

Remark 1.13. The defect of the decomposition of $I^\sharp(\widehat{Y})$ in Theorem 1.10 is that currently we do not know if it is independent of the choice of $\widehat{K}$ inside $\widehat{Y}$.

For integral surgeries on a null-homologous knot, we obtain more results inspired from the large surgery formula of Heegaard Floer homology (*c.f.* Ozsváth and Szabó [OS04a]). For a null homologous knot K in a closed 3-manifold Y, let the basis of $\partial Y(K)$ be formed by the meridian of K and the boundary of a Seifert surface.

Proposition 1.14. *Suppose Y is a closed 3-manifold and $K \subset Y$ is a null-homologous knot. Suppose $\widehat{Y}$ is obtained from Y by performing the $q/1$ surgery along K, with $q \geqslant 2g(K) + 1$. Then for any integer $i \in [0, q - 2g(K) - 1]$ and $i = q - 1$, we have*

$$I^\sharp(\widehat{Y}, i) \cong I^\sharp(Y).$$

Proposition 1.15. *Suppose Y is a closed 3-manifold and $K \subset Y$ is a null-homologous knot. Suppose $\widehat{Y}_q$ is obtained from Y by performing the $q/1$ surgery along K. If $q \geqslant 2g(K) + 1$, then for any integer $i \in [0, q-1]$, we have*

$$I^\sharp(\widehat{Y}_q, i) \cong I^\sharp(\widehat{Y}_{q+1}, i + 1).$$

Remark 1.16. Proposition 1.14 states that for large integral surgeries, all but $(2g-1)$ many summands are isomorphic to the $I^\sharp(Y)$. Proposition 1.15 then states that all those summands will stabilize when performing large integral surgeries with increasing slopes.

The proofs of Theorem 1.10, Proposition 1.11, Proposition 1.14 and Proposition 1.15 make use of $SHI(\widehat{Y}(\widehat{K}),\gamma)$ for some special suture $\gamma \subset \partial\widehat{Y}(\widehat{K})$. In [Li19, Section 3], the first author constructed a grading, *i.e.*, a decomposition of sutured instanton Floer homology associated to a properly embedded surface with some mild conditions. The Seifert surface plays role of this properly embedded surface, which decomposes $SHI(Y(K),\gamma)$ for any suture γ. Then we are able to identify some direct summands of the decomposition with $I^\sharp(\widehat{Y})$. Explicitly, we can construct the decomposition by the following proposition.

Proposition 1.17. *Suppose Y is a closed 3-manifold and $K \subset Y$ is a null-homologous knot. Suppose $\widehat{Y}$ is obtained from Y by performing a q/p-surgery along K with $q>0$. Then there is a set $\mathcal{G}$ of sutures on the boundary of the knot complement $Y(K)$, determined by the integer q and the genus $g(K)$ of K so that the followings hold.*

(1) For any suture $\gamma \in \mathcal{G}$, sutured instanton Floer homology carries a $\mathbb{Z}$ grading, i.e.,

$$\underline{\mathrm{SHI}}(-Y(K),\gamma) = \bigoplus_{i\in\mathbb{Z}} \underline{\mathrm{SHI}}(-Y(K),\gamma,i).$$

(2) For any suture $\gamma \in \mathcal{G}$, there is an integer i_γ so that there is an isomorphism

$$f_{K,\gamma} : I^\sharp(-\widehat{Y}) \xrightarrow{\cong} \bigoplus_{j=i_\gamma}^{i_\gamma+q-1} \underline{\mathrm{SHI}}(-Y(K),\gamma,j).$$

(3) For any two sutures $\gamma_1,\gamma_2 \in \mathcal{G}$, there is an isomorphism with respect to gradings

$$g_{K,\gamma_1,\gamma_2} : \bigoplus_{j=i_{\gamma_1}}^{i_{\gamma_1}+q-1} \underline{\mathrm{SHI}}(-Y(K),\gamma_1,j) \xrightarrow{\cong} \bigoplus_{j=i_{\gamma_2}}^{i_{\gamma_2}+q-1} \underline{\mathrm{SHI}}(-Y(K),\gamma_2,j).$$

(4) For any two sutures $\gamma_1,\gamma_2 \in \mathcal{G}$, there is a commutative diagram

$$\begin{tikzcd}
\bigoplus_{j=i_{\gamma_1}}^{i_{\gamma_1}+q-1} \underline{\mathrm{SHI}}(-Y(K),\gamma_1,j) \arrow[rr, "g_{K,\gamma_1,\gamma_2}"] & & \bigoplus_{j=i_{\gamma_2}}^{i_{\gamma_2}+q-1} \underline{\mathrm{SHI}}(-Y(K),\gamma_2,j) \\
& I^\sharp(-\widehat{Y}) \arrow[ul, "f_{K,\gamma_1}"] \arrow[ur, "f_{K,\gamma_2}"'] &
\end{tikzcd}$$

The technique to construct the decomposition has other applications on instanton Floer homology. Let $\underline{\mathrm{KHI}}^-$ be the minus version of instanton knot homology and let $\tau_I(K)$ be the instaton tau invariant of a knot $K \subset S^3$, constructed by the first author [Li19]. We prove the following proposition.

Proposition 1.18. *Suppose $K \subset S^3$ is a knot. Let $\bar{K}$ be the mirror knot of K. Then*

$$\tau_I(\bar{K}) = -\tau_I(K).$$

Also, using this technique, we can obtain a dimension formula of framed instanton Floer homology for large surgeries on a null-homologous knot in a 3-manifold.

Theorem 1.19. *Suppose Y is a closed 3-manifold and $K \subset Y$ is a null-homologous knot. Suppose $\widehat{Y}_{q_1/p_1}$, $\widehat{Y}_{q_2/p_2}$, and $\widehat{Y}_{q_3/p_3}$ are obtained from Y by surgeries along K with slopes q_1/p_1, q_2/p_2, and q_3/p_3, respectively. Suppose further that $q_i > 0$ for $i \in \{1,2,3\}$ and these surgeries fit into a surgery exact triangle*

$$\begin{tikzcd} I^\sharp(\widehat{Y}_{q_1/p_1}) \arrow[rr, "f_1"] & & I^\sharp(\widehat{Y}_{q_2/p_2}) \arrow[dl, "f_2"] \\ & I^\sharp(\widehat{Y}_{q_3/p_3}) \arrow[ul, "f_3"] & \end{tikzcd} \tag{1.3}$$

Conside the indices mod 3. If there exists $i \in \{1,2,3\}$ so that $q_i > 2g(K)$, then $f_{i+1} = 0$, and hence the exact triangle (1.3) splits.

Corollary 1.20. *Suppose Y is a closed 3-manifold and $K \subset Y$ is a null-homologous knot. Suppose $\widehat{Y}_{q/p}$ is the 3-manifold obtained from Y by a q/p-surgery along K. Let $n = \lfloor q/p \rfloor$ be the maximal integer satisfying $n \leqslant q/p$. Then there exist maps $d, s : \mathbb{Z} \to \mathbb{Z}$, so that for $n \geqslant g(K)$, we have*

$$\dim_{\mathbb{C}} I^\sharp(\widehat{Y}_{q/p}) = d(n) \cdot p + s(n) \cdot q.$$

Furthermore, there exists an integer $N < 2g(K)$, so that for $n \geqslant N$, we have

$$d(n+1) = d(n), \text{ and } s(n) = \dim_{\mathbb{C}} I^\sharp(Y).$$

Proof. Suppose $r_3 = q_3/p_3 = q/p$ is the surgery slope and $\lfloor r \rfloor = n$. Then it is well known (*c.f.* Kronheimer, Mrowka, Ozsváth, and Szabó [KMOS07] or Baldwin and Sivek [BS20]) that there exist slopes $r_1 = q_1/p_1$ and $r_2 = q_2/p_2$ so that

(1) $p_1, p_2, q_1, q_2 > 0$.
(2) $q_3 = q_1 + q_2$, $p_3 = p_1 + p_2$.
(3) For $i \in \{1,2\}$, if $p_i = 1$, then q_i is either n or $n+1$. If $p_i \neq 1$, then $\lfloor r_i \rfloor = n$.
(4) r_1, r_2, and r_3 form a surgery exact triangle as in Theorem 1.19.

When $n \geqslant g(K)$, we know that $q_3 = q_1 + q_2 \geqslant 2g(K) + 1$. The hypothesis of Theorem 1.19 is satisfied, and we have

$$\dim_{\mathbb{C}}(I^\sharp(\widehat{Y}_{q_3/p_3}) = \dim_{\mathbb{C}} I^\sharp(\widehat{Y}_{q_1/p_1}) + \dim_{\mathbb{C}} I^\sharp(\widehat{Y}_{q_2/p_2}).$$

Hence from (2) we know that if $\dim_{\mathbb{C}} I^\sharp(\widehat{Y}_{q_1/p_1})$ and $\dim_{\mathbb{C}} I^\sharp(\widehat{Y}_{q_2/p_2})$ both satisfy the same linear relation, so is $\dim_{\mathbb{C}} I^\sharp(\widehat{Y}_{q_3/p_3})$. Then (1) and (2) ensure that we can apply an induction and (3) implies that the base cases are $r_1 = n$ and $r_2 = n+1$. We can fix the coefficients $r(n)$ and $d(n)$ by considering $\dim_{\mathbb{C}} I^\sharp(\widehat{Y}_n)$ and $\dim_{\mathbb{C}} I^\sharp(\widehat{Y}_{n+1})$.

When n is a large enough integer, we can apply Theorem 1.19 again and conclude

$$\dim_{\mathbb{C}} I^\sharp(\widehat{Y}_{n+1}) = \dim_{\mathbb{C}} I^\sharp(\widehat{Y}_n) + \dim_{\mathbb{C}} I^\sharp(Y).$$

Hence we obtain the desired result. □

Remark 1.21. In Theorem 1.19, the assumptions are only on the numerators of slopes. In principle, we could also write down some formula for surgeries of large denominators and small slopes. However, it would be much more complicated. On the other hand, the proof of Theorem 1.19 not only states the splitting of the exact triangle but also indicates some information about the decomposition (in the sense of Theorem 1.10) of $I^\sharp(\widehat{Y}_{q_i/p_i})$ for $i = \{1,2,3\}$.

Remark 1.22. For the dimension formula of surgeries, the special case for knots in S^3 has recently been resolved by Baldwin and Sivek [BS20]. However, their technique depends strongly on the fact that $I^\sharp(S^3)$ is 1-dimensional, and it is difficult to generalize the technique to general 3-manifolds.

Conventions. If not mentioned, homology groups and cohomology groups are with $\mathbb{Z}$ coefficient. (Sutured) Heegaard Floer homology is with $\mathbb{Z}$ coefficient, while (sutured) instanton Floer homology is with $\mathbb{C}$ coefficients.

If not mentioned, all manifolds are smooth, connected and oriented. Suppose M is an oriented manifold. Let $-M$ denote the same manifold with the reverse orientation, called the **mirror manifold** of M. If not mentioned, we do not consider orientations of knots. Suppose K is a knot in a 3-manifold M. Then $(-M, K)$ is the **mirror knot** in the mirror manifold. In S^3, the mirror knot is also denoted by $\bar{K}$.

For a manifold M, let $\mathrm{int}(M)$ denote its interior. For a submanifold A in a manifold Y, let $N(A)$ denote the tubular neighborhood. The knot complement of K in Y is denoted by $Y(K) = Y \backslash \mathrm{int}(N(K))$.

For a simple closed curve on a surface, we do not distinguish between its homology class and itself. The algebraic intersection number of two curves α and β on a surface is denoted by $\alpha \cdot \beta$, while the number of intersection points between α and β is denoted by $|\alpha \cap \beta|$. A basis (m, l) of $H_1(T^2; \mathbb{Z})$ satisfies $m \cdot l = -1$. The **surgery** means the Dehn surgery and the slope (in some basis) is usually denoted by q/p (rather than p/q).

A **rational homology sphere** is a closed 3-manifold whose homology groups with rational coefficients are isomorphic to those of S^3. The **integral homology sphere** is defined similarly. A knot $K \subset Y$ is called **null-homologous** if it represents trivial homology class in $H_1(Y; \mathbb{Z})$, while it is called **rationally null-homologous** if it represents trivial homology class in $H_1(Y; \mathbb{Q})$.

For $r \in \mathbb{R}$, let $\lceil r \rceil$ and $\lfloor r \rfloor$ be the minimal integer and the maximal integer satisfying $\lceil r \rceil \geqslant r$ and $\lfloor r \rfloor \leqslant r$, respectively. An argument holds for **large enough** n if there exists a fixed $N \in \mathbb{Z}$ so that the argument holds for any integer $n > N$. An argument holds for **small enough** n if there exists a fixed $N \in \mathbb{Z}$ so that the argument holds for any integer $n < N$.

Organization. The paper is organized as follows. In Section 2, we review some backgrounds, including instanton Floer homology of closed manifolds, Heegaard Floer homology and instanton Floer homology of balanced sutured manifolds, and bypass attachments on balanced sutured manifolds. In Section 3, we prove Theorem 1.2 and Proposition 1.3 by a dimension inequality from tangles in balanced sutured manifolds. Then we deal with (1,1)-knots and prove Theorem 1.4 and Theorem 1.8. In Section 4, we prove Theorem 1.10 by Proposition 1.17, and then prove Proposition 1.11, Proposition 1.18, Proposition 1.14, Proposition 1.15, and Theorem 1.19. Finally, in Section 5, we discuss some possible future directions.

Acknowledgement. The first author is partially supported by his Ph.D. advisor Tomasz Mrowka's NSF Grant 1808794. The second author would like to thank his supervisor Jacob Rasmussen for patient guidance and helpful comments, and thank his parents for support and constant encouragement. The authors would also like to thank Ciprian Manolescu for helpful comments on the draft of the paper, and thank John Baldwin, Qilong Guo, Joshua Wang, Yi Xie, and Ruide Fu for helpful conversations. The second author is also grateful to Yi Liu for inviting him to Peking University, and Zekun Chen, Honghuai Fang, and Longke Tang for the company when he was writing this paper.

2. Instanton Floer homology and sutured manifolds

2.1. Instanton Floer homology.

Definition 2.1. Suppose Y is a closed 3-manifold and ω is a closed 1-submanifold in Y. Suppose further that there is a close oriented surface $\Sigma \subset Y$ of genus at least one so that the algebraic intersection number $\omega \cdot \Sigma$ is odd. Then the pair (Y, ω) is called an **admissible pair**.

We can construct instanton Floer homology for admissible pairs.

Theorem 2.2 (Floer [Flo90])**.** *Suppose (Y, ω) is an admissible pair. Then there is a finite-dimensional complex vector space $I^\omega(Y)$ called the **instanton Floer homology** of (Y, ω).*

Suppose (Y, ω) and (Y', ω') are two admissible pairs and suppose (W, ν) is a cobordism from (Y, ω) to (Y', ω'). Then there exists a complex-linear map

$$I(W, \nu) : I^\omega(Y) \to I^{\omega'}(Y'),$$

*called the **cobordism map** of (W, ν).*

For an admissible pair (Y, ω), any homology class $\alpha \in H_*(Y)$ induces a complex-linear action on the instanton Floer homology:

$$\mu(\alpha) : I^\omega(Y) \to I^\omega(Y).$$

For any two homology classes $\alpha_1, \alpha_2 \in H_*(Y)$, we have

$$\mu(\alpha_1 + \alpha_2) = \mu(\alpha_1) + \mu(\alpha_2), \text{ and } \mu(\alpha_1)\mu(\alpha_2) = (-1)^{\deg(\alpha_1)\deg(\alpha_2)}\mu(\alpha_2)\mu(\alpha_1).$$

If $b_2(Y) > 0$, we can pick a basis $\beta_1, \dots, \beta_n$ of $H_2(Y; \mathbb{Q})$ and consider the simultaneous generalized eigenspaces of all the actions $\mu(\beta_1), \dots, \mu(\beta_n)$. The simultaneous eigenvalues, as a tuple $(\lambda_1, \dots, \lambda_n)$, can be viewed as a linear map from $H_2(Y; \mathbb{Q})$ to $\mathbb{Q}$. This linear map is the analog of (the first Chern classes of) non-torsion spin^c structures in Heegaard Floer homology. However, if $\alpha = 0 \in H_2(Y; \mathbb{Q})$, then $\mu(\alpha) = 0$. Hence for a rational homology sphere Y, all μ-actions associated to second homology classes are zero, and we cannot obtain an effective decomposition from μ.

Definition 2.3 (Kronheimer and Mrowka [KM10b])**.** Suppose (Y, ω) is an admissible pair, R is a closed surface of genus at least one, and $\omega \cdot R$ is odd. Let $I^\omega(Y|R)$ be the $(2g(R) - 2, 2)$-generalized eigenspaces of the pair of actions $((\mu(R), \mu(\mathrm{pt}))$ on $I^\omega(Y)$, where pt is any fixed basepoint on Y.

Suppose M is a compact oriented 3-manifold with torus boundary. Suppose $\omega \subset M$ is a closed 1-submanifold so that there exists a closed surface Σ of genus at least one with $\omega \cdot \Sigma$ odd. Let $i : \partial M \to M$ be the inclusion, and let

$$i_* : H_1(\partial M; \mathbb{Q}) \to H_1(M; \mathbb{Q}). \tag{2.1}$$

be the induced map on homology. Let $\gamma_1, \gamma_2, \gamma_3$ be three oriented simple closed curves on ∂M so that

$$\gamma_1 \cdot \gamma_2 = \gamma_2 \cdot \gamma_3 = \gamma_3 \cdot \gamma_1 = -1.$$

For $i \in \{1, 2, 3\}$, let Y_i be the closed 3-manifold obtained by Dehn filling along γ_i:

$$Y_i = M \underset{\gamma_i = \{1\} \times \partial D^2}{\cup} S^1 \times D^2.$$

Then clearly for $i \in \{1, 2, 3\}$, (Y_i, ω) are all admissible pairs. We have the following theorem.

Theorem 2.4 ([Flo90])**.** *There is an exact triangle*

$$\begin{tikzcd} I^\omega(Y_1) \arrow[rr, "f_1"] & & I^\omega(Y_2) \arrow[dl, "f_2"] \\ & I^\omega(Y_3) \arrow[ul, "f_3"] & \end{tikzcd} \tag{2.2}$$

Furthermore, all maps in the exact triangle (2.2) are induced by cobordism maps.

Remark 2.5. Note in Floer [Flo90] or Scaduto [Sca15], one has to add some extra component to ω in one of the 3-manifolds Y_1, Y_2, and Y_3 to make the exact triangle hold. However, from [BS20, Section 2.2], Baldwin and Sivek showed that one could wisely choose some other 1-submanifold ω' to start with. After adding the extra component coming from the original exact triangle, we finally arrive at a 1-submanifold representing the same homology class as ω for all three 3-manifolds.

2.2. **Balanced sutured manifolds.**

Definition 2.6 ([Juh06, KM10b])**.** A **balanced sutured manifold** (M, γ) consists of a compact oriented 3-manifold M with non-empty boundary together with a closed 1-submanifold γ on ∂M. Let $A(\gamma) = [-1, 1] \times \gamma$ be an annular neighborhood of $\gamma \subset \partial M$ and let $R(\gamma) = \partial M \backslash \mathrm{int}(A(\gamma))$. They satisfy the following properties.

(1) Neither M nor $R(\gamma)$ has a closed component.
(2) If $\partial A(\gamma) = -\partial R(\gamma)$ is oriented in the same way as γ, then we require this orientation of $\partial R(\gamma)$ induces one on $R(\gamma)$. The induced orientation on $R(\gamma)$ is called the **canonical orientation**.
(3) Let $R_+(\gamma)$ be the part of $R(\gamma)$ so that the canonical orientation coincides with the induced orientation on ∂M, and let $R_-(\gamma) = R(\gamma) \backslash R_+(\gamma)$. We require that $\chi(R_+(\gamma)) = \chi(R_-(\gamma))$. If γ is clear in the contents, we simply write $R_\pm = R_\pm(\gamma)$.

For a balanced sutured manifold (M, γ), Juhász constructed sutured (Heegaard) Floer homology, and Kronheimer and Mrowka constructed sutured instanton Floer homology.

Definition 2.7 ([Juh06])**.** A **balanced diagram** is a triple (Σ, α, β) so that the followings hold.

(1) We have Σ being a connected compact oriented surface with boundary.
(2) We have $\alpha = \{\alpha_1, \ldots, \alpha_n\}$ and $\beta = \{\beta_1, \ldots, \beta_n\}$ being two sets of pair-wise disjoint simple closed curves in the interior of Σ. We do not distinguish between the set and the union of curves.
(3) The maps $\pi_0(\partial\Sigma) \to \pi_0(\Sigma \backslash \alpha)$ and $\pi_0(\partial\Sigma) \to \pi_0(\Sigma \backslash \beta)$ are surjective.

Let M be the 3-manifold obtained from $\Sigma \times [-1, 1]$ by attaching 3–dimensional 2–handles along $\alpha_i \times \{-1\}$ and $\beta_i \times \{1\}$ for each integer $i \in [1, n]$ and let $\gamma = \partial\Sigma \times \{0\}$. A balanced diagram (Σ, α, β) is called **compatible** with a balanced sutured manifold (N, ν) if the sutured manifold (M, γ) is diffeomorphic to (N, ν). A sutured manifold (N, ν) is called a **product sutured manifold** if it is compatible with $(\Sigma, \varnothing, \varnothing)$ for some Σ.

Theorem 2.8 (Juhász [Juh06])**.** *Suppose (M, γ) is a balanced sutured manifold. There is a balanced diagram compatible with (M, γ). Then there is a module $SFH(M, \gamma)$ over $\mathbb{Z}$, which is called the* ***sutured (Heegaard) Floer homology*** *of (M, γ).*

Remark 2.9. Sutured Heegaard Floer homology generalizes Heegaard Floer homology [OS04b], and knot Flor homology [OS04a, Ras03]. Suppose Y is a closed 3-manifold and $K \subset Y$ is a knot. Let

$Y(1)$ be obtained from Y by removing a 3-ball and let δ be a simple closed curve on $\partial Y(1)$. Let γ consist of two meridians of K. Then there are canonical isomorphisms

$$SFH(Y(1),\delta) \cong \widehat{HF}(Y), \text{ and } SFH(Y(K),\gamma) \cong \widehat{HFK}(Y,K).$$

Theorem 2.10 (Kronheimer and Mrowka [KM10b])**.** *For a balanced sutured manifold (M,γ), one can associate a triple (Y,R,ω), called a* ***closure*** *of (M,γ), so that the followings hold.*

(1) We have Y being a closed 3-manifold so that M is an oriented submanifold of Y.
(2) We have $R \subset Y$ being a connected closed oriented surface of genus at least one so that $R_+(\gamma)$ is an oriented submanifold of R and $R \cap \mathrm{int}(M) = \varnothing$.
(3) We have $\omega \subset Y$ being a simple closed curve so that it intersects R transversely at one point and $\omega \cap \mathrm{int}(M) = \varnothing$.

Moreover, the isomorphism class of $I^\omega(Y|R)$ as in Definition 2.3 is independent of the choices of the triple (Y,R,ω) and is a topological invariant of (M,γ).

Definition 2.11. For a balanced sutured manifold (M,γ), the vector space $I^\omega(Y|R)$ for a closure (Y,R,ω) of (M,γ) is called the **sutured instanton Floer homology** of (M,γ). It is also denoted by $SHI(M,\gamma)$ to stress the independence of choices of closures as claimed in Theorem 2.10.

One of the most important properties of these two kinds of sutured Floer homologies is that they detect the tautness of balanced sutured manifolds.

Definition 2.12 ([Juh06])**.** A sutured manifold (M,γ) is called **taut** if M is irreducible and $R_\pm(\gamma)$ are both incompressible and Thurston norm-minimizing in the homology class they represent in $H_2(M,\gamma)$.

Theorem 2.13 (Juhász [Juh06, Juh08], Kronheimer and Mrowka [KM10b])**.** *Suppose (M,γ) is a balanced sutured manifold so that M is irreducible, then the following three statements are equivalent.*

- *(M,γ) is taut.*
- *$SFH(M,\gamma) \neq 0$.*
- *$SHI(M,\gamma) \neq 0$.*

Another important property is about the product manifold.

Theorem 2.14 (Ni [Ni07], Kronheimer and Mrowka [KM10b])**.** *Suppose (M,γ) is a sutured manifold and a homology product. Then the following three statements are equivalent.*

- *(M,γ) is a product sutured manifold.*
- *$SFH(M,\gamma) \cong \mathbb{Z}$.*
- *$SHI(M,\gamma) \cong \mathbb{C}$.*

In Theorem 2.10, only the isomorphism class of SHI is well-defined. Later, Baldwin and Sivek improved the naturality of SHI, making it possible to discuss about elements in SHI. Similar work is done by Juhász, Thurston and Zemke [JTZ12] for SFH over $\mathbb{Z}_2$, and Kutluhan, Sivek, and Taubes [KST13] for sutured ECH.

Theorem 2.15 (Baldwin and Sivek [BS15])**.** *For a balanced sutured manifold (M,γ) and any two closures (Y_1,R_1,ω_1) and (Y_2,R_2,ω_2), there is an isomorphism*

$$\Phi_{1,2} : I^{\omega_1}(Y_1|R_1) \xrightarrow{\cong} I^{\omega_2}(Y_2|R_2)$$

that is well-defined up to multiplication by a unit, i.e., a nonzero complex number. Furthermore, the isomorphism Φ satisfies the following two conditions.

(1) If $(Y_1, R_1, \omega_1) = (Y_2, R_2, \omega_2)$*, then*

$$\Phi_{1,2} \doteq \mathrm{id},$$

where $\doteq$ *means equal up to multiplication by a unit.*

(2) If there is a third closure (Y_3, R_3, ω_3)*, then we have*

$$\Phi_{1,3} \doteq \Phi_{2,3} \circ \Phi_{1,2} : I^{\omega_1}(Y_1|R_1) \to I^{\omega_3}(Y_3|R_3).$$

From Theorem 2.15, for a balanced sutured manifold (M, γ), Baldwin and Sivek constructed a projective transitive system based on the vector space $I^{\omega}(Y|R)$ coming from different closures of (M, γ) and the canonical maps Φ between them. There is then a canonical module associated to the projective transitive system, which we write as

$$\underline{\mathrm{SHI}}(M, \gamma).$$

We can regard it as a complex vector space well defined up to multiplication by a unit. From now on, we will write $\underline{\mathrm{SHI}}(M, \gamma)$ for sutured instanton Floer homology.

Definition 2.16 ([KM10b, Section 7])**.** Suppose Y is a closed 3-manifold and $K \subset Y$ is a knot. Let $(Y(1), \delta)$ and $(Y(K), \gamma)$ be sutured manifolds defined as in Remark 2.9. The **framed instanton Floer homology** of Y is defined by

$$I^{\sharp}(Y) := I^{S^1}(Y \sharp (S^1 \times T^2)|\{1\} \times T^2) \cong \underline{\mathrm{SHI}}(Y(1), \delta).$$

The **instanton knot homology** of (Y, K) is defined by

$$KHI(Y, K) := \underline{\mathrm{SHI}}(Y(K), \gamma).$$

Remark 2.17. In [BS15], in order to make the definition of KHI independent of different choices of knot complements and the position of the meridianal suture, Baldwin and Sivek also added a basepoint to the data. However, since in this paper we only care about the dimension of KHI, we overlook this ambiguity and omit the basepoint from our notation. Also, the definition of $\underline{\mathrm{SHI}}(Y(1), \delta)$ depends on a choice of basepoint. By the same reason we omit the basepoint.

The surgery exact triangle in Theorem 2.4 can easily be generalized for KHI. Suppose M is a compact oriented 3-manifold with torus boundary and $K \subset M$ is a knot. Let $\gamma_1, \gamma_2, \gamma_3$ be three oriented simple closed curves on ∂M so that

$$\gamma_1 \cdot \gamma_2 = \gamma_2 \cdot \gamma_3 = \gamma_3 \cdot \gamma_1 = -1.$$

For $i \in \{1, 2, 3\}$, let Y_i be a closed 3-manifold obtained by Dehn filling along γ_i and let K_i be the knot induced by K:

$$(Y_i, K_i) = (M, K) \underset{\gamma_i = \{1\} \times \partial D^2}{\cup} S^1 \times D^2.$$

Theorem 2.18. *There is an exact triangle*

$$\begin{tikzcd} KHI(Y_1, K_1) \arrow[rr, "f_1"] & & KHI(Y_2, K_2) \arrow[dl, "f_2"] \\ & KHI(Y_3, K_3) \arrow[ul, "f_3"] & \end{tikzcd} \tag{2.3}$$

Furthermore, all maps in the exact triangle (2.3) are induced by cobordism maps between corresponding closures of sutured manifolds induced by (Y_i, K_i)*.*

Suppose (M,γ) is a balanced sutured manifold and $S \subset M$ is a properly embedded surface. We state results about the decomposition of $\underline{\mathrm{SHI}}(M,\gamma)$ associated to S.

Definition 2.19 ([GL19, Definition 2.25])**.** Suppose (M,γ) is a balanced sutured manifold and $S \subset (M,\gamma)$ is a properly embedded surface in M. The surface S is called an **admissible surface** if the followings hold.

(1) Every boundary component of S intersects γ transversely and non-trivially.
(2) We require that $\frac{1}{2}|S \cap \gamma| - \chi(S)$ is an even integer.

For an admissible surface $S \subset (M,\gamma)$, there is a well-defined $\mathbb{Z}$ grading on $\underline{\mathrm{SHI}}(M,\gamma)$.

Theorem 2.20 (Li [Li19])**.** *Suppose (M,γ) is a balanced sutured manifold and $S \subset (M,\gamma)$ is an admissible surface with $n = \frac{1}{2}|S \cap \gamma|$. Then there exists a closure (Y,R,ω) of (M,γ) so that S extends to a closed surface $\bar{S} \subset Y$ with $\chi(\bar{S}) = \chi(S) - n$. Let $SHI(M,\gamma,S,i)$ denote the $(2i)$-generalized eigenspace of $\mu(\bar{S})$ acting on $SHI(M,\gamma) = I^{\omega}(Y|R)$. Then $SHI(M,\gamma,S,i)$ is preserved by the canonical maps in Theorem 2.15. Thus, the vector space*

$$\underline{\mathrm{SHI}}(M,\gamma,S,i)$$

is well-defined. Furthermore, the followings hold.

(1) If $|i| > \frac{1}{2}(n - \chi(S))$, then $\underline{\mathrm{SHI}}(M,\gamma,S,i) = 0$.
(2) If there is a sutured manifold decomposition $(M,\gamma) \overset{S}{\rightsquigarrow} (M',\gamma')$ in the sense of Gabai [Gab83], then we have

$$\underline{\mathrm{SHI}}(M,\gamma,S,\frac{1}{2}(n-\chi(S))) \cong \underline{\mathrm{SHI}}(M',\gamma').$$

(3) For any $i \in \mathbb{Z}$, we have

$$\underline{\mathrm{SHI}}(M,\gamma,S,i) = \underline{\mathrm{SHI}}(M,\gamma,-S,-i).$$

Remark 2.21. In [Li19], the grading was only constructed for an admissible surface with a connected boundary. When generalizing it to admissible surfaces with multiple boundary components, more choices arise in the construction of the grading. This new ambiguity was reduced to a combinatorial problem as discussed in [Li19] and was then resolved in [Kav19].

Remark 2.22. The first statement in Theorem 2.20 comes from an adjunction inequality from instanton Floer homology [KM10b, Proposition 7.5]. The second statement in Theorem 2.20 is a restatement of [KM10b, Proposition 7.11].

Suppose (M,γ) is a balanced sutured manifold, and $S \subset M$ is a properly embedded surface. If S is not admissible, then we can perform isotopies on S to make it admissible.

Definition 2.23. Suppose (M,γ) is a balanced sutured manifold, and S is a properly embedded oriented surface. A **stabilization** of S is an isotopy of S to a surface S' in the following sense. This isotopy creates a new pair of intersection points:

$$\partial S' \cap \gamma = (\partial S \cap \gamma) \cup \{p_+, p_-\}.$$

We require that there are arcs $\alpha \subset \partial S'$ and $\beta \subset \gamma$, oriented in the same way as $\partial S'$ and γ, respectively, so that the followings hold.

(1) We have $\partial\alpha = \partial\beta = \{p_+, p_-\}$.
(2) The curves α and β cobound a disk D so that $\mathrm{int}(D) \cap (\gamma \cup \partial S') = \varnothing$.

The stabilization is called **negative** if D can be oriented so that $\partial D = \alpha \cup \beta$ as oriented curves. It is called **positive** if $\partial D = (-\alpha) \cup \beta$. See Figure 1. We denote by $S^{\pm k}$ the result of performing k many positive or negative stabilizations of S.

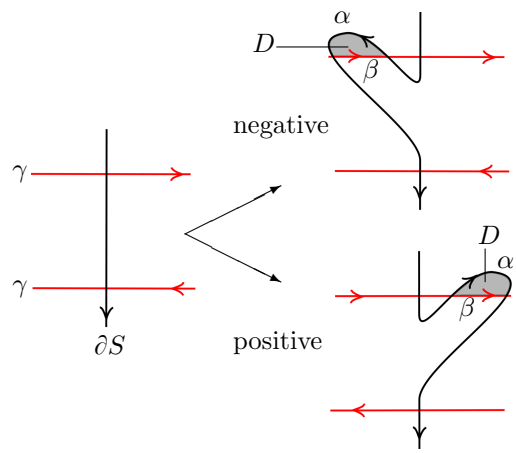


FIGURE 1. The positive and negative stabilizations of S.

The following lemma is straightforward.

Lemma 2.24. *Suppose (M, γ) is a balanced sutured manifold, and S is a properly embedded oriented surface. Suppose S^+ and S^- are the results of performing a positive and a negative stabilization on S, respectively. Then, we have the following.*

(1) If we decompose (M, γ) along S or S^+, then the resulting two balanced sutured manifolds are diffeomorphic.

(2) If we decompose (M, γ) along S^-, then the resulting balanced sutured manifold (M', γ') is not taut, as $R_\pm(\gamma')$ would both become compressible.

Remark 2.25. Note the definition of stabilizations of a surface depends on the orientations of the suture and the surface. If the orientation of the suture reverses, then positive and negative stabilizations switch between each other. Similarly, if we switch the orientation of the surface, then the positive and negative stabilizations also switch between each other.

One important result that will be useful is to relate the gradings associated to different stabilizations of the same surface.

Theorem 2.26 (Li [Li19, Proposition 4.3] and Wang [Wan20, Proposition 4.17])**.** *Suppose (M, γ) is a balanced sutured manifold and S is a properly embedded surface in M, which intersects the suture γ transversely. Suppose S has a distinguished boundary component so that all the stabilizations mentioned below are performed on this boundary component. Then, for any $p, k, l \in \mathbb{Z}$ so that the stabilized surfaces S^p and S^{p+2k} are both admissible, we have*

$$\underline{\mathrm{SHI}}(M, \gamma, S^p, l) = \underline{\mathrm{SHI}}(M, \gamma, S^{p+2k}, l + k).$$

Note S^p is a stabilization of S as introduced in Definition 2.23, and, in particular, $S^0 = S$.

Remark 2.27. The original form of Theorem 2.26 in [Li19] was stated for a Seifert surface in the case of a knot complement. Though it is straightforward to generalize the proof to the case of a general admissible surface in a general balanced sutured manifold, given the condition that the decompositions along S and $-S$ are both taut. This extra condition on taut decompositions was then dropped due to the work in [Wan20].

2.3. **Bypass attachments.**

In this subsection, we review an important tool for sutured manifolds, *i.e.*, the bypass map.

Definition 2.28 ([Hon00])**.** Suppose (M, γ) is a balanced sutured manifold. An arc $\alpha \subset \partial M$ is called a **bypass arc** if the arc intersects the suture γ transversely at three points, including two endpoints.

For a bypass arc α, let its three intersection points with γ be P_0, P_1, and P_2, ordered by an arbitrary orientation of α. For $i = 0, 1, 2$, let γ_i be the component of γ containing P_i. If $\gamma_0 = \gamma_1 \neq \gamma_2$ or $\gamma_1 = \gamma_2 \neq \gamma_0$, then α is called a **wave bypass**. If $\gamma_0 = \gamma_2 \neq \gamma_1$, then α is called an **anti-wave bypass**.

Remark 2.29. The names of wave and anti-wave follows from [GL16, Section 7], where waves and anti-waves are arcs whose endpoints are on the same curve. After removing the component of γ containing only one intersection point with α, the arc α becomes a wave or an anti-wave.

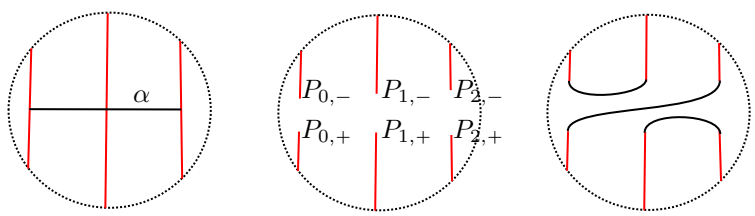


FIGURE 2. The bypass arc and the bypass attachment.

Given a bypass arc α on a balanced sutured manifold (M, γ_1), we can change the suture locally as follows. Let $D \subset \partial M$ be a neighborhood of the arc $\alpha \subset \partial M$, which is a disk containing α and intersecting γ in three arcs. There are six endpoints after cutting γ along α. We label them corresponding to the orientation of ∂M: if $P_{i,-}$ and $P_{i,+}$ are two endpoints according to P_i, we label them so that the oriented arc α, together with the arc-component of $\gamma \cap D$ from $P_{i,-}$ to $P_{i,+}$, gives an oriented framing of the boundary ∂M. Then we connect the newly created six endpoints according to the 'left-handed-principle': in D, a new suture γ_2 is obtained by connecting $P_{0,-}$ to $P_{1,-}$, connecting $P_{2,-}$ to $P_{0,+}$, and connecting $P_{2,+}$ to $P_{1,+}$. See Figure 2 for an example of a bypass arc and the corresponding new sutures.

Proposition 2.30 (Honda [Hon02])**.** *Suppose (M, γ) is a balanced sutured manifold and α is a bypass arc. Suppose γ_2 is the new suture described as above. Then (M, γ_2) is still a balanced sutured manifold.*

If α is a wave bypass, the suture γ_2 is obtained from γ_1 via a 'mystery move'. If α is an anti-wave bypass, the suture γ_2 is obtained from γ_1 via a positive Dehn twist. In both cases, the numbers of components of γ_1 and γ_2 are the same.

Definition 2.31. The change from (M, γ_1) to (M, γ_2) is called a **bypass attachment** along α.

Remark 2.32. The definition of a bypass attachment is due to [Hon00, Section 3.4]. Originally, a bypass attachment is a thickened half-disk, carrying some particular contact structure, attached to a contact 3-manifold M along an arc $\alpha \subset \partial M$. The dividing set on the boundary ∂M can be thought of as equivalent to the suture. After the half-disk-attachment, the dividing set, or the suture, is changed in the way described in Definition 2.31. For our purpose, we do not require the balanced suture manifold (M, γ_1) to carry a contact structure, while we can still perform an abstract bypass attachment.

A bypass attachment induces a map

$$\psi_1 : \underline{\mathrm{SHI}}(-M, -\gamma_1) \to \underline{\mathrm{SHI}}(-M, -\gamma_2).$$

This map can be explained in the following two ways.

Contact handle decomposition. By Ozbagci [Ozb11, Section 3], the half-disk-attachment can be decomposed into two contact handle attachments. First, one can attach a contact 1-handle along two endpoints of the bypass arc α. Then one can attach a contact 2-handle along a circle that is the union of α and an arc on the attached 1-handle. Topologically, the 1-handle and the 2-handle form a canceling pair, so the homeomorphism type of the 3-manifold is the same. However, the contact structure is changed, and the suture γ_1 is replaced by γ_2. In [BS16b, Section 5], Baldwin and Sivek constructed contact handle attaching maps for $\underline{\mathrm{SHI}}$. Following Ozbagci's idea, they defined the map

$$\psi_1 : \underline{\mathrm{SHI}}(-M, -\gamma_1) \to \underline{\mathrm{SHI}}(-M, -\gamma_2)$$

to be the composition of the corresponding contact 2-handle and 1-handle attaching maps.

Contact gluing maps. The half-disk attachment can be re-interpreted as follows. We start with (M, γ_1). Then pick $[1, 2] \times \partial M$ to be a collar of the boundary, carrying a particular contact structure ξ specified by the bypass attachment, so that the boundary $\{1, 2\} \times \partial M$ is convex and the dividing set is $\gamma_1 \cup \gamma_2$, with $\gamma_i \subset \{i\} \times \partial M$ for $i \in \{1, 2\}$. Then we glue $[1, 2] \times \partial M$ to M by the identification $\{1\} \times \partial M = \partial M$. The new 3-manifold is diffeomorphic to M, while the suture γ_1 is replaced by γ_2. In [HKM08], Honda, Kazez, and Matić defined a gluing map for SFH

$$\Phi_\xi : SFH(-M, -\gamma_1) \to SFH(-M, -\gamma_2),$$

which was then re-visited by Juhász and Zemke [JZ12]. Later, the first author defined a similar gluing map for $\underline{\mathrm{SHI}}$ in [Li18], and we can define

$$\psi_1 = \Phi_\xi : \underline{\mathrm{SHI}}(-M, -\gamma_1) \to \underline{\mathrm{SHI}}(-M, -\gamma_2).$$

These two points of view are equivalent due to [Li18, Section 4]. They will both be useful. Some immediate corollaries are the followings.

Lemma 2.33. *Suppose (M, γ) is a balanced sutured manifold and $\alpha, \beta \subset \partial M$ are two bypass arcs with $\alpha \cap \beta = \varnothing$. Let ψ_α and ψ_β be the bypass attaching maps associated to α and β, respectively. Let (M, γ') be the resulting balanced sutured manifold after bypass attachments along both α and β. Then we have*

$$\psi_\alpha \circ \psi_\beta = \psi_\beta \circ \psi_\alpha : \underline{\mathrm{SHI}}(-M, -\gamma) \to \underline{\mathrm{SHI}}(-M, -\gamma').$$

Proof. Consider bypasses as the compositions of contact handle attachments. Since $\alpha \cap \beta = \varnothing$, the contact handles associated to α and β are attached to disjoint regions on ∂M. Then it follows immediately that the corresponding contact handle attaching maps commute with each other. □

Lemma 2.34. *Suppose (M,γ) is a balanced sutured manifold and $\alpha_0,\alpha_1 \subset \partial M$ are two bypass arcs. Suppose further that these two arcs are isotopic as bypass arcs, i.e., there is a smooth family α_t, for $t \in [0,1]$, of bypass arcs. Then, α_1 and α_2 lead to isotopic balanced sutured manifold (M,γ'), and the bypass maps ψ_{α_1} and ψ_{α_2} are the same:*

$$\psi_{\alpha_1} = \psi_{\alpha_2} : \underline{\mathrm{SHI}}(-M,-\gamma) \to \underline{\mathrm{SHI}}(-M,-\gamma').$$

Proof. It follows from the contact handle decomposition interpretation of the bypass attachments. □

Remark 2.35. On the level of contact geometry, Honda has already proved Lemma 2.33 and Lemma 2.34 in [Hon00]. Thus, these two lemmas can also be proved by combining Honda's results with the functoriality of gluing maps Φ_ξ in [Li18].

One of the most important properties of bypass attachments is the bypass exact triangle. Suppose (M,γ) is a balanced sutured manifold, and α is a bypass arc. Suppose D is a neighborhood of $\alpha \subset \partial M$ and (M,γ_2) is the result of the bypass attachment along α. As shown in Figure 3, after attaching a bypass along α, there is an obvious bypass arc $\beta \subset D$. If we do the bypass attachement along β, then we obtain another balanced sutured manifold (M,γ_3), and it still carries an obvious bypass arc θ. When we further do the bypass attachement along θ, we obtain (M,γ_1) again. Let ψ_1, ψ_2, and ψ_3 be the bypass maps associated to α, β, and θ, respectively. We have the following.

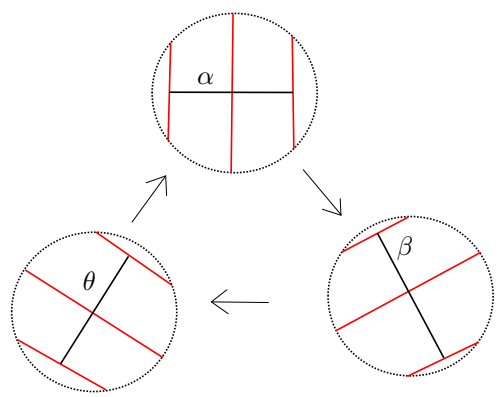


FIGURE 3. The bypass triangle.

Theorem 2.36 (Baldwin and Sivek [BS18, Theorem 1.20])**.** *There exists an exact triangle*

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-M,-\gamma_1) \arrow[rr, "\psi_1"] & & \underline{\mathrm{SHI}}(-M,-\gamma_2) \arrow[dl, "\psi_2"] \\
& \underline{\mathrm{SHI}}(-M,-\gamma_3) \arrow[ul, "\psi_3"] &
\end{tikzcd}$$

3. Instanton Floer homology and Heegaard diagrams

3.1. Sutured manifold with tangles and Heegaard diagrams.

Definition 3.1 ([XZ19, Definition 1.1])**.** Suppose (M,γ) is a balanced sutured manifold. A **tangle** $T\subset(M,\gamma)$ is a properly embedded 1-submanifold so that $T\cap A(\gamma)=\varnothing$. A tangle T is called **balanced** if

$$|T\cap R_+(\gamma)|=|T\cap R_-(\gamma)|.$$

A component α of T is called **vertical** if α is an arc,

$$T\cap R_+(\gamma)\neq\varnothing,\ \text{and}\ T\cap R_-(\gamma)\neq\varnothing.$$

A tangle is called **vertical** if every component of T is vertical.

Theorem 3.2 (Xie and Zhang [XZ19])**.** *For a balanced sutured manifold (M,γ) and a balanced tangle $T\subset(M,\gamma)$, there is a finite-dimensional complex vector space $SHI(M,\gamma,T)$ associated to them, whose isomorphism class is a topological invariant of the triple (M,γ,T).*

Suppose $T\subset(M,\gamma)$ is a vertical tangle, one can associate a new balanced sutured manifold (M_T,γ_T) to it. Let $M_T=M\backslash N(T)$ and let the suture γ_T be the union of γ and one meridian for each component of T. We have the following theorem.

Theorem 3.3 ([XZ19, Lemma 7.10])**.** *For a vertical tangle $T\subset(M,\gamma)$, there is an isomorphism*

$$SHI(M,\gamma,T)\cong SHI(M_T,\gamma_T).$$

Then we introduce Heegaard diagrams of closed 3-manifolds and knots.

Definition 3.4. A **(genus g) diagram** is a triple (Σ,α,β) so that the followings hold.

(1) We have Σ being a connected closed surface of genus g.
(2) We have $\alpha=\{\alpha_1,\dots,\alpha_m\}$ and $\beta=\{\beta_1,\dots,\beta_n\}$ being two sets of pair-wise disjoint simple closed curves on Σ. We do not distinguish the set and the union of curves.

Let N_0 be the manifold obtained from $\Sigma\times[-1,1]$ by attaching 3–dimensional 2–handles along $\alpha_i\times\{-1\}$ and $\beta_j\times\{1\}$ for each integer $i\in[1,m]$ and each integer $j\in[1,n]$. Let N be the manifold obtained from N_0 by capping off spherical boundaries. A diagram (Σ,α,β) is called **compatible** with a 3-manifold M if $M\cong N$. In such case, we also write M is compatible with (Σ,α,β), or (Σ,α,β) is a diagram of M.

Definition 3.5. A **(genus g) Heegaard diagram** is a (genus g) diagram (Σ,α,β) satisfying the following conditions.

(1) We have $|\alpha|=|\beta|=g$, *i.e.*, there are g many curves in either tuple.
(2) The complements $\Sigma\backslash\alpha$ and $\Sigma\backslash\beta$ are connected.

Given a Heegaard diagram (Σ,α,β), the manifolds compatible with $(\Sigma,\alpha,\varnothing)$ and $(\Sigma,\varnothing,\beta)$ are called the **α-handlebody** and the **β-handlebody**, respectively.

Definition 3.6. A **basepoint** in a Heegaard diagram (Σ,α,β) is a point in $\Sigma\backslash(\alpha\cup\beta)$. A **(genus g) doubly-pointed Heegaard diagram** $(\Sigma,\alpha,\beta,z,w)$ is a (genus g) Heegaard diagram with two basepoints z and w. Let $a\subset\Sigma\backslash\alpha$ and $b\subset\Sigma\backslash\beta$ be two arcs connecting z and w. Suppose a' and b' are obtained from a and b by pushing them into α-handlebody and β-handlebody, respectively. A doubly-pointed Heegaard diagram $(\Sigma,\alpha,\beta,z,w)$ is called **compatible** with a knot K in a closed 3-manifold Y if (Σ,α,β) is compatible with Y and the union $a'\cup b'$ is isotopic to K.

Definition 3.7. Suppose (Σ, α, β) is a Heegaard diagram of a closed 3-manifold Y. A knot $K \subset Y$ is called the **core knot** of β_i for some $\beta_i \subset \beta$ if it is constructed as follows. First, let M be the manifold compatible with the diagram $(\Sigma, \alpha, \beta \backslash \beta_i)$. It has a torus boundary and β_i induces a simple closed curve on ∂M. Dehn filling M along $\beta_i \subset \partial M$ gives Y. Second, let K be the image of $S^1 \times 0 \subset S^1 \times D^2$ under the filling map, where $S^1 \times D^2$ is the filling solid torus.

The following is a basic fact in 3-dimensional topology.

Theorem 3.8. *For any closed 3-manifold Y and any knot $K \subset Y$, there is a Heegaard diagram compatible with Y and a doubly-pointed Heegaard diagram compatible with (Y, K).*

Lemma 3.9. *Suppose Y is a closed 3-manifold and $K \subset Y$ is a knot. Suppose $(\Sigma, \alpha, \beta, z, w)$ is a genus g doubly-pointed Heegaard diagram compatible with (Y, K). Then there is a genus $(g+1)$ Heegaard diagram $(\Sigma', \alpha', \beta')$ compatible with Y so that β_{g+1} is a meridian of K. Hence K is the core knot of β_{g+1}.*

Proof. Consider the manifold obtained from $\Sigma \times [-1, 1]$ by attaching a 3-dimensional 1-handle along $\{z, w\} \times \{1\}$. This manifold has two boundaries. Let Σ' be the boundary that is not diffeomorphic to Σ. Let $\alpha_{g+1} \subset \Sigma'$ be the curve running from z to w and then back over the 1-handle. Let $\beta_{g+1} \subset \Sigma'$ be a small circle around z. Then $\alpha' = \alpha \cup \{\alpha_{g+1}\}$ and $\beta' = \beta \cup \{\beta_{g+1}\}$ are desired. □

Lemma 3.10. *Suppose Y is a closed 3-manifold and (Σ, α, β) is a Heegaard diagram compatible with Y. Then we can construct a vertical tangle $T \subset (Y(1), \delta)$, where $(Y(1), \delta)$ is defined as in Remark 2.9. Furthermore, the balanced sutured manifold $(Y(1)_T, \gamma_T)$ has the following properties:*

(1) The 3-manifold $Y(1)_T$ is a handlebody with boundary Σ.
(2) The suture δ_T contains all curves in β.
(3) $R_+(\delta_T)$ and $R_-(\delta_T)$ are both spheres with $(g+1)$ many punctures.

Proof. The manifold $Y(1)$ can be obtained from the α-handlebody by attaching 3–dimensional 2-handles along β_i for each integer $i \in [1, g]$. To pick the tangle, first note that a 3-dimensional 2-handle can be thought of as $[-1, 1] \times D^2$ attached along $[-1, 1] \times \partial D^2$. Let $\theta_i = [-1, 1] \times \{0\}$ be the co-core of the 2-handle attached along β_i, and let $T = \theta_1 \cup \cdots \cup \theta_g$. Then T is properly embedded in $Y(1)$. To obtain a balanced sutured manifold as well as a vertical tangle, we can pick a simple closed curve $\delta \subset \partial Y(1)$ so that for any i, two endpoints of θ_i lie on two different sides of δ. Since T is chosen to be the co-core of the 2-handles attached along β_i, it is straightforward to check that $Y(1)_T$ is a handlebody and all β_i are components of δ_T. Other than β_i, the suture δ_T contains another component, namely δ. Thus, $R_+(\gamma_T)$ and $R_-(\gamma_T)$ can be obtained from $\Sigma \backslash \beta$ by cutting along δ. Thus, they are both spheres with $(g+1)$ many punctures. □

Lemma 3.11. *Suppose H is a handlebody, and γ is a suture on ∂H so that $R_+(\gamma)$ and $R_-(\gamma)$ are both spheres with $(g+1)$ many punctures. We can find a closed 3-manifold Y and a Heegaard diagram (Σ, α, β) compatible with Y so that for the vertical tangle $T \subset (Y(1), \delta)$ constructed in the proof of Lemma 3.10, we have $(Y(1)_T, \delta_T) = (H, \gamma)$.*

Proof. Let $\Sigma = \partial H$. Suppose Σ has genus g. Let $\alpha_1, \ldots, \alpha_g$ be boundaries of g compressing disks $D_1, \ldots, D_g$ so that $H \backslash (D_1 \cup \cdots \cup D_g)$ is a 3-ball. Since $R_+(\gamma)$ and $R_-(\gamma)$ are both spheres with $(g+1)$ many punctures, the suture γ has $(g+1)$ many components. We can take arbitrary g many of them to form β. Different choices of such g many curves from γ are clearly related to each other by a finite sequence of handle slides, so they lead to the same closed 3-manifold Y up to diffeomorphism.

Let δ be the remaining component of γ. Suppose $Y(1)$ is obtained by attaching 3-dimensional 2-handles along β curves. Then δ becomes the desired suture on $\partial Y(1)$. Then it is straightforward to check the final argument. □

Lemma 3.12. *Suppose (Σ, α, β) is a Heegaard diagram compatible with a closed 3-manifold Y. Then there exists a balanced sutured handlebody (H, γ) satisfying the hypothesis of Lemma 3.11. Furthermore, the manifold obtained by applying Lemma 3.11 is precisely Y.*

Proof. Let H be the α-handleboby. Suppose Σ has genus g. For any integer $i \in [1, g]$, let β_i be oriented arbitrarily and let β_i' be the curve obtained by pushing off β_i to the right with respect to the orientation. Suppose β_i' is oriented reversely. Let β_{g+1} be the curve obtained from β_i' by band sums with respect to orientations so that β_{g+1} is disjoint from $\beta_1, ..., \beta_g$. Let $\gamma = \beta_1 \cup \cdots \cup \beta_{g+1}$. It is straightforward to check that (H, γ) is the desired sutured handlebody. □

Remark 3.13. Suppose Y is a closed 3-manifold and $K \subset Y$ is a knot. Suppose $(\Sigma, \alpha, \beta, z, w)$ is a genus $(g-1)$ doubly-pointed Heegaard diagram of (Y, K). It induces a genus g Heegaard diagram $(\Sigma', \alpha', \beta')$ of Y by Lemma 3.9 and then a balanced sutured handlebody (H, γ) by Lemma 3.12. Note the diagram $(\Sigma', \alpha', \beta)$ is compatible with the knot complement $Y(K)$. Since $\beta_g \cap \beta_{g+1} = \varnothing$ and β_{g+1} is the meridian of K, the suture β_g becomes a parallel copy of the meridian of K with the reverse orientation.

3.2. **A dimension inequality for tangles.**

In this subsection, we prove the following proposition.

Proposition 3.14. *Suppose (M, γ) is a balanced sutured manifold and T is a vertical tangle. Suppose $\alpha \subset T$ is a component of T so that $[\alpha] = 0 \in H_1(M, \partial M; \mathbb{Q})$. Let $T' = T \backslash \alpha$. Then we have*

$$\dim_{\mathbb{C}} SHI(-M, -\gamma, T') \leqslant \dim_{\mathbb{C}} SHI(-M, -\gamma, T).$$

Proof. We prove this propositions in followings steps.

Step 1. We construct an auxiliary manifold M_{T_0} with a family of sutures Γ_n for $n \in \mathbb{N} \cup \{-, +\}$ on ∂M_{T_0}.

Since $[\alpha] = 0 \in H_1(M, \partial M; \mathbb{Q})$, there exist $q, k \in \mathbb{Z}$ and arcs $\beta_1, ..., \beta_k \subset \partial M$ so that there exists a properly embedded surface S in M with ∂S consisting of β_1,..., β_k and q copies of α. Suppose components of T are $\alpha_1, \alpha_2, ..., \alpha_m$, with $\alpha_1 = \alpha$.

As in Subsection 3.1, we form a new balanced sutured manifold (M_T, γ_T) as follows. Let $N(T)$ be a tubular neighborhood of T in M. Then $\partial N(T)$ intersects either $R_+(\gamma)$ or $R_-(\gamma)$ in m many disks and intersects $\mathrm{int}(M)$ in m many cylinders. Let $C_1, \ldots, C_m$ be the cylinders corresponding to $\alpha_1, \ldots, \alpha_m$, respectively. Let $M_T = M \backslash \mathrm{int}(N(T))$. For any integer $i \in [1, n]$, let $\gamma_i \subset C_i$ be a simple closed curve representing the generator of $H_1(C_i)$. Let α_i be oriented from $R_+(\gamma)$ to $R_-(\gamma)$. Then γ_i has an induced orientation from α_i. Let

$$\gamma_T = \gamma \cup \gamma_1 \cup ... \cup \gamma_m.$$

The surface S can be modified into a properly embedded surface S_T in M_T as follows. First, for the part of ∂S, which are q copies of α, we isotop them to be on C_1. Then β_j for any integer $j \in [1, k]$ can be viewed as an arc on $\partial M_T \backslash C_1$. We can isotop S if necessary to make S intersect α_i for any integer $i \in [2, m]$ transversely. When removing $N(T)$ from M, we remove disks, viewed as neighborhoods of $S \cap T'$, from S. Thus, we obtain $S_T \subset M_T$. Note $S_T \cap C_1$ consists of q many arcs, each intersecting γ_1 transversely at one point. Also note for any integer $i \in [2, m]$, $S_T \cap C_i$ is a (possibly empty) collection of circles that are parallel to γ_i.

Note β_1 does not intersect any γ_i, but intersects the original suture γ. Since the arc $\beta_1 \subset \partial S_T \subset \partial M_T$ has one endpoint in $R_+(\gamma_T)$ and the other in $R_-(\gamma_T)$, its intersection with γ must be odd. Let β_1 be oriented from $R_+(\gamma_T)$ to $R_-(\gamma_T)$ and let the intersection points between β_1 and γ be $p_1, \dots, p_l$ with l odd, ordered by the orientation of β_1. Let β_1' be a perturbation of β_1 so that β_1' and β_1 meet at endpoints. Suppose the intersection points between β_1' and γ be $q_1, \dots, q_l$ so that q_i is near p_i for integer $i \in [1, l]$.

If $l = 1$, then (M_T, γ_T) is enough for the proof. If $l > 1$, we have to perform the following modification on (M_T, γ_T). We attach a contact 1-handle to (M, γ) along q_1 and q_{l-1} in the sense of Baldwin and Sivek [BS16a, Section 4.2.2], or equivalently, attach a product 1-handle in the sense of Kronheimer and Mrowka [KM10b, Section 6]. They both proved that the balanced sutured manifolds before and after attaching such a 1-handle have exactly the same closure, which is also compatible with Xie and Zhang's construction. As a result, after attaching the 1-handle, the sutured instanton Floer homology does not change. Thus, we will still use (M, γ) and (M_T, γ_T) to denote sutured manifolds after attaching the 1-handle. The convenience of having an extra 1-handle attached is that now there exists an arc ζ with the following significance.

(1) Endpoints of ζ are contained in ∂C_1.
(2) The arc ζ intersects γ transversely at one point.
(3) The arc ζ is disjoint from S_T.

The arc ζ can be obtained by first going along β_1' until reaching q_1, then going along the 1-handle, and going back to β_1' at the point q_{l-1} and then keeping going along β_1'. Finally, we slightly perturb this arc to make it disjoint from S_T. See Figure 4.

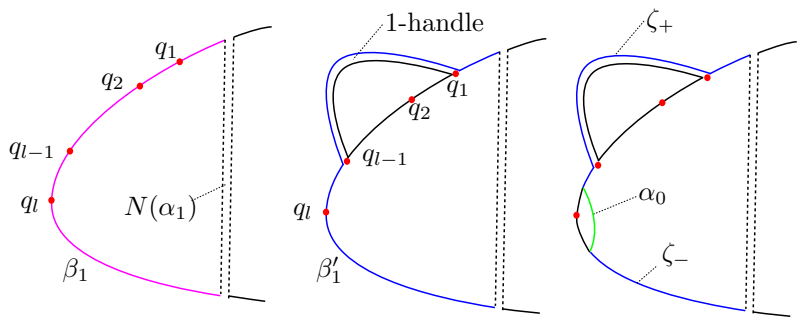


FIGURE 4. Left, the arc β_1; middle, the 1-handle on β_1'; right, the arcs ζ_+, α_0, and ζ_-.

Let α_0 be the arc obtained by pushing a neighborhood of q_l on ζ into the interior of M_T. Suppose the endpoints of α_0 are still in ζ and α_0 is disjoint from S_T. The arc α_0 can be regarded as a vertical tangle in M_T and hence in the original manifold M. Let $T_0 = T \cup \alpha_0$, and $T_0' = T' \cup \alpha_0$. Let (M_{T_0}, γ_{T_0}) be obtained similarly as (M_T, γ_T). Since $M_{T_0} = M_T \backslash N(\alpha_0)$, the cylinder C_0 and the suture γ_0 are defined similarly as C_i and γ_i for any integer $i \in [1, m]$. A sketch is shown in Figure 5. Let $\zeta_\pm$ be two parts of ζ contained in $R_\pm(\gamma_{T_0})$, respectively.

Now, we pick a family of sutures Γ_n on M_{T_0} so that Γ_n is the suture obtained from γ_{T_0} by replacing γ_0 and γ_1 by other two sutures. We describe the two new sutures in details as follows. For Γ_0, the new sutures are depicted in Figure 5. Note that the part of the new sutures in $R_\pm(\gamma_{T_0})$ consists of two parallel copies of $\zeta_\pm$, respectively. To obtain other sutures Γ_n, we perform twists along $-\gamma_1$ for n times, as shown in Figure 6.

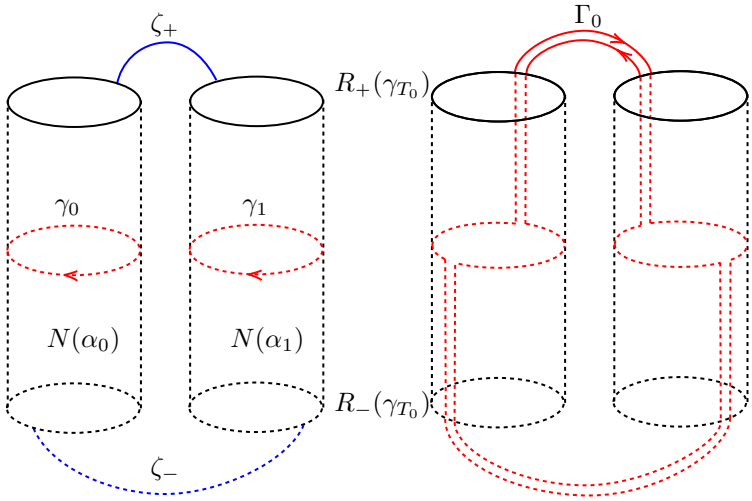


FIGURE 5. Left, a sketch of the balanced sutured manifold (M_{T_0}, γ_{T_0}) with the arcs ζ_+ and ζ_-; right, the suture Γ_0.

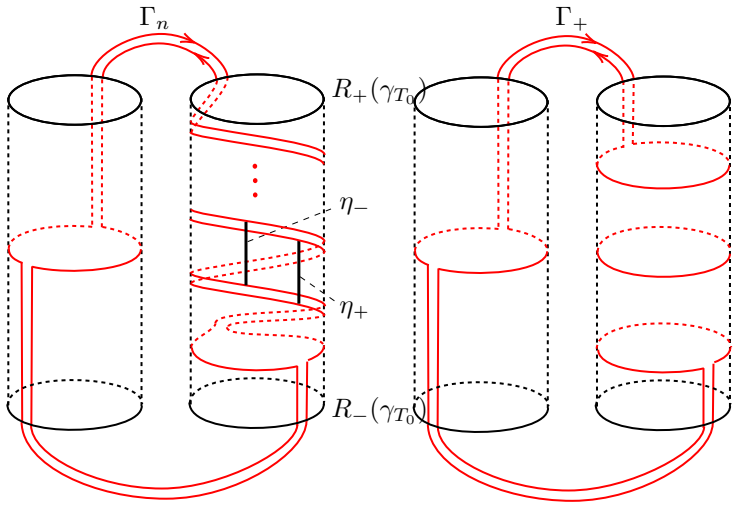


FIGURE 6. Left, the suture Γ_n, and the bypass arcs η_+ and η_-; right, the suture Γ_+.

There are two obvious bypass arcs in Figure 6, denoted by η_+ and η_-, respectively. It is straightforward to check that these two bypass arcs induce bypass exact triangles from Theorem 2.36:

$$\begin{tikzcd} \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_{n-1}) \arrow[rr, "\psi^{n-1}_{+,n}"] & & \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n) \arrow[dl, "\psi^{n}_{+,+}"] \\ & \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_+) \arrow[ul, "\psi^{+}_{+,n-1}"] & \end{tikzcd} \tag{3.1}$$

and

$$\begin{tikzcd} \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_{n-1}) \arrow[rr, "\psi^{n-1}_{-,n}"] & & \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n) \arrow[dl, "\psi^{n}_{-,-}"] \\ & \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_-) \arrow[ul, "\psi^{-}_{-,n-1}"] & \end{tikzcd} \tag{3.2}$$

Here Γ_- is the same as the original suture γ_{T_0}, and Γ_+ is the suture as depicted in Figure 6.

One thing we would like to remark here is about the bypass attachment. The bypasses are attached to η_+ and η_- from the exterior of the 3-manifold M_{T_0}, though the point of view in Figure 6 is from the interior of the manifold. So readers have to take extra care when performing these bypass attachments.

Step 2. We use bypass maps and bypass triangles to derive an inequality for M_{T_0}.

Recall we have a properly embedded surface $S_T \subset M_T$. Since $\alpha_0 \cap S_T = \varnothing$, we can regard S_T as being properly embedded in M_{T_0}. Let S_T be oriented so that the orientation of ∂S_T coincides with that of α $(= \alpha_1)$. Note that ζ_+ and ζ_- are both disjoint from S_T. We can perform stabilizations on S_T so that the followings hold.

(1) S_T is admissible with respect to the suture Γ_- $(= \gamma_{T_0})$.
(2) For any $n \in \mathbb{Z}_{\geqslant 0} \cup \{-,+\}$, S_T has minimal possible number of intersection points with the suture $\Gamma_n \backslash (\gamma \cup \gamma_2 \cup ... \cup \gamma_m)$.
(3) S_T is disjoint from a neighborhood of $\zeta_+ \cup \zeta_-$.

The surface after stabilizations is still denoted by S_T. After obtaining the surface S_T that satisfies the above three conditions, we further perform stabilizations on it as follows. For any $n \in \mathbb{N} \cup \{-,+\}$, if S_T is already admissible with respect to Γ_n, then let S_n be the surface S_T without any further change. If S_T is not admissible with respect to Γ_n, then perform a negative stabilization on S_T within C_1 to make it admissible, and call the resulting surface S_n. Equivalently, define a map

$$\tau : \mathbb{N} \cup \{-,+\} \mapsto \{0, -1\}$$
$$\tau(n) = \begin{cases} 0 & \text{If } S_T \cap \Gamma_n \cap C_1 \text{ is odd} \\ -1 & \text{If } S_T \cap \Gamma_n \cap C_1 \text{ is even} \end{cases}$$

Then we take $S_n = S_T^{\tau(n)}$, where the superscript on the surface S_T follows Definition 2.23. Note all the stabilizations are with respect to Γ_n.

By Theorem 2.20, for any $n \in \mathbb{N} \cup \{-,+\}$, there is a closure (Y_n, R_n, ω_n) for (M_{T_0}, Γ_n) so that S_n extends to a closed surface $\bar{S}_n$. Let

$$i^n_{max} = -\frac{1}{2}\chi(\bar{S}_n), \text{ and } i^n_{min} = \frac{1}{2}\chi(\bar{S}_n) - \tau(n).$$

Lemma 3.15. *If $i > i^n_{max}$ or $i < i^n_{min}$, then* $\underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n, S_n, i) = 0$.

Proof. If $\tau(n) = 0$, then the lemma follows directly from term (1) of Theorem 2.20. If $\tau(n) = -1$, then term (1) of Theorem 2.20 only implies that for $i < i^n_{min} - 1$,

$$\underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n, S_n, i) = 0.$$

For the remaining case when $i = i^n_{min} - 1$, from term (3) of Theorem 2.20, we have

$$\underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n, S_n, i) = \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n, -S_n, -i) = \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n, -S_n, -\frac{\chi(\bar{S}_n)}{2}).$$

From the construction of S_n, we know that $-S_n$ is obtained from S_T by a negative stabilization with respect to the suture $-\Gamma_n$. Hence we can apply Lemma 2.24 and term (2) of Theorem 2.20 to obtain the desired result. □

Remark 3.16. Note we do not know if the corresponding $\underline{\mathrm{SHI}}$ is non-vanishing at the gradings i^n_{max} and i^n_{min}, though this does not make a difference in the proof of Proposition 3.14.

Next, we will derive a graded version of bypass exact triangles (3.1) and (3.2). To do so, we will discuss more about the surface S_n. Since ∂S_T contains q copies of α_1, for any $n \in \mathbb{N}$, we have

$$|S_T \cap \Gamma_+ \cap C_1| = q,\ |S_T \cap \Gamma_- \cap C_1| = 3q,\ \text{and}\ |S_T \cap \Gamma_n \cap C_1| = (2n+1)q.$$

From Theorem 2.20, we know that for any $n \in \mathbb{N}$,

$$\chi(\bar{S}_-) = \chi(\bar{S}_+) - q + \tau(-),\ \text{and}\ \chi(\bar{S}_n) = \chi(\bar{S}_+) - nq + \tau(n). \tag{3.3}$$

From Lemma 3.15, we have

$$\lim_{n\to+\infty} i^n_{max} = +\infty,\ \text{and}\ \lim_{n\to+\infty} i^n_{min} = -\infty. \tag{3.4}$$

To present the grading shifting property better, we make the following definition.

Definition 3.17. Suppose (M,γ) is a balanced sutured manifold and S is an admissible surface in (M,γ). For any $i,j \in \mathbb{Z}$, define

$$\underline{\mathrm{SHI}}(M,\gamma,S,i)[j] = \underline{\mathrm{SHI}}(M,\gamma,S,i-j).$$

Lemma 3.18. *For any $n \in \mathbb{N}$, we have two exact triangles*

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-M_{T_0},-\Gamma_n,S_n)[i^{n+1}_{min}-i^n_{min}] \arrow[r, "\psi^n_{+,n+1}"] & \underline{\mathrm{SHI}}(-M_{T_0},-\Gamma_{n+1},S_{n+1}) \arrow[dl, "\psi^{n+1}_{+,+}"] \\
\underline{\mathrm{SHI}}(-M_{T_0},-\Gamma_+,S_+)[i^{n+1}_{max}-i^+_{max}] \arrow[u, "\psi^+_{+,n}"] &
\end{tikzcd}$$

and

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-M_{T_0},-\Gamma_n,S_n)[i^{n+1}_{max}-i^n_{max}] \arrow[r, "\psi^n_{-,n+1}"] & \underline{\mathrm{SHI}}(-M_{T_0},-\Gamma_{n+1},S_{n+1}) \arrow[dl, "\psi^{n+1}_{-,-}"] \\
\underline{\mathrm{SHI}}(-M_{T_0},-\Gamma_-,S_-)[i^{n+1}_{min}-i^-_{min}] \arrow[u, "\psi^-_{-,n}"] &
\end{tikzcd}.$$

Furthermore, all maps in the above two exact triangles are grading preserving.

Proof. We only prove the grading shifting behavior of the map $\psi^{n+1}_{+,+}$ in the triangle (3.1), and the proof for all other maps are similar. For simplicity, we also assume that $\tau(n) = \tau(+) = 0$ (Note $\tau(+) = 0$ by construction), and other cases are similar. From Subsection 2.3, we know bypass maps are constructed via contact handle maps and ultimately via cobordisms maps associated to Dehn surgeries. It is obvious that the surface S_{n+1} is disjoint from the bypass arc, so we conclude that the following map is grading preserving:

$$\psi^{n+1}_{+,+} : \underline{\mathrm{SHI}}(-M_{T_0},-\Gamma_{n+1},S_{n+1}) \to \underline{\mathrm{SHI}}(-M_{T_0},-\Gamma_+,S_{n+1}).$$

From Figure 7, it is straightforward to check that S_{n+1} is obtained from S_+ by

$$\frac{1}{2}(|S_{n+1} \cap \Gamma_+ \cap C_1| - |S_+ \cap \Gamma_+ \cap C_1|) = 2(i^{n+1}_{max} - i^+_{max})$$

many negative stabilizations, with respect to the suture Γ_+. Hence they become positive stabilizations with respect to $-\Gamma_+$. Then Theorem 2.26 applies and we obtain the desired result. □

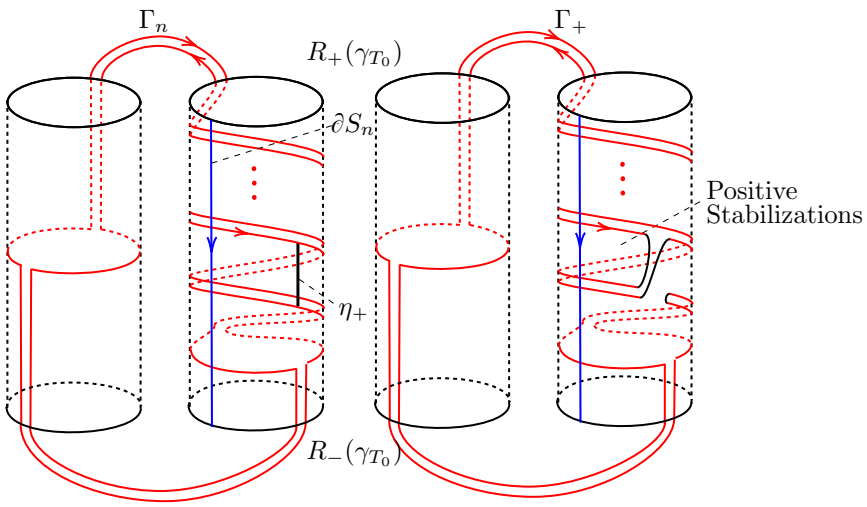


FIGURE 7. Left, the suture Γ_n, the surface S_n, and the bypass η_+; right, the suture Γ_+ after the bypass attachment along η_+.

Remark 3.19. The statement of Lemma 3.18 can be illustrated in Figure 8, where vector spaces are denoted by the related sutures and maps are denoted by horizontal arrows. The heights of the blocks depend on $i_{max} - i_{min}$. This illustration is also useful for statements in Section 4.

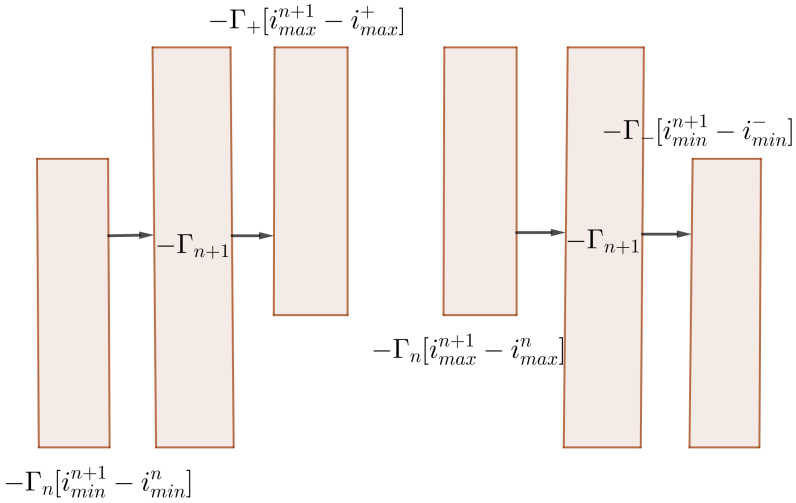


FIGURE 8. Illustration of Lemma 3.18.

Equipped with Lemma 3.18, we are able to prove the following lemma.

Lemma 3.20. *The map*

$$\psi^{n,j}_{+,n+1} : \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n, S_n, j) \to \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_{n+1}, S_{n+1}, j - (i^n_{min} - i^{n+1}_{min}))$$

is an isomorphism if

$$j < i^{n+1}_{max} + (i^n_{min} - i^{n+1}_{min}) - (i^+_{max} - i^+_{min}).$$

Similarly, the map

$$\psi^{n,j}_{-,n+1} : \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n, S_n, j) \to \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_{n+1}, S_{n+1}, j + (i^{n+1}_{max} - i^n_{max}))$$

is an isomorphism if

$$j > i^{n+1}_{min} - (i^{n+1}_{max} - i^n_{max}) + (i^-_{max} - i^-_{min}).$$

Proof. We only prove the first statement. The proof of the second is similar. Suppose

$$i = j - (i^{n+1}_{min} - i^{n+1}_{min}).$$

Then we know that there is a map

$$\psi^{n+1,i}_{+,+} : \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_{n+1}, S_{n+1}, i) \to \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_+, S_+, i - i^{n+1}_{max} + i^+_{max}).$$

By assumption, we have

$$\begin{aligned} i - i^{n+1}_{max} + i^+_{max} &= j - (i^n_{min} - i^{n+1}_{min}) - i^{n+1}_{max} + i^+_{max} \\ &< i^{n+1}_{max} + (i^n_{min} - i^{n+1}_{min}) - (i^+_{max} - i^+_{min}) - (i^n_{min} - i^{n+1}_{min}) - i^{n+1}_{max} + i^+_{max} \\ &= i^+_{min}. \end{aligned}$$

Hence it follows from Lemma 3.15 that $\psi^{n+1,i}_{+,+} = 0$. Then it follows from Lemma 3.18 that $\psi^{n,j}_{+,n+1}$ is surjective. The proof that it is injective is similar. □

Lemma 3.21. *For any $n \in \mathbb{Z}$, there is an exact triangle*

$$\begin{tikzcd} \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n) \arrow[rr] & & \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_{n+1}) \arrow[dl, "F_{n+1}"] \\ & \underline{\mathrm{SHI}}(-M_{T'_0}, -\gamma_{T'_0}) \arrow[ul, "G_n"] & \end{tikzcd} \tag{3.5}$$

Furthermore, we have two commutative diagrams related to $\psi^n_{+,n+1}$ and $\psi^n_{-,n+1}$, respectively

$$\begin{tikzcd} \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n) \arrow[rr, "\psi^n_{\pm,n+1}"] & & \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_{n+1}) \\ & \underline{\mathrm{SHI}}(-M_{T'_0}, -\gamma_{T'_0}) \arrow[ul, "G_n"] \arrow[ur, "G_{n+1}"'] & \end{tikzcd}$$

Proof. Let γ'_1 be the curve obtained by pushing γ_1 into the interior of M_{T_0}. The surgery triangle associated to γ'_1 is the following.

$$\begin{tikzcd} ((-M_{T_0})_1, -\Gamma_{n+1}) \arrow[rr] & & ((-M_{T_0})_\infty, -\Gamma_{n+1}) \arrow[dl] \\ & ((-M_{T_0})_0, -\Gamma_{n+1}) \arrow[ul] & \end{tikzcd}$$

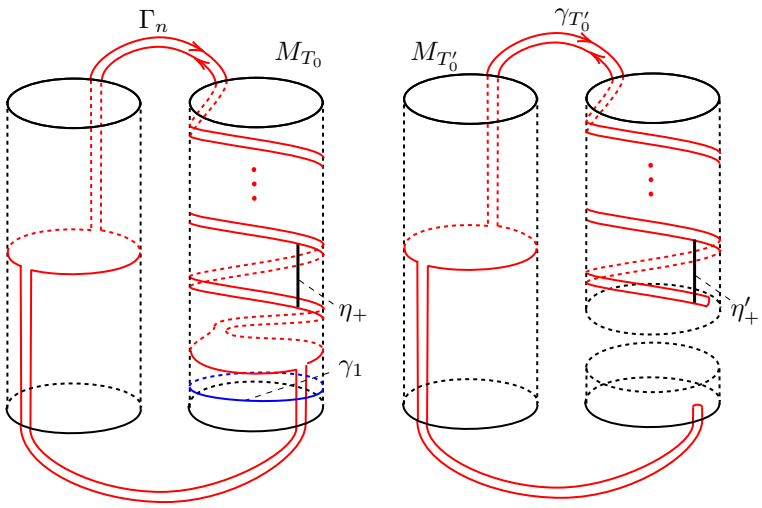


FIGURE 9. Left, the suture Γ_n, the meridian γ_1, and the bypass η_+; right, the balanced sutured manifold $(M_{T_0'}, \gamma_{T_0'})$ after attaching a contact 2-handle along γ_1.

The slopes ∞, 0, and 1 are with respect to the surface framing of γ_1' induced by ∂M_{T_0}. Since γ_1' is in the interior of M_{T_0}, the surgeries do not influence the procedure of closing up the balanced sutured manifolds. Hence from Theorem 2.4 we have an exact triangle

$$\begin{tikzcd}
\underline{\mathrm{SHI}}((-M_{T_0})_1, -\Gamma_{n+1}) \arrow[rr] & & \underline{\mathrm{SHI}}((-M_{T_0})_\infty, -\Gamma_{n+1}) \arrow[dl] \\
& \underline{\mathrm{SHI}}((-M_{T_0})_0, -\Gamma_{n+1}) \arrow[ul] &
\end{tikzcd}$$

The ∞-surgery does not change anything, so

$$((-M_{T_0})_\infty, -\Gamma_{n+1}) \cong (-M_{T_0}, -\Gamma_{n+1}).$$

The 1-surgery is equivalent to a local twist along γ_1'. It does not change the underlying 3-manifold, while the suture Γ_{n+1} is replaced by Γ_n:

$$((-M_{T_0})_1, -\Gamma_{n+1}) \cong (-M_{T_0}, -\Gamma_n).$$

Finally, for the 0-surgery, from Baldwin and Sivek [BS16b], we know that on the level of closures, performing a 0-surgery is equivalent to attaching a contact 2-handle along $\gamma_1 \subset \partial M_{T_0}$. Attaching such a contact 2-handle changes (M_{T_0}, Γ_{n+1}) to $(M_{T_0'}, \gamma_{T_0'})$. Hence we derive the desired exact triangle.

To prove two commutative diagrams, first note that the curve γ_1' is disjoint from the bypass arc η_+. As a result, the related maps commute with each other:

$$\psi^n_{+,n+1} \circ G_n = G_{n+1} \circ \psi_{\eta_+'},$$

where η_+' is the bypass arc as shown in Figure 9. It is straightforward to check that the bypass along η_+' is a trivial bypass, and hence from [Hon02] it does not change the contact structure. In Subsection 2.3, the bypass maps can be re-interpreted as contact gluing maps, and by the functoriality of instanton contact gluing maps in [Li18], we know that the bypass map ψ_{η_+} corresponding to the

trivial bypass η_+ is the identity map. Hence we conclude that

$$\psi^n_{+,n+1} \circ G_n = G_{n+1} \circ \psi_{\eta'_+} = G_{n+1} \circ \mathrm{id} = G_{n+1}.$$

The other commutative diagram involving $\psi^n_{-,n+1}$ can be proved similarly. □

Lemma 3.22. *For a large enough integer n, the map G_n in Lemma 3.21 is zero.*

Proof. Suppose the contrary. Then for any n, there exists $x \in \underline{\mathrm{SHI}}(-M_{T'_0}, -\gamma_{T'_0})$ so that

$$y = G_n(x) \neq 0 \in \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n).$$

Suppose

$$y = \sum_{j\in\mathbb{Z}} y_j, \text{ where } y_j \in \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n, S_n, j),$$

$$j_{max} = \max_{y_j\neq 0} j, \text{ and } j_{min} = \min_{y_j \neq 0} j.$$

By assumption j_{max} and j_{min} both exist and $j_{max} \geqslant j_{min}$. Suppose

$$z = G_{n+1}(x) \in \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_{n+1}),$$

and similarly

$$z = \sum_{j\in\mathbb{Z}} z_j, \text{ where } z_j \in \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_{n+1}, S_{n+1}, j).$$

From facts (3.3) and (3.4), we know that for a large enough integer n, we have

$$i^{n+1}_{max} + (i^n_{min} - i^{n+1}_{min}) - (i^+_{max} - i^+_{min}) > i^{n+1}_{min} - (i^{n+1}_{max} - i^n_{max}) + (i^-_{max} - i^-_{min}).$$

Hence at least one of the following two statements must be true.

(1) $j_{max} > i^{n+1}_{min} - (i^{n+1}_{max} - i^n_{max}) + (i^-_{max} - i^-_{min})$
(2) $j_{min} < i^{n+1}_{max} + (i^n_{min} - i^{n+1}_{min}) - (i^+_{max} - i^+_{min})$

We only work with the case when the first statement is true, and the other case is similar. From Lemma 3.21, we have

$$z = \psi^n_{+,n+1}(y) = \psi^n_{-,n+1}(y).$$

Suppose

$$i = j_{max} = j_{max} + (i^n_{max} - i^{n+1}_{max}),$$

and

$$j' = i + (i^n_{min} - i^{n+1}_{min}).$$

By Lemma 3.18, we have

$$\psi^{n,j'}_{+,n+1}(y_{j'}) = z_i = \psi^{n,j_{max}}_{-,n+1}(y_{j_{max}}).$$

Since $j' > j_{max}$, we have $z_i = 0$. By Lemma 3.20, the first statement implies $\psi^{n,j_{max}}_{-,n+1}$ is an isomorphism. Hence $y_{j_{max}} = 0$, which contradicts with the assumpotion of j_{max}. □

Suppose n is large enough. From the exact triangle (3.5), the map G_n is zero, and this implies

$$\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-M_{T'_0}, -\gamma_{T'_0}) = \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_{n+1}) - \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n).$$

From the exact triangle (3.2) and the fact that $\Gamma_- = \gamma_{T_0}$, we have

$$\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-M_{T_0}, -\gamma_{T_0}) \leqslant \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_{n+1}) - \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-M_{T_0}, -\Gamma_n).$$

Step 3. We obtain the desired inequality from the above equality and inequality.

Note that α_0 is an arc obtained by pushing a neighborhood of q_l in ζ into the interior of M_T, and M_{T_0} is obtained from M_T by removing a tubular neighborhood of α_0. There is an embedded disk

D in M_{T_0} whose boundary is the union of α_0 and the neighborhood of q_l in ζ. The boundary ∂D intersects γ_{T_0} at two points, one of which is q_l and the other is in γ_0. By Kronheimer and Mrowka [KM10b, Section 6], decomposing the sutured manifold (M_{T_0}, γ_{T_0}) along D does not change the isomorphism class of the sutured instanton Floer homology (D is a product disk in the sense of [Juh08, Definition 2.8]). It is straightforward to check the sutured manifold after the sutured manifold decomposition is exactly (M_T, γ_T). Then we have

$$\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-M_{T_0}, -\gamma_{T_0}) = \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-M_T, -\gamma_T).$$

Similarly, we have

$$\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-M_{T_0'}, -\gamma_{T_0'}) = \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-M_{T'}, -\gamma_{T'}).$$

Thus, we conclude

$$\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-M_T, -\gamma_T) \leqslant \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-M_{T'}, -\gamma_{T'}).$$

Then Proposition 3.14 follows from Theorem 3.3. □

Proof of Theorem 1.2. Consider the Heegaard diagram $(\Sigma', \alpha', \beta')$ compatible with Y obtained from Lemma 3.9 and the tangle T obtained from Lemma 3.13. Since Y is a rational homology sphere, we have

$$H_1(Y(1), \partial Y(1); \mathbb{Q}) = 0,$$

where $(Y(1), \delta)$ is defined as in Remark 2.9. In particular, any component of T has trivial rational homology class. Then the theorem follows from Proposition 3.14, Theorem 3.3, and Remark 3.13. □

Remark 3.23. Suppose (Σ, α, β) is a Heegaard diagram of a rational homology sphere Y and K is the core knot of β_i for some $\beta_i \subset \beta$. Suppose (H, γ) is the sutured handlebody obtained from (Σ, α, β) by Lemma 3.12 and T is the tangle obtained from Lemma 3.13. Then the proof of Theorem 1.2 applies without change and we conclude the same inequality.

Proof of Proposition 1.3. Similar to the proof of Theorem 1.2, since the knot K has trivial rational homology class, the corresponding tangle has trivial homology class in

$$H_1(Y(1), \partial Y(1); \mathbb{Q}).$$

□

Corollary 3.24. *Suppose (Σ, α, β) is a genus g Heegaard diagram of a rational homology sphere Y. Let (H, γ) be a sutured handlebody obtained by Lemma 3.12. Suppose*

$$n = \frac{1}{2} \sum_{i=1}^{g} |\alpha_i \cap \gamma|.$$

Then we have

$$\dim_{\mathbb{C}} I^{\sharp}(-Y) \leqslant 2^{n-g}.$$

Proof. By Lemma 3.10 and Theorem 1.2, we know that

$$\dim_{\mathbb{C}} I^{\sharp}(Y) \leqslant \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-H, -\gamma).$$

Hence it suffices to prove that

$$\underline{\mathrm{SHI}}(-M, -\gamma) \leqslant 2^{n-g}.$$

First note that if $n < g$, then there exists an integer $i \in [1, g]$ so that $\alpha_i \cap \gamma = \varnothing$. Hence the balanced sutured manifold $(-H, -\gamma)$ is not taut, and by Theorem 2.13 we know that $\underline{\mathrm{SHI}}(-M, -\gamma) = 0$.

When $g = n$, then either there exists an integer $i \in [1, g]$ so that $\alpha_i \cap \gamma = \varnothing$, and then $\underline{\mathrm{SHI}}(-M, -\gamma) = 0$ as above, or for each $i \in \{1, ..., g\}$, α_i intersects γ precisely at two points. Thus the disk $D_i \subset H$ bounded by α_i is a product disk. Let $D = D_1 \cup ... \cup D_g$. We can perform a sutured manifold decomposition $(-H, -\gamma) \overset{D}{\rightsquigarrow} (D^3, \delta)$, where D^3 is a 3-ball and δ is a suture on ∂D^3. From [KM10b, Section 6], we know that

$$\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-H, -\gamma) = \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(D^3, \delta) = 1.$$

We prove other cases by induction. Assume that the statement has been proved for $g \leqslant n = k-1$. For the case of $n = k$, we proceed as follows. Since $k > g$, there is a curve α_i satisfying $\alpha_i \cap \gamma \geqslant 4$. Suppose $\eta \subset \alpha_i$ is an arc so that $\partial\eta \subset \gamma$, and the interior of η intersects with γ transversely once. Then we can do the bypass attachement along η and form an exact triangle by Theorem 2.36:

$$\begin{tikzcd}
 & \underline{\mathrm{SHI}}(-H, -\gamma) \arrow[dr] & \\
\underline{\mathrm{SHI}}(-H, -\gamma_2) \arrow[ur] & & \underline{\mathrm{SHI}}(-H, -\gamma_1) \arrow[ll]
\end{tikzcd}$$

It is straightforward to check that

$$\sum_{i=1}^{n} |\gamma_1 \cap \alpha_i| \leqslant 2k-2, \text{ and } \sum_{i=1}^{n} |\gamma_2 \cap \alpha_i| \leqslant 2k-2.$$

So by the induction hypothesis and the exactness, we conclude that

$$\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-H, -\gamma) \leqslant \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-H, -\gamma_1) + \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-H, -\gamma_2) \leqslant 2^{k-g}.$$

Hence we complete the induction. □

Example 3.25. Suppose we have a Heegaard diagram (Σ, α, β) as in Figure 10, and let Y be the corresponding 3-manifold. It is straightforward to check that

$$H_1(Y) = \mathbb{Z}_2 \oplus \mathbb{Z}_2.$$

We can apply Lemma 3.12 to obtain a sutured handlebody (H, γ). See Figure 11. By Lemma 3.10 and Theorem 1.2, we know that

$$\dim_{\mathbb{C}} I^{\sharp}(-Y) \leqslant \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-H, -\gamma). \tag{3.6}$$

So it remains to bound $\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-H, -\gamma)$. If we apply Corollary 3.24 directly, we obtain

$$\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-H, -\gamma) \leqslant 64.$$

However, we can improve this bound by examining the bypass exact triangles more carefully in following steps.

Step 1. We can do the bypass attachement along the arc η_1 as shown in Figure 11. By Theorem 2.36, there exists an exact triangle

$$\begin{tikzcd}
 & \underline{\mathrm{SHI}}(-H, -\gamma) \arrow[dr] & \\
\underline{\mathrm{SHI}}(-H, -\gamma_2) \arrow[ur] & & \underline{\mathrm{SHI}}(-H, -\gamma_1) \arrow[ll]
\end{tikzcd} \tag{3.7}$$

The sutures γ_1 and γ_2 are depicted in Figure 12 and Figure 13, respectively.

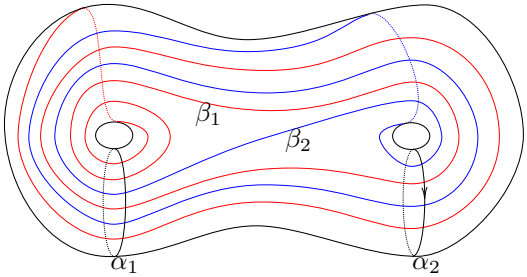


FIGURE 10. A Heegaard diagram.

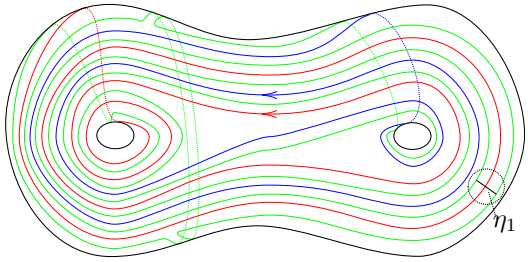


FIGURE 11. The sutured manifold (H, γ).

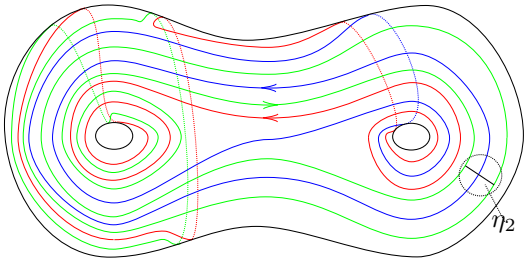


FIGURE 12. The sutured manifold (H, γ_1).

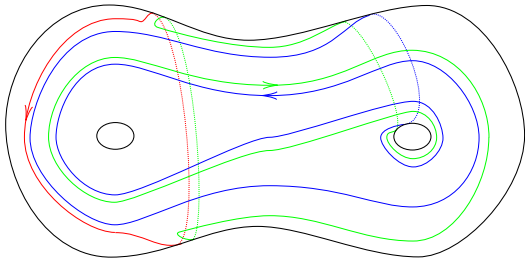


FIGURE 13. The sutured manifold (H, γ_2).

Step 2. We compute $\underline{\mathrm{SHI}}(-H, -\gamma_2)$. The curve α_2 bounds a disk $D \subset H$. It then induces a grading on $\underline{\mathrm{SHI}}(-H, -\gamma_2)$. Note D is not admissible in the sense of Definition 2.19, so we perform a negative stabilization on D as in Definition 2.23, and call the result D^-. By Theorem 2.20, for $|i| > 1$, we have $\underline{\mathrm{SHI}}(-H, -\gamma_2, D^-, i) = 0$. We can perform a sutured manifold decomposition

$$(-H, -\gamma_2) \overset{D^-}{\rightsquigarrow} (V, \gamma_2'),$$

where V is a solid torus and γ_1' is depicted as in Figure 14. From Theorem 2.20 and Theorem 2.13, we know that

$$\underline{\mathrm{SHI}}(-H, -\gamma_2, D^-, 1) \cong \underline{\mathrm{SHI}}(V, \gamma_2') = 0.$$

By Lemma 2.24, Theorem 2.20, and Theorem 2.13, we know

$$\begin{aligned}\underline{\mathrm{SHI}}(-H, -\gamma_2, D^-, -1) =& \underline{\mathrm{SHI}}(-H, -\gamma_2, -D^-, 1)\\ =& \underline{\mathrm{SHI}}(-H, -\gamma_2, (-D)^+, 1)\\ =& 0.\end{aligned}$$

By Theorem 2.20, Theorem 2.26, and Theorem 2.13, we know

$$\begin{aligned}\underline{\mathrm{SHI}}(-H, -\gamma_2, D^-, 0) =& \underline{\mathrm{SHI}}(-H, -\gamma_2, -D^-, 0)\\ =& \underline{\mathrm{SHI}}(-H, -\gamma_2, (-D)^+, 0)\\ =& \underline{\mathrm{SHI}}(-H, -\gamma_2, (-D)^-, 1)\\ =& \underline{\mathrm{SHI}}(V, \gamma_2'').\end{aligned}$$

Here, (V, γ_2'') is obtained from $(-H, -\gamma_2)$ by decomposing along $(-D)^-$. V is a solid torus and γ_2'' is depicted as in Figure 15. From [Li19, Proposition 1.4], we know that

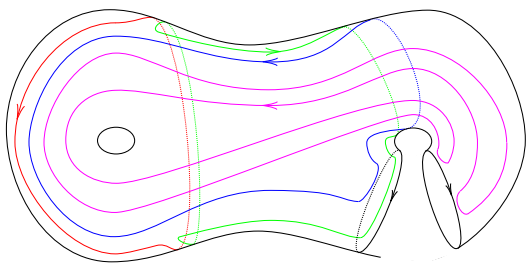


FIGURE 14. The sutured manifold (V, γ_1').

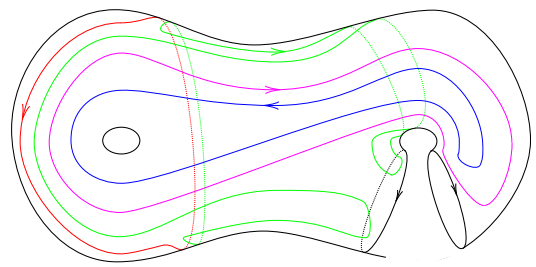


FIGURE 15. The sutured manifold (V, γ_2'').

$$\underline{\mathrm{SHI}}(V, \gamma_2'') \cong \mathbb{C}^2.$$

Hence we conclude that

$$\underline{\mathrm{SHI}}(-H, -\gamma_2) \cong \mathbb{C}^2. \tag{3.8}$$

Step 3. We can perform a second bypass along the arc η_2 as shown in Figure 12 on (H, γ_1) and obtain an exact triangle

$$\begin{tikzcd} & \underline{\mathrm{SHI}}(-H, -\gamma_1) \arrow[dr] & \\ \underline{\mathrm{SHI}}(-H, -\gamma_4) \arrow[ur] & & \underline{\mathrm{SHI}}(-H, -\gamma_3) \arrow[ll, "\psi_4^3"'] \end{tikzcd} \tag{3.9}$$

It is straightforward to check that both γ_3 and γ_4 intersect the disk D at four points, so we can compute in the same way as we did for $\underline{\mathrm{SHI}}(-H, -\gamma_2)$ that

$$\underline{\mathrm{SHI}}(-H, -\gamma_3) \cong \mathbb{C}^3, \text{ and } \underline{\mathrm{SHI}}(-H, -\gamma_4) \cong \mathbb{C}^5. \tag{3.10}$$

From (3.6), (3.7), (3.8), (3.9), and (3.10), we know that

$$\dim_{\mathbb{C}} I^\sharp(-Y) \leqslant 10.$$

3.3. **The instanton knot homology of (1,1)-knots.**

In this subsection, we make use of Theorem 1.2 to prove Theorem 1.4.

Definition 3.26. Suppose integers p and q satisfy $p \geqslant 1, 0 \leqslant q < p$ and $\gcd(p, q) = 1$. Let $\tilde{\alpha}$ and $\tilde{\beta}$ be two straight lines in $\mathbb{R}^2$ passing the origin with slopes 0 and p/q, respectively, and let $r : \mathbb{R}^2 \to T^2$ be the quotient map induced by $(x, y) \to (x + m, y + n)$ for $m, n \in \mathbb{Z}$. Suppose $\alpha = r(\tilde{\alpha})$, and $\beta = r(\tilde{\beta})$. Then the manifold compatible with the Heegaard diagram (T^2, α, β) is called a **lens space** and is denoted by $L(p, q)$. Furthermore, the Heegaard diagram (T^2, α, β) is called the **standard diagram** of the lens space. In particular, we regard S^3 as a lens space.

The lens space is oriented so that the orientation on the α-handlebody is induced from the standard embedding of $S^1 \times D^2$ in $\mathbb{R}^3$. With this convention, the lens space $L(p, q)$ comes from the p/q-surgery on the unknot in S^3.

Definition 3.27. A proper embedded arc γ in a handlebody H is called a **trivial arc** if there is an embedded disk $D \subset H$ satisfying $\partial D = \gamma \cup (D \cap \partial H)$. The disk D is called the **cancelling disk** of γ. A knot K in a closed 3-manifold Y admits a **(1,1)-decomposition** if the followings hold.

(1) Y admits a splitting $Y = H_1 \cup_{T^2} H_2$ so that $H_1 \cong H_2 \cong S^1 \times D^2$.
(2) $K \cap H_i$ is a properly embedded trivial arc in H_i for $i \in \{1, 2\}$.

In this case, Y is either a lens space or $S^1 \times S^2$. A knot K admitting a (1,1)-decomposition is called a **(1,1)-knot**.

Proposition 3.28 (Rasmussen [Ras05], Goda, Matsuda and Morifuji [GMM05])**.** *For $p, q, r, s \geqslant 0$ satisfying $2q + r \leqslant p$ and $s < p$, a (1,1)-decomposition of a knot determines and is determined by a doubly-pointed diagram. After isotopy, such a diagram becomes $(T^2, \alpha, \beta, z, w)$ in Figure 16, where p is the total number of intersection points, q is the number of strands around either basepoint, r is the number of strands in the middle band, and the i-th point on the right-hand side is identified with the $(i + s)$-th point on the left-hand side.*

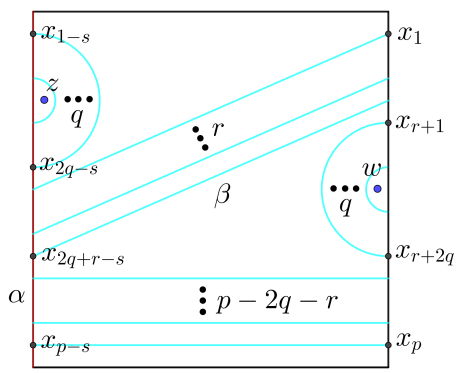


FIGURE 16. (1,1)-diagram.

Definition 3.29. A simple closed curve β on (T^2, α, z, w) is called **reduced** if the number of intersection points between α and β is minimal. The doubly-pointed diagram in Figure 16 is called the **(1,1)-diagram** of type (p, q, r, s), which is denoted by $W(p, q, r, s)$. Strands around basepoints are called **rainbows** and strands in the bands are called **stripes**.

If for some parameters (p, q, r, s), the **(1,1)-diagram** of $W(p, q, r, s)$ is a Heegaard diagram, or equivalently, β has one component and represents a nontrivial homology class in $H_1(T^2)$, then the corresponding knot is also denoted by $W(p, q, r, s)$.

A (1,1)-knot whose (1,1)-diagram does not have rainbows is called a **simple knot** (*c.f.* Rasmussen [Ras07]). For simple knots, let $K(p, q, k) = W(p, 0, k, q)$.

Proposition 3.30. *The mirror knot of a (1,1)-knot $W(p, q, r, s)$ is*

$$W(p, q, p - 2q - r, p - s + 2q).$$

Proof. The Heegaard diagram of the mirror knot of $W(p, q, r, s)$ is obtained by the (1,1)-diagram of $W(p, q, r, s)$ by vertical reflection. Redrawing the Heegaard diagram so that the lower band becomes the middle band and the middle band becomes the lower band, this proposition is straightforward from definition. □

According to [GMM05], for the $\widehat{HFK}$ of a (1,1)-knot, the generators of the chain complexes are intersection points of α and β in the (1,1)-diagram and there is no differential. Thus, the following proposition holds.

Proposition 3.31 ([GMM05])**.** *For a (1,1)-knot $K = W(p,q,r,s)$ in Y, we have*

$$\widehat{HFK}(Y,K) \cong \mathbb{Z}^p.$$

We restate the construction in Remark 3.13 more carefully.

Definition 3.32. Suppose $(T^2, \alpha, \beta, z, w)$ is the (1,1)-diagram of $W(p,q,r,s)$. We construct a sutured handlebody (H,γ), called the **(1,1)-sutured-handlebody** of $W(p,q,r,s)$, as follows.

(1) Let Σ be the genus-two boundary of the manifold obtained from $[-1,1] \times T^2$ by attaching a 3-dimensional 1-handle along $\{1\} \times \{z,w\}$. For simplicity, when drawing the diagram, the attached 1-handle will still be denoted by two basepoints z and w.
(2) Let α_1 and β_1 denote the curves on Σ induced from α and β, respectively. Let β be oriented so that the innermost rainbow around z is oriented clockwise, which induces an orientation of β_1. If there is no rainbow, let β be oriented so that each stripe goes from left to right in Figure 16.
(3) Consider the straight arc connecting z and w in Figure 16. It induces a simple closed curve α_2 on Σ by going along the 1-handle. Let β_2 be the curve on Σ induced by a small circle around z, oriented counterclockwise.
(4) Let γ_1 and γ_2 be obtained by pushing off β_1 and β_2 to the right with respect to the orientation. Suppose they are oriented reversely. Let a_0 be a straight arc connecting the innermost rainbow of β around z and the above small circle. It induces an arc connecting γ_1 and γ_2, still denoted by a_0. Let γ_3 be obtained by a band sum of γ_1 and γ_2 along a_1, with the induced orientation.
(5) Let H be the handlebody compatible with the diagram $(\Sigma, \{\alpha_1, \alpha_2\}, \varnothing)$ and let

$$\gamma = \gamma_1 \cup \gamma_2 \cup \gamma_3.$$

Rainbows and stripes are defined similarly for sutures.

The main goal is to prove the following theorem.

Theorem 3.33. *For a (1,1)-sutured-handlebody (H,γ) of $W(p,q,r,s)$, we have*

$$\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-H,-\gamma) \leqslant p.$$

Before proving this theorem, we first use it to derive Theorem 1.4.

Proof of Theorem 1.4. Combining Theorem 1.2, Proposition 3.30, Proposition 3.31, and Theorem 3.33, for a (1,1)-knot $K = W(p,q,r,s)$ in a lens space Y, we have

$$\dim_{\mathbb{C}} KHI(-Y,K) \leqslant \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-H,-\gamma) \leqslant p = \mathrm{rk}_{\mathbb{Z}} \widehat{HFK}(-Y,K).$$

The theorem follows from the fact that the mirror knot of a (1,1)-knot is still a (1,1)-knot. □

Proof of Theorem 3.33. We prove the theorem by induction on p for any (1,1)-diagram of $W(p,q,r,s)$ where β has only one component. This includes the case that β represents trivial homology class.

For the base case, consider $p = 1$. The curves $\alpha_1, \alpha_2, \beta_1, \beta_2$ in Definition 3.32 satisfy

$$|\alpha_1 \cap \beta_1| = |\alpha_2 \cap \beta_2| = 1.$$

It is straightforward to check (H,γ) is a product sutured manifold, so is $(-H,-\gamma)$. Then Theorem 2.14 implies

$$\dim_{\mathbb{C}} SHI(-H,-\gamma) = 1.$$

Now we deal with the case when $p > 1$. In Definition 3.32, the innermost rainbow around z, if exists, is oriented clockwise. Suppose δ_1 is either the innermost rainbow around z, or a stripe that is closest to z with z on its right-hand side. Suppose δ_2 is another rainbow or stripe that is closest to δ_1 and is to the left of δ_1. See Figure 17 for all possible cases. Compared to Figure 16, we have rotated the square counterclockwise by 90 degrees for the purpose of a better display.

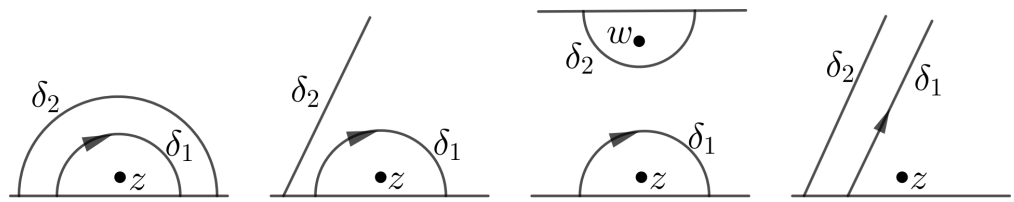


FIGURE 17. Several cases of δ_1 and δ_2.

We consider two different cases about the orientation of δ_2.

Case 1. Suppose δ_1 and δ_2 are oriented parallelly.

We use $W(6,2,1,3)$ shown in Figure 18 as an example to carry out the proof, and the general case is similar. In this example, two innermost rainbows around z are oriented parallelly. By construction, the suture γ_3 is parallel (regardless of orientations) to γ_1 outside the neighborhood of the band-sum arc a_0. Thus, there exists a unique rainbow of γ_3 between δ_1 and δ_2 around z. Let a_1 be an anti-wave bypass arc cutting these three rainbows, as shown in Figure 18. Suppose γ' and γ'' are the other two sutures involved in the bypass triangle associated to a_2, respectively.

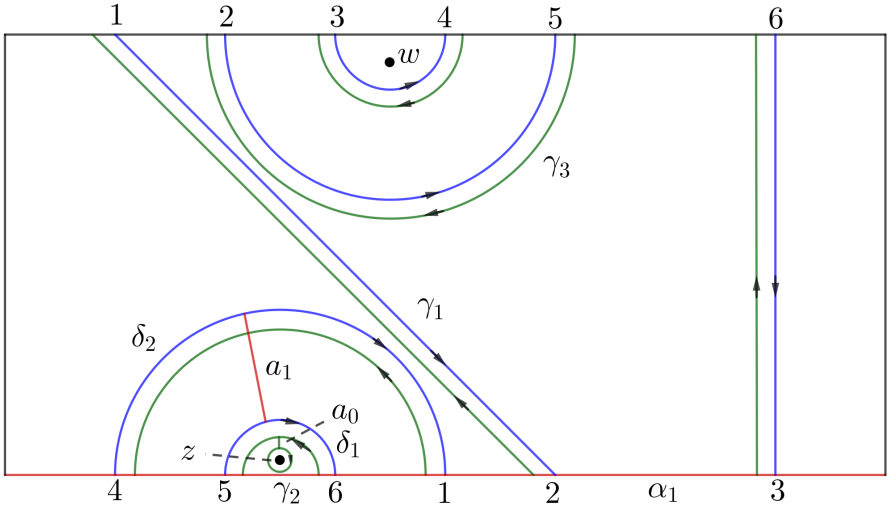


FIGURE 18. Sutures related to $W(6,2,1,3)$ and the anti-wave bypass arc.

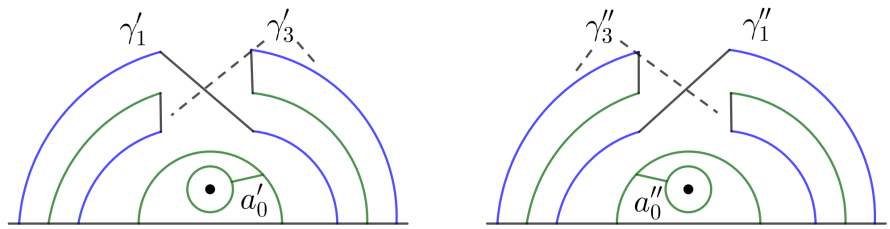


FIGURE 19. Local diagrams after bypass attachments.

From Proposition 2.30, we can describe the sutures γ' and γ'' as follows. First, let γ_1^1 and γ_1^2 be two components of $\gamma_1 \backslash \partial a_1$. Suppose

$$\gamma_1' = \gamma_1^1 \cup a_1, \text{ and } \gamma_1'' = \gamma_1^2 \cup a_1$$

as shown in Figure 19. Second, γ' is obtained from γ by a Dehn twist along γ_1'', and γ'' is obtained from γ by a Dehn twist along γ_1'.

There is a more direct way to describe γ' and γ''. First, note that the suture γ_2 is disjoint from both Dehn-twist curves γ_1' and γ_1'', so γ_2 remains the same in γ' and γ''. Second, it is straightforward to check the result of γ_1 under the Dehn twist along γ_1'' is γ_1', and the result of γ_1 under the Dehn twist along γ_1' is γ_1''. Thus, γ_1' is a component of γ' and γ_1'' is a component of γ''.

To figure out the image γ_3' of γ_3 under the Dehn twist along γ_1'', we first observe that we can isotop the band-sum arc a_0 to a new position a_0' so that its endpoints $\partial a_0'$ lie on $\gamma_1' \cap \gamma_1$ and γ_2, as shown in the left sub-figure of Figure 19. Thus, the facts that a_0' is disjoint from γ_1'' and that γ_1' is the image of γ_1 under the Dehn twist along γ_1'' imply that performing a Dehn twist along γ_1'' and performing the band sum along a_0' commute with each other. Thus, we conclude that γ_3' can be obtained from a band sum on γ_1' and γ_2 along the arc a_0'. Similarly we can describe the image γ_3'' of γ_3 under the Dehn twist along γ_1'. Thus, we have described the sutures

$$\gamma' = \gamma_1' \cup \gamma_2 \cup \gamma_3', \text{ and } \gamma'' = \gamma_1' \cup \gamma_2 \cup \gamma_3''$$

explicitly, and it follows that (H, γ') and (H, γ'') are both (1,1)-sutured handlebodies. Suppose they are associated to $W(p', q', r', s')$ and $W(p'', q'', r'', s'')$, respectively.

From the above description, both γ_1' and γ_1'' are reduced. We have

$$p' + p'' = |\gamma_1' \cap \alpha_1| + |\gamma_1'' \cap \alpha_1| = |\gamma_1 \cap \alpha_1| = p.$$

Thus, the induction applies.

Case 2. Suppose δ_1 and δ_2 are oriented oppositely.

An example $W(10, 3, 1, 5)$ is shown in Figure 20. By construction, there is a rainbow of γ_3 to the right of δ_2. Let a_2 be a wave bypass arc cutting δ_1, δ_2, and this rainbow as shown in Figure 20. Suppose γ' and γ'' are the other two sutures involved in the bypass triangle associated to a_2, respectively.

To describe the sutures γ' and γ'' more explicitly, note that the arc a_2 cuts γ_1 into two parts γ_1^1 and γ_1^2. Suppose that near a_2, γ_1^1 is to the left of a_2 and γ_1^2 is to the right of a_2. For γ', it consists of three components:

$$\gamma' = \gamma_1' \cup \gamma_2 \cup \gamma_3',$$

where γ_2 is as before, γ_1' is obtained by cutting γ_3 open by a_2 and gluing it to γ_1^2, and γ_3' is obtained by gluing a copy of a_2 to γ_1^1. They are depicted as in the left sub-figure of Figure 21. Note the curve

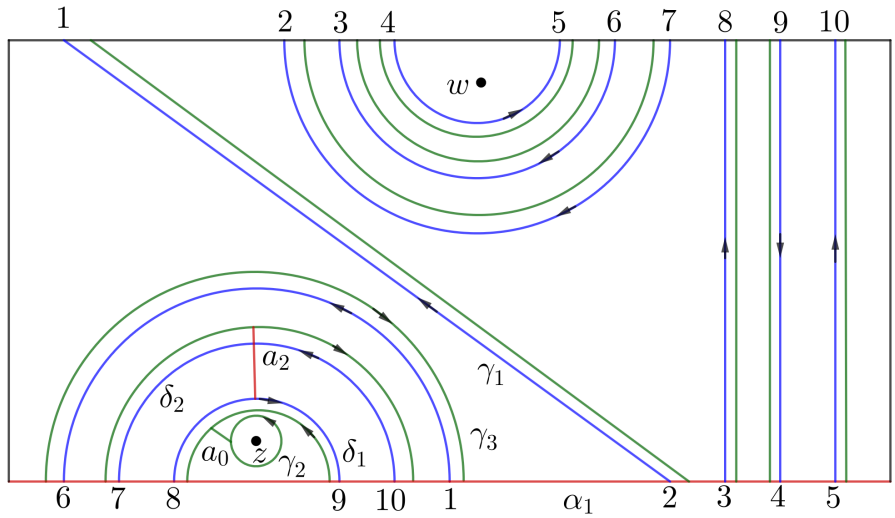


FIGURE 20. Sutures related to $W(10, 3, 1, 5)$ and the wave bypass arc.

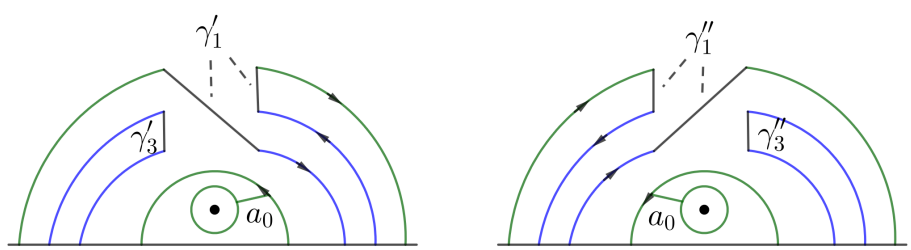


FIGURE 21. Local diagrams after bypass attachments.

FIGURE 22. Local diagrams after isotopy.

γ_1' is not reduced. We can isotop the curve along the arc γ_1^2 into a reduced curve. The orientations of sutures imply this reduced curve is depicted as in the left sub-figure of Figure 22. Note γ_3' is also not reduced. However, from Figure 22 it is straightforward to check that γ_3' can be thought of as obtained from γ_2 and γ_1' by a band sum along the arc a_0'. Also, it is clear that

$$|\gamma_1' \cap \alpha_1| = |\gamma_1^1 \cap \alpha_1|.$$

Similarly, γ'' consists of three components:

$$\gamma'' = \gamma_1'' \cup \gamma_2 \cup \gamma_3'',$$

where γ_2 is as before, γ_1'' is obtained by cutting γ_3 open by a_2 and gluing it to γ_1^1, and γ_3'' is obtained by gluing a copy of a_2 to γ_1^2. They are depicted as in the right sub-figure of Figure 21. Considering the orientations, we can isotop γ_2'' along γ_2^1 to the position shown in the right sub-figure of Figure 22. Then γ_3'' can be thought of as obtained from γ_1 and γ_2'' by a band sum along a_0''. Also,

$$|\gamma_1'' \cap \alpha_1| = |\gamma_1^2 \cap \alpha_1|.$$

Hence we conclude that γ' and γ'' both come from some (1,1)-diagrams, and

$$|\gamma_2' \cap \alpha_1| + |\gamma_2'' \cap \alpha_1| = |\gamma_{2,1} \cap \alpha_1| + |\gamma_{2,2} \cap \alpha_1| = |\gamma_2 \cap \alpha_1|.$$

Thus, the induction applies.

□

3.4. **Large surgeries on simple knots.**

In this subsection, we generalize the idea about the anti-wave bypass arc in the proof of Theorem 3.33 to prove Theorem 1.8.

For a simple knot $K = K(p, q, k) \subset Y$, consider its (1,1)-diagram $(T^2, \alpha_1, \beta_1, z, w)$ and the Heegaard diagram $(\Sigma, \{\alpha_1, \alpha_2\}, \{\beta_1, \beta_2\})$ of Y from Lemma 3.9. Suppose $\alpha = \{\alpha_1, \alpha_2\}$ and $m = \beta_2$. Let l be an arc connecting z to w in $T^2 - \beta$, which induces a curve on Σ, still denoted by l. Then $(\Sigma, \alpha, \{\beta_1\})$ is a diagram of $Y(K)$ and (m, l) forms a basis of $H_1(\partial Y(K))$.

Given $p, q \in \mathbb{Z}$ so that $\gcd(p, q) = 1$, let β_2' be the curve on Σ obtained by resolving intersection points of $|q|$ parallel copies of m and $|p|$ parallel copies of l. Then $(\Sigma, \alpha, \{\beta_1, \beta_2'\})$ is a Heegaard diagram of the manifold obtained from the q/p-surgery on K.

Remark 3.34. There are two ways of resolutions, namely the positive resolution and the negative resolution. The choice of the ways depends on orientations of Σ and curves, and signs of p and q. However, the goal in Theorem 1.8 is for any large enough surgery slope without regard of the sign. So the choice here is not important.

Definition 3.35 (Greene and Levine [GL16])**.** An arc a in a Heegaard diagram (Σ, α, β) is called an **anti-wave** if it satisfies the following conditions.

(1) It is properly embedded in a component R of $\Sigma \backslash (\alpha \cup \beta)$.
(2) Its endpoints lie on the interior of distinct arcs r_1, r_2 of ∂R, where r_1 and r_2 are subsets of the same curve $\alpha_i \subset \alpha$ or $\beta_i \subset \beta$ for some i.
(3) The local signs of intersection at two endpoints are the same.

For the simple knot K, there exist anti-waves in each component of $T^2 - \alpha \cup \beta$. It is possible to choose l so that $l \cap R = \varnothing$ for some component R of $\Sigma \backslash (\alpha \cup \beta)$. Let a be an anti-wave in R. It induces an anti-wave in $(\Sigma, \alpha, \{\beta_1, \beta_2'\})$ which is still denoted by a.

Lemma 3.36. *Given the surgery slope q/p, consider the Heegaard diagram $(\Sigma, \alpha, \{\beta_1, \beta_2'\})$ and the anti-wave a defined as above. Let β_1^1 and β_1^2 denote two arcs of $\beta_1 \backslash \partial a$ and let $\beta_{1,i} = \beta_1^i \cup a$ for $i \in \{1, 2\}$. Suppose Y_i is the manifold compatible with the Heegaard diagram $(\Sigma, \alpha, \{\beta_{1,i}, \beta_2'\})$ and K_i is the core knot of β_2'. Suppose $(Y_0, K_0) = (Y, K)$. Then there exists a knot $K' \subset Y$ so that (Y_i, K_i), for $i = 0, 1, 2$, satisfy the exact triangle associated to K' in Theorem 2.18.*

Proof. Consider neighborhoods $N(\beta_1)$ and $N(\beta_2')$ of β_1 and β_2' on Σ, respectively. The manifold

$$\Sigma\backslash \mathrm{int}(N(\beta_1')) \cup \mathrm{int}(N(\beta_2))$$

is diffeomorphic to $S^2\backslash \cup_{i=1}^4 D_i$, where D_i are pair-wise disjoint disks. Suppose ∂D_1 and ∂D_2 are images of $\partial N(\beta_2)$ under the diffeomorphism and a becomes an arc connecting ∂D_1 and ∂D_2. There exists a curve $\beta_3 \subset S^2$ separates $D_1 \cup D_2$ and $D_3 \cup D_4$. It induces a null-homologous curve on Σ which is still denoted by β_3. By construction, β_3 is disjoint from β_1, β_2' and $\beta_{1,i}$ for $i \in \{1,2\}$.

The 3-manifold compatible with the diagram $(\Sigma, \alpha, \{\beta_3\})$ has two toroidal boundary components. It can be regarded the complement of K and K' in Y for some knot K'. Equivalently, K is the core knot of β_2 and K' is the core knot of β_1 and the way how K and K' is linked is determined by β_3. After isotopy, $\beta_2, \beta_{2,1}$ and $\beta_{2,2}$ intersect pair-wise at one point. Then Theorem 2.18 applies in this case for K'. □

Lemma 3.37. *There exists $N_0 > 0$ so that for any surgery slope q/p satifying $|q/p| \geqslant N_0$, the manifolds Y_i in Lemma 3.36 corresponding to q/p, for $i = 0, 1, 2$, satisfy*

$$|H_1(Y_0)| = |H_1(Y_1)| + |H_1(Y_2)|.$$

Proof. For $i \in \{1,2\}$, it is straightforward to check that $\beta_{1,i}$ is a reduced curve without rainbows. Suppose orientations of curves are chosen so that

$$|\beta_{1,i} \cap \alpha_1| = \beta_{1,i} \cdot \alpha_1 = p_i, \text{ and } |\beta_{1,i} \cap \alpha_2| = \beta_{1,i} \cdot \alpha_2 = k_i.$$

By construction, we have

$$p_1 + p_2 = p_0, \text{ and } k_1 + k_2 = k_0.$$

It is clear that K_i is the dual knot of some surgery on $K(p_i, q_i, k_i)$ for some q_i. Suppose

$$l \cdot \alpha_1 = x, \text{ and } l \cdot \alpha_2 = y.$$

Suppose the orientation of m is chosen so that $m \cdot \alpha_2 = 1$. Then we have

$$\beta_2' \cdot \alpha_1 = px, \text{ and } \beta_2' \cdot \alpha_2 = q + py.$$

Thus, for $i \in \{1,2\}$, we have

$$|H_1(Y_0)| = |pk_0 - (q+py)p_0|, \text{ and } |H_1(Y_i)| = |pk_i - (q+py)p_i|.$$

One of these orders is the sum of other two. If $|q/p|$ is large enough, the order

$$|H_1(Y_0)| = p|k_0 - (q/p + y)p_0|$$

is greater than $|H_1(Y_i)|$ for $i \in \{1,2\}$. Hence we conclude the lemma. □

Proof of Theorem 1.8. We prove the theorem by induction on p_0 for simple knots $K(p_0, q_0, k_0)$.

For the base case, consider $p_0 = 1$. The simple knot is the unknot in S^3. The dual knot is the core knot of β for the standard diagram of a lens space, which is also a simple knot (*c.f.* Rasmussen [Ras07]). The theorem follows from Proposition 1.7.

When $p_0 > 1$, consider knots K_i and the related simple knots $K(p_i, q_i, k_i)$ for $i \in \{1,2\}$ in Lemma 3.36 and Lemma 3.37. By induction hypothesis, there exist N_i for $K(p_i, q_i, k_i)$ so that r surgery with $|r| \geqslant N$ induces an instanton Floer simple knot. Equivalently,

$$\dim_{\mathbb{C}} KHI(Y_1, K_1) = |H_1(Y_1)|, \text{ and } \dim_{\mathbb{C}} KHI(Y_2, K_2) = |H_1(Y_2)|.$$

We have to discuss about the basis of the homology at first. The basis (m, l_0) of $H_1(\partial Y(K_0))$ is chosen with respect to the anti-wave in the Heegaard diagram corresponding to K_0, which induces a basis on $H_1(\partial Y_i(K_i))$. However, in the proofs of Lemma 3.36 and Lemma 3.37, another basis (m, l_i)

is chosen with respect to the anti-wave in the Heegaard diagram corresponding to K_i. Suppose $l_i = x_i m + l$. Then

$$qm + pl = (q - px_i)m + pl_i.$$

Suppose $N = \max\{N_1 + |x_1|, N_2 + |x_2|, N_0\}$. Then Lemma 3.37 implies

$$|H_1(Y_0)| = |H_1(Y_1)| + |H_1(Y_2)|.$$

Combining Theorem 2.18 and Lemma 3.36, we have

$$\dim_{\mathbb{C}} KHI(Y_0, K_0) \leqslant \dim_{\mathbb{C}} KHI(Y_1, K_1) + \dim_{\mathbb{C}} KHI(Y_2, K_2).$$

Hence we have

$$\dim_{\mathbb{C}} KHI(Y_0, K_0) \leqslant |H_1(Y_0)|.$$

Combining Theorem 1.2 and [Sca15, Corollary 1.4], we have

$$\dim_{\mathbb{C}} KHI(Y_0, K_0) \geqslant \dim_{\mathbb{C}} I^{\sharp}(Y_0) \geqslant |H_1(Y_0)|.$$

Thus,

$$\dim_{\mathbb{C}} KHI(Y_0, K_0) = |H_1(Y_0)|.$$

and the induction applies. □

Remark 3.38. For the above proof, the induction based on $\underline{\mathrm{SHI}}(-H, -\gamma)$ does not work any more because the inequality

$$\dim_{\mathbb{C}} KHI(-Y, K) \leqslant \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-H, -\gamma)$$

might not always be sharp. That is the reason why we switch to surgery exact triangles instead of sticking to bypass exact triangles.

4. Instanton Floer homology and the decomposition

4.1. **Basic setups.**

Suppose Y is a closed 3-manifold and $K \subset Y$ is a null-homologous knot. Let $Y(K)$ be the knot complement $Y \backslash \mathrm{int}(N(K))$. Any Seifert surface S of K gives rise to a framing on $\partial Y(K)$: the longitude λ can be picked as $S \cap \partial Y(K)$ with the induced orientation from S, and the meridian μ can be picked as the meridian of the solid torus $N(K)$ with the orientation so that $\mu \cdot \lambda = -1$. The 'half lives and half dies' fact for 3-manifolds implies that the following map has a 1-dimensional image:

$$\partial_* : H_2(Y(K), \partial Y(K); \mathbb{Q}) \to H_1(\partial Y(K); \mathbb{Q}).$$

Hence any two Seifert surfaces lead to the same framing on $\partial Y(K)$.

Definition 4.1. The framing (μ, λ) defined as above is called the **canonical framing** of (Y, K). With respect to this canonical framing, let

$$\widehat{Y}_{q/p} = Y(K) \cup_{\phi} S^1 \times D^2$$

be the 3-manifold obtained from Y by a q/p surgery along K, *i.e.*,

$$\phi(\{1\} \times \partial D^2) = q\mu + p\lambda.$$

When the surgery slope is understood, we also write $\widehat{Y}_{q/p}$ simply as $\widehat{Y}$. Let $\widehat{K}$ be the dual knot, *i.e.*, the image of $S^1 \times \{0\} \subset S^1 \times D^2$ in $\widehat{Y}$ under the gluing map.

Convention. Throughout this section, we will always assume that $\gcd(p,q) = 1$ and $q > 0$ or $(p,q) = (1,0)$ for a Dehn surgery. Especially, the original pair (Y,K) can be thought of as a pair $(\widehat{Y},\widehat{K})$ obtained from (Y,K) by the $1/0$ surgery. Moreover, we will always assume that the knot complement $Y(K)$ is irreducible.

Remark 4.2. The reason to use

Next, we describe various families of sutures on the knot complement. Suppose $K \subset Y$ is a null-homologous knot and the pair $(\widehat{Y},\widehat{K})$ is obtained from (Y,K) by a q/p surgery. Note we can identify the complement of $K \subset Y$ with that of $\widehat{K} \subset \widehat{Y}$:

$$\widehat{Y}(\widehat{K}) = Y(K).$$

On $\partial Y(K)$, there are two framings: One comes from K, and we write longitude and meridian as λ and μ, respectively. The other comes from $\widehat{K}$. Note only the meridian $\hat{\mu}$ of $\widehat{K}$ is well-defined, and by definition, it is

$$\hat{\mu} = q\mu + p\lambda.$$

Remark 4.3. The reason using q and p for coefficients of μ and λ, respectively, is to be consistent with the definition of 'slope' in [Hon00].

Definition 4.4. If $p = 0$, then $q = 1$ and $\hat{\mu} = \mu$. We can take $\hat{\lambda} = \lambda$. If $(q,p) = (0,1)$, then we take $\hat{\lambda} = -\mu$. If $p, q \neq 0$, then we take $\hat{\lambda} = q_0\mu + p_0\lambda$, where (p_0, q_0) is the unique pair of integers so that the following conditions are true.

(1) $0 \leqslant |p_0| < |p|$ and $p_0 p \leqslant 0$.
(2) $0 \leqslant |q_0| < |q|$ and $q_0 q \leqslant 0$.
(3) $p_0 q - p q_0 = 1$.

For a homology class $x\lambda + y\mu$, let $\gamma_{(x,y)} = \gamma_{x\lambda+y\mu}$ be the suture consists of two disjoint simple closed curves representing $\pm(x\lambda + y\mu)$ on $\partial Y(K)$. Furthermore, for $n \in \mathbb{Z}$, define

$$\widehat{\Gamma}_n(q/p) = \gamma_{\hat{\lambda}-n\hat{\mu}}$$

Suppose (p_n, q_n) is the unique pair of integers so that $q_n \geqslant 0$ and $\widehat{\Gamma}_n(q/p) = \gamma_{(p_n,q_n)}$. Finally, define

$$\widehat{\Gamma}_\mu(q/p) = \gamma_{\hat{\mu}}.$$

When $\hat{\lambda}$ and $\hat{\mu}$ are understood, we omit the slope q/p and simply write $\widehat{\Gamma}_n$ and $\widehat{\Gamma}_\mu$.

Remark 4.5. Since the two components of the suture must be given opposite orientations, the notations $\gamma_{(p,q)}$ and $\gamma_{(-p,-q)}$ represent the same suture on the knot complement $Y(K)$. Our choice makes $q_{n+1} \leqslant q_n$ for $n < -1$ and $q_{n+1} \geqslant q_n$ for $n \geqslant 0$.

In the following three subsections, we prove Proposition 1.17.

4.2. **Bypasses on knot complements.**

Suppose Y is a closed 3-manifold and $K \subset Y$ is a null-homologous knot. Let (μ,λ) be the canonical framing on $Y(K)$. Suppose y_3/x_3 is a surgery slope with $y_3 \geqslant 0$. According to Honda [Hon00], there are two basic bypasses on the balanced sutured manifold $(Y(K), \gamma_{(x_3,y_3)})$, whose arcs are depicted as in Figure 23. The sutures involved in the bypass triangles were described explicitly in Honda [Hon00, Section 4.4.4].

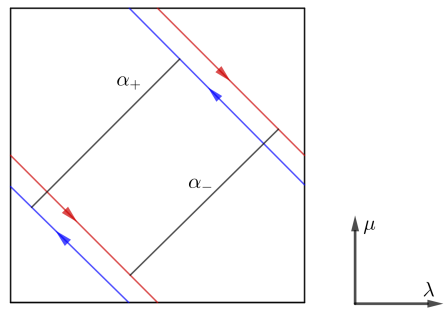


FIGURE 23. Bypass arcs on $\gamma_{(1,-1)}$.

Definition 4.6. For a surgery slope y_3/x_3 with $y_3 \geqslant 0$, suppose its continued fraction is

$$\frac{y_3}{x_3} = [a_0, a_1, ..., a_n] = a_0 - \frac{1}{a_1 - \frac{1}{\cdots - \frac{1}{a_n}}},$$

where integers $a_i < -1$. If $y_3 > -x_3 > 0$, let

$$\frac{y_1}{x_1} = [a_0, ..., a_{n-1}] \text{ and } \frac{y_2}{x_2} = [a_0, ..., a_n + 1].$$

When $a_i = -2$ for integer $i \in (k, n]$ and $a_k \neq -2$, we can set

$$[a_0, ..., a_n + 1] = [a_0, ..., a_k + 1]$$

for simplicity. If $-x_3 > y_3 > 0$, we do the same thing for $x_3/(-y_3)$. If $y_3 > x_3 > 0$, we do the same thing for $y_3/(-x_3)$. If $x_3 > y_3 > 0$, we do the same thing for $x_3/(-y_3)$. If $y_3/x_3 = 1/0$, then $y_1/x_1 = 0/1$ and $y_2/x_2 = 1/(-1)$. If $y_3/x_3 = 0/1$, let $y_1/x_1 = 1/(-1)$ and $y_1/x_1 = 0/1$. We always require that $y_1 \geqslant 0$ and $y_2 \geqslant 0$.

Remark 4.7. It is straightforward to use induction to verify that for $y_3 > -x_3 > 0$,

$$x_3 = x_1 + x_2, \text{ and } y_3 = y_1 + y_2.$$

Then the bypass exact triangle in Theorem 2.36 becomes the following.

Proposition 4.8. *Suppose $K \subset Y$ is a null-homologous knot, and suppose the surgery slopes y_i/x_i for $i \in \{1, 2, 3\}$ are defined as in Definition 4.6. Suppose the indices are considered mod 3. Let $\psi^{y_i/x_i}_{+,y_{i+1}/x_{i+1}}$ and $\psi^{y_i/x_i}_{-,y_{i+1}/x_{i+1}}$ be bypass maps from two different bypasses, respectively. Then there are two exact triangles related to $\psi^{y_i/x_i}_{+,y_{i+1}/x_{i+1}}$ and $\psi^{y_i/x_i}_{-,y_{i+1}/x_{i+1}}$, respectively.*

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\gamma_{(x_2,y_2)}) \arrow[rr, "\psi^{y_2/x_2}_{\pm,y_3/x_3}"] & & \underline{\mathrm{SHI}}(-Y(K), -\gamma_{(x_3,y_3)}) \arrow[dl, "\psi^{y_3/x_3}_{\pm,y_1/x_1}"] \\
& \underline{\mathrm{SHI}}(-Y(K), -\gamma_{(x_1,y_1)}) \arrow[ul, "\psi^{y_1/x_1}_{\pm,y_2/x_2}"] &
\end{tikzcd}$$

Remark 4.9. Note there are two different bypasses, so two different exact triangles, while both of them involve the same set of balanced sutured manifolds.

Convention. We will use $\psi^*_{+,*}$ and $\psi^*_{-,*}$ to denote bypass maps with respect to some slopes.

Next, we discuss the special case when the bypasses are performed on sutures that arise from Dehn surgeries.

Proposition 4.10. *Suppose $K \subset Y$ is a null-homologous knot and the pair $(\widehat{Y}, \widehat{K})$ being obtained from (Y, K) by a q/p surgery. Suppose further that the sutures $\widehat{\Gamma}_n$ and $\widehat{\Gamma}_\mu$ are defined as in Definition 4.4. Then there are two exact triangles related to $\psi^*_{+,*}$ and $\psi^*_{-,*}$, respectively.*

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n) \arrow[rr, "\psi^n_{\pm,n+1}"] & & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}) \arrow[dl, "\psi^{n+1}_{\pm,\mu}"] \\
& \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_\mu) \arrow[ul, "\psi^{\mu}_{\pm,n}"] &
\end{tikzcd} \tag{4.1}$$

Proof. If $\widehat{\Gamma}_{n+1} = \gamma_{(x_3,y_3)}$ and $y_3 > -x_3 > 0$ and , then it is straight forward to check that

$$\gamma_{(x_1,y_1)} = \widehat{\Gamma}_\mu, \text{ and } \gamma_{(x_2,y_2)} = \widehat{\Gamma}_n.$$

Then the exact triangles follows from Proposition 4.8. The similar proof applies for other cases. □

Similar to Lemma 3.21, we have the following proposition.

Lemma 4.11 ([GLW19, Section 3])**.** *Suppose $K \subset Y$ is a null-homologous knot and the pair $(\widehat{Y}, \widehat{K})$ being obtained from (Y, K) by a q/p surgery. Suppose further that the sutures $\widehat{\Gamma}_n$ are defined as in Definition 4.4. Then, there is an exact triangle*

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n) \arrow[rr] & & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}) \arrow[dl, "F_{n+1}"] \\
& \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta) \arrow[ul, "G_n"] &
\end{tikzcd} \tag{4.2}$$

Here, the balanced sutured manifold $(\widehat{Y}(1), \delta)$ is defined as in Remark 2.9.

Furthermore, we have four commutative diagrams related to $\psi^n_{+,n+1}$ and $\psi^n_{-,n+1}$, respectively

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n) \arrow[rr, "\psi^n_{\pm,n+1}"] & & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}) \\
& \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta) \arrow[ul, "G_n"] \arrow[ur, "G_{n+1}"'] &
\end{tikzcd}$$

and

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n) \arrow[rr, "\psi^n_{\pm,n+1}"] \arrow[dr, "F_n"'] & & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}) \arrow[dl, "F_{n+1}"] \\
& \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta) &
\end{tikzcd}$$

The bypass maps in (4.1) behave well under the gradings on $\underline{\mathrm{SHI}}$ associated to a fixed Seifert surface of K. To provide more details, let us fix a minimal genus Seifert surface S of K.

Convention. We will always assume by default that the Seifert surface S has minimal possible intersections with any suture $\gamma_{(p,q)}$ we are dealing with.

Definition 4.12. Suppose $K \subset Y$ is a null-homologous knot and $\gamma_{(x,y)}$ is a suture on $\partial Y(K)$ with $y \geqslant 0$. Suppose further that S is a minimal genus Seifert surface of K. Write $S^{\tau(y)}$ to be a negative stabilization of S if y is even and to be the original S if y is odd. When the suture $\gamma_{(x,y)}$ is understood, we simply write S^τ. More explicitly, we define a map $\tau : \mathbb{N} \mapsto \{0, -1\}$ as

$$\tau(y) = \begin{cases} 0 & y \text{ is odd} \\ -1 & y \text{ is even} \end{cases}$$

Remark 4.13. It is straightforward to check that $S^\tau \subset (M, \gamma_{(x,y)})$ is admissible. Note that the negative stabilization is with respect to the suture $\gamma_{(x,y)}$ rather than $-\gamma_{(x,y)}$. This is important because later we will incorporate this definition of S^τ with the bypass maps, where the orientations of the sutures are reverse (*c.f.* Remark 2.25).

Convention. Note in Subsection 3.2, we also define another function τ. Since the old definition will no longer be used, and the new tau function serves for the same purpose as the old, we keep using the same notation. From now on, we use the new definition of τ as in Definition 4.12.

Lemma 4.14. *Suppose $K \subset Y$ is a null-homologous knot and $\gamma_{(x,y)}$ is a suture on $\partial Y(K)$ with $y \geqslant 0$. Suppose further that S is a minimal genus Seifert surface of K. Suppose the maximal and minimal nonzero gradings of $\underline{\mathrm{SHI}}(-Y(K), -\gamma_{(x,y)}, S^\tau)$, are i_{max} and i_{min}, respectively. Then we have*

$$i_{max} = \frac{1}{2}(y - 1 - \tau(y)) + g(S) = \lceil \frac{y-1}{2} \rceil + g(S)$$

and

$$i_{min} = -\frac{1}{2}(y - 1 + \tau(y)) - g(S) = \lceil -\frac{y-1}{2} \rceil - g(S).$$

Proof. The proof is similar to that of Lemma 3.15, though in the current lemma, we can also identify the top and bottom non-trivial gradings by making use of sutured manifold decompositions. Note we have assumed that the knot complement $Y(K)$ is irreducible, and S is a minimal genus Seifert surface of K, so the decomposition of $(Y(K), \gamma)$ along S and $-S$ are both taut.

When y is odd, we have $S^\tau = S$. Then it follows directly from Theorem 2.20 that

$$i_{max} = \frac{y-1}{2} + g(S), \text{ and } i_{min} = -\frac{y-1}{2} - g(S).$$

When y is even, we have $S^\tau = S^-$. Then it follows directly from Theorem 2.20 that

$$i_{max} = \frac{y}{2} + g(S).$$

To figure out the grading i_{min}, note

$$\begin{aligned} \underline{\mathrm{SHI}}(-Y(K), -\gamma_{(x,y)}, S^-, -i_{\max}) =& \underline{\mathrm{SHI}}(-Y(K), -\gamma_{(x,y)}, -(S^-), i_{\max}) \\ =& \underline{\mathrm{SHI}}(-Y(K), -\gamma_{(x,y)}, (-S)^+, i_{\max}) \\ =& 0 \end{aligned}$$

The last equality follows from Lemma 2.24 and the fact that the positive stabilization on $(-S)$ is with respect to $\gamma_{(x,y)}$ and becomes a negative one with respect to $-\gamma_{(x,y)}$.

We also have

$$\begin{aligned}\underline{\mathrm{SHI}}(-Y(K),-\gamma_{(x,y)},S^-,1-i_{\max}) =&\underline{\mathrm{SHI}}(-Y(K),-\gamma_{(x,y)},-(S^-),i_{\max}-1)\\ =&\underline{\mathrm{SHI}}(-Y(K),-\gamma_{(x,y)},(-S)^+,i_{\max}-1)\\ =&\underline{\mathrm{SHI}}(-Y(K),-\gamma_{(x,y)},(-S)^-,i_{\max})\end{aligned}$$

Note the last equality follows from Theorem 2.26. Now, from Lemma 2.24 and Theorem 2.20, we know that

$$\underline{\mathrm{SHI}}(-Y(K),-\gamma_{(x,y)},(-S)^-,i_{\max}) \cong \underline{\mathrm{SHI}}(-M',-\gamma') \neq 0,$$

where (M',γ') is the taut balanced sutured manifold obtained from $(Y(K),\gamma_{(x,y)})$ by decomposing along $-S$. Hence we conclude that

$$i_{min} = 1 - i_{max} = 1 - \frac{y}{2} - g(S)$$

and we obtain the desired result. □

Definition 4.15. For any integer $y \in \mathbb{N}$, define

$$i^y_{max} = \lceil \frac{y-1}{2} \rceil + g(S), \text{ and } i^y_{min} = \lceil -\frac{y-1}{2} \rceil + g(S).$$

For the suture $\widehat{\Gamma}_n = \gamma_{(p_n,q_n)}$, define

$$\hat{i}^n_{max} = i^{q_n}_{max}, \text{ and } \hat{i}^n_{min} = i^{q_n}_{min}.$$

Convention. Note we use the similar notations as in Subsection 3.2, while from now on, we use the new definitions of i^y_{max} and i^y_{min} as in Definition 4.15. We will use $i^*_{max}, \hat{i}^*_{max}$ and $i^*_{min}, \hat{i}^*_{min}$ to denote the maximal and minimal gradings for the slope specified by $*$.

In [Li19, Section 5], a graded version of the bypass exact triangles in Proposition 4.8 is proved, which is similar to Lemma 3.18.

Proposition 4.16 ([Li19, Proposition 5.5])**.** *Suppose $K \subset Y$ is a null-homologous knot and the pair $(\widehat{Y},\widehat{K})$ is obtained from (Y,K) by a q/p surgery. Suppose further that the sutures $\widehat{\Gamma}_n$ and $\widehat{\Gamma}_\mu$ are defined as in Definition 4.4 and S is a minimal genus Seifert surface of K. Then the followings hold.*

(1) For $n \in \mathbb{Z}$ so that $q_{n+1} = q_n + q$, i.e., $n \geqslant 0$, there are two bypass exact triangles:

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_n,S^\tau)[\hat{i}^{n+1}_{min}-\hat{i}^n_{min}] \arrow[r, "\psi^n_{+,n+1}"] & \underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_{n+1},S^\tau) \arrow[dl, "\psi^{n+1}_{+,\mu}"] \\
\underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_\mu,S^\tau)[\hat{i}^{n+1}_{max}-\hat{i}^\mu_{max}] \arrow[u, "\psi^\mu_{+,n}"] &
\end{tikzcd}$$

and

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_n,S^\tau)[\hat{i}^{n+1}_{max}-\hat{i}^n_{max}] \arrow[r, "\psi^n_{-,n+1}"] & \underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_{n+1},S^\tau) \arrow[dl, "\psi^{n+1}_{-,\mu}"] \\
\underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_\mu,S^\tau)[\hat{i}^{n+1}_{min}-\hat{i}^\mu_{min}] \arrow[u, "\psi^\mu_{-,n}"] &
\end{tikzcd}$$

(2) For $n \in \mathbb{Z}$ so that $q_{n+1} = q_n - q$, i.e., $n < -1$, there are two bypass exact triangles:

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau) \arrow[r, "\psi^n_{+,n+1}"] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau)[\hat{i}^n_{max} - \hat{i}^{n+1}_{max}] \arrow[dl, "\psi^{n+1}_{+,\mu}"] \\
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_\mu, S^\tau)[\hat{i}^n_{min} - \hat{i}^\mu_{min}] \arrow[u, "\psi^\mu_{+,n}"] &
\end{tikzcd}$$

and

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau) \arrow[r, "\psi^n_{-,n+1}"] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau)[\hat{i}^n_{min} - \hat{i}^{n+1}_{min}] \arrow[dl, "\psi^{n+1}_{-,\mu}"] \\
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_\mu, S^\tau)[\hat{i}^n_{max} - \hat{i}^\mu_{max}] \arrow[u, "\psi^\mu_{-,n}"] &
\end{tikzcd} \tag{4.3}$$

(3) For $n \in \mathbb{Z}$ so that $q_{n+1} + q_n = q$, i.e., $n = -1$, there are two bypass exact triangles:

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau)[\hat{i}^\mu_{min} - \hat{i}^n_{min}] \arrow[r, "\psi^n_{+,n+1}"] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau)[\hat{i}^\mu_{max} - \hat{i}^{n+1}_{max}] \arrow[dl, "\psi^{n+1}_{+,\mu}"] \\
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_\mu, S^\tau) \arrow[u, "\psi^\mu_{+,n}"] &
\end{tikzcd} \tag{4.4}$$

and

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau)[\hat{i}^\mu_{max} - \hat{i}^n_{max}] \arrow[r, "\psi^n_{-,n+1}"] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau)[\hat{i}^\mu_{min} - \hat{i}^{n+1}_{min}] \arrow[dl, "\psi^{n+1}_{-,\mu}"] \\
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_\mu, S^\tau) \arrow[u, "\psi^\mu_{-,n}"] &
\end{tikzcd} \tag{4.5}$$

Furthermore, all maps involved in the above bypass exact triangles are grading preserving.

Remark 4.17. The above proposition can be understood by Remark 3.19. Alternatively, we can understand the above proposition by the following method, which is inspired by the curve invariant introduced by Henselman, Rasmussen and Waston [HRW16, HRW18].

(1) Consider the lattice $\mathbb{Z}^2 \subset \mathbb{R}^2$. A surgery slope $y/x \in \mathbb{Q} \cup \{\infty\}$ corresponds to a straight arc connecting two lattice points in $\mathbb{R}^2 \backslash \mathbb{Z}^2$.
(2) Suppose the sutures $\gamma_{(x_1,y_1)}$, $\gamma_{(x_2,y_2)}$, and $\gamma_{(x_3,y_3)}$ defined as in Definition 4.6. Then it is easy to see the arcs corresponding to these three sutures bound a triangle containing no lattice point in the interior. There are two different triangles up to translation, which correspond to two different bypass triangles. All bypass maps are clockwise in $\mathbb{R}^2$. Rotation around the origin by 180 degrees will switch the roles of $\psi^*_{+,*}$ and $\psi^*_{-,*}$.
(3) The height of the middle point of the straight arc indicates the grading before stabilization (so there are gradings of half integers). If top endpoints of two arcs are the same, the grading shift is about $\hat{i}^*_{min}$. If bottom endpoints of two arcs are the same, the grading shift is about $\hat{i}^*_{max}$.

4.3. **Decomposing framed instanton Floer homology.**

In this subsection, we prove term (2) of Proposition 1.17. Throughout this subsection, let $K \subset Y$ be a null-homologous knot and let the pair $(\widehat{Y}, \widehat{K})$ be obtained from (Y, K) by a q/p surgery with $q > 0$. We have the sutures $\widehat{\Gamma}_n = \gamma_{(p_n, q_n)}$ and $\widehat{\Gamma}_\mu$ defined as in Definition 4.4 and S being a minimal genus Seifert surface of K. The stabilization S^τ of S is chosen according to Definition 4.12. The maximal and minimal gradings of the involved sutured instanton Floer homology are described in Lemma 4.14. For any $i \in \mathbb{Z}$, let

$$\psi^{n,i}_{\pm,n+1} = \psi^{n}_{\pm,n+1}|_{\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, i)}$$

be the restriction.

Lemma 4.18. *Suppose $n \in \mathbb{Z}$ so that $q_{n+1} = q_n + q$. Then the map*

$$\psi^{n,i}_{+,n+1} : \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, i) \to \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau, i - \hat{i}^n_{min} + \hat{i}^{n+1}_{min})$$

is an isomorphism if $i \leqslant \hat{i}^n_{max} - 2g(S)$. Similarly, the map

$$\psi^{n,i}_{-,n+1} : \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, i) \to \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau, i - \hat{i}^n_{max} + \hat{i}^{n+1}_{max})$$

is an isomorphism if $i \geqslant \hat{i}^n_{min} + 2g(S)$.

Proof. The proof the lemma is similar to that of Lemma 3.20. □

Lemma 4.19. *For any $n \in \mathbb{Z}$ so that $q_{n+1} - q = q_n \geqslant 2g$, the map*

$$G_n : \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta) \to \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n)$$

defined as in (4.2) is the zero map.

Proof. The proof of the lemma is similar to that of Lemma 3.22. □

Corollary 4.20. *We have*

$$\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_\mu) \geqslant \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta).$$

Proof. Since for a large enough integer n, we have $G_n = 0$, we know that

$$\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta) = \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}) - \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n).$$

Then the corollary follows directly from Proposition 4.10. □

Corollary 4.21. *Suppose L_1 is a non-empty link in $\widehat{Y}$ that is disjoint from $\widehat{K}$. Let $L_2 = L_1 \cup \widehat{K}$. Consider the link complements $-\widehat{Y}(L_1)$ and $-\widehat{Y}(L_2)$. For the link complement, let $\widehat{\Gamma}_\mu$ be the suture consisting of two meridians for each component of the link. We have*

$$\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-\widehat{Y}(L_2), -\widehat{\Gamma}_\mu) \geqslant 2 \cdot \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-\widehat{Y}(L_1), -\widehat{\Gamma}_\mu)$$

Proof. The same argument to prove Corollary 4.20 can be applied verbatim to verify the following inequality:

$$\dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-\widehat{Y}(L_2), -\widehat{\Gamma}_\mu) \geqslant \underline{\mathrm{SHI}}(-\widehat{Y}(L_1)(1), -\widehat{\Gamma}_\mu \cup -\delta),$$

where $\widehat{Y}(L_1)(1)$ is obtained from $-\widehat{Y}(L_1)$ by removing a 3-ball disjoint from L_1, and δ is a simple closed curve on the new spherical boundary component. Then it follows from [BS16b, Lemma 4.14] that

$$\underline{\mathrm{SHI}}(-\widehat{Y}(L_1)(1), -\widehat{\Gamma}_\mu \cup -\delta) = 2 \cdot \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-\widehat{Y}(L_1), -\widehat{\Gamma}_\mu).$$

□

Lemma 4.22. *Suppose $n \in \mathbb{Z}$ satisfies $q_{n+1} - q = q_n \geqslant q + 2g(S)$, and suppose $i, j \in \mathbb{Z}$ with*

$$\hat{i}^n_{min} + 2g(S) \leqslant i, j \leqslant \hat{i}^n_{max} - 2g(S), \text{ and } i - j = q.$$

Then we have

$$\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, i) \cong \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, j).$$

Proof. Since $i \geqslant \hat{i}^n_{min} + 2g(S)$, by Lemma 4.18, we know

$$\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, i) \cong \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau, i - \hat{i}^n_{min} + \hat{i}^{n+1}_{min}).$$

Similarly, since $j \leqslant \hat{i}^n_{max} - 2g(S)$, we know that

$$\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, j) \cong \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau, j - \hat{i}^n_{max} + \hat{i}^{n+1}_{max}).$$

Note also that

$$\begin{aligned} i - \hat{i}^n_{min} + \hat{i}^{n+1}_{min} =& j - \hat{i}^n_{max} + \hat{i}^{n+1}_{max} + q + (\hat{i}^n_{max} - \hat{i}^n_{min}) - (\hat{i}^{n+1}_{max} - \hat{i}^{n+1}_{min}) \\ =& j - \hat{i}^n_{max} + \hat{i}^{n+1}_{max} + q + (q_n - 1 + 2g(S)) - (q_{n+1} - 1 + 2g(S)) \\ =& j - \hat{i}^n_{max} + \hat{i}^{n+1}_{max}. \end{aligned}$$

Hence we obtain the desired result. □

Definition 4.23. Suppose $n \in \mathbb{Z}$ satisfies $q_{n+1} - q = q_n \geqslant q + 2g(S)$. Define

$$\mathcal{I}_+(-\widehat{Y}, \widehat{K}, i) = \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, \hat{i}^n_{max} - 2g - i),$$

and

$$\mathcal{I}_+(-\widehat{Y}, \widehat{K}) = \bigoplus_{i=0}^{q-1} \mathcal{I}_+(-\widehat{Y}, \widehat{K}, i).$$

Remark 4.24. From Lemma 4.18, the definition of $\mathcal{I}_+(-\widehat{Y}, \widehat{K})$ is independent of the choice of the integer n satisfying the required condition. Also, by Lemma 4.22, the definition of $\mathcal{I}_+(-\widehat{Y}, \widehat{K})$ would be the same (up to a $\mathbb{Z}_q$ grading shift) if we consider arbitrary q many consecutive gradings within the range $[\hat{i}^n_{min} + 2g(S), \hat{i}^n_{max} - 2g(S)]$.

Next, our goal is to show that there is an isomorphism

$$\mathcal{I}_+(-\widehat{Y}, \widehat{K}) \cong \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta).$$

To do so, we first introduce some notations for performing computations.

Definition 4.25. Suppose $n \in \mathbb{Z}$. The direct sum of some consecutive gradings of

$$\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau)$$

is called a **block**. For a block A, the number of gradings involved is called the **size** of A.

Example 4.26. Suppose $n \in \mathbb{Z}$ satisfies $q_n \geqslant q + 2g(S)$. Let A, B, C and D be the blocks consisting of the top $2g(S)$ gradings, the next q gradings, the next $q_n - q - 2g(S)$ gradings, and the last $2g(S)$ gradings of $\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau)$, respectively. We write

$$\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau) = \begin{pmatrix} A \\ B \\ C \\ D \end{pmatrix}.$$

Note from Definition 4.23, we know that $\mathcal{I}_+(-\widehat{Y},\widehat{K})$ is itself a block and in fact

$$\mathcal{I}_+(-\widehat{Y},\widehat{K}) = B.$$

Also, we can write

$$\underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_n,S^\tau) = \begin{pmatrix} A \\ E \\ F \\ D \end{pmatrix},$$

where E and F are of size $q_n - q - 2g(S)$ and q, respectively. By comparing the gradings, we have

$$\begin{pmatrix} B \\ C \end{pmatrix} = \begin{pmatrix} E \\ F \end{pmatrix}$$

Note we do not have $B = E$ and $C = F$ since they have different sizes. However, when putting together, the total size of B and C equals that of E and F.

Lemma 4.27. *Let $\mathcal{I}_+(-\widehat{Y},\widehat{K})$ be defined as in Definition 4.23. Then we know that*

$$\dim_{\mathbb{C}} \mathcal{I}_+(-\widehat{Y},\widehat{K}) = \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-\widehat{Y}(1),-\delta).$$

Proof. Suppose $n \in \mathbb{Z}$ satisfies $q_n \geqslant q + 2g(S)$. We can apply Proposition 4.16. Using blocks, we have the following. (There is no enough room for writing down the whole notation for the graded sutured instanton Floer homology, so we will only write down the sutures to denote them.)

size	$-\widehat{\Gamma}_\mu$	$\xrightarrow{\psi^{\mu}_{+,n}}$	$-\widehat{\Gamma}_n$	$\xrightarrow{\psi^{n}_{+,n+1}}$	$-\widehat{\Gamma}_{n+1}$	$\xrightarrow{\psi^{n+1}_{+,\mu}}$	$-\widehat{\Gamma}_\mu$
q	G				X_1		G
$2g(S)$	H		A		X_2		H
$q_n - q - 2g(S)$			E		X_3		
q			F		X_4		
$2g(S)$			D		X_5		

From the exactness, we know that

$$X_1 = G,\ X_3 = E,\ X_4 = F,\ \text{and } X_5 = D.$$

There is another bypass exact triangle, and similarly we have

size	$-\widehat{\Gamma}_\mu$	$\xrightarrow{\psi^{\mu}_{-,n}}$	$-\widehat{\Gamma}_n$	$\xrightarrow{\psi^{n}_{-,n+1}}$	$-\widehat{\Gamma}_{n+1}$	$\xrightarrow{\psi^{n+1}_{-,\mu}}$	$-\widehat{\Gamma}_\mu$.
$2g(S)$			A		A		
q			B		B		
$q_n - q - 2g(S)$			C		C		
$2g(S)$	I		D		X_6		I
q	J				J		J

Comparing the two expressions of $\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau)$, we have

$$\begin{pmatrix} G \\ X_2 \\ E \\ F \\ D \end{pmatrix} = \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau) = \begin{pmatrix} A \\ B \\ C \\ X_6 \\ J \end{pmatrix}.$$

Taking sizes into consideration, we know that

$$\begin{pmatrix} G \\ X_2 \end{pmatrix} = \begin{pmatrix} A \\ B \end{pmatrix}, \ E = C, \text{ and } \begin{pmatrix} F \\ D \end{pmatrix} = \begin{pmatrix} X_6 \\ J \end{pmatrix}.$$

Thus, we know that

$$\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau) = \begin{pmatrix} A \\ B \\ E \\ F \\ D \end{pmatrix}.$$

Compare this expression with the expression of $\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau)$ in Example 4.26, we have

$$\begin{aligned} \dim_{\mathbb{C}} \mathcal{I}_+(-\widehat{Y}, \widehat{K}) &= \dim_{\mathbb{C}} B \\ &= \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau) - \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau) \\ &= \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta). \end{aligned}$$

Note the last equality follows from Lemma 3.22. Hence we obtain the desired result. □

Proposition 4.28. *Suppose $n \in \mathbb{Z}$ satisfies $q_{n+1} - q = q_n \geqslant q + 2g(S)$. Then the map F_n restricted to $\mathcal{I}_+(-\widehat{Y}, \widehat{K})$ is an isomorphism.*

$$F_n|_{\mathcal{I}_+(-\widehat{Y}, \widehat{K})} : \mathcal{I}_+(-\widehat{Y}, \widehat{K}) \xrightarrow{\cong} \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta)$$

Proof. It suffices to show that the restriction

$$F_n|_{\mathcal{I}_+(-\widehat{Y}, \widehat{K})}$$

is surjective. Since $q_n \geqslant q + 2g(S)$, we have $q_{n-1} = q_n - q \geqslant 2g(S)$. By Lemma 4.19, we know that $G_{n-1} = 0$, so F_n is surjective. Then it suffices to show that F_n remains surjective when restricted to $\mathcal{I}_+(-\widehat{Y}, \widehat{K})$. For any $x \in \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta)$, let $y \in \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n)$ be an element so that $F_n(y) = x$. Suppose

$$y = \sum_{j \in \mathbb{Z}} y_j, \text{ where } y_j \in \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, j).$$

For any y_j, we want to find $y'_j \in \mathcal{I}_+(-\widehat{Y}, \widehat{K})$ so that $F_n(y_j) = F_n(y'_j)$.

To do this, we first assume that $j > \hat{i}^n_{max} - 2g(S)$. Then there exists an integer m so that

$$\hat{i}^n_{max} - 2g(S) - q + 1 \leqslant j - mq \leqslant \hat{i}^n_{max} - 2g(S).$$

We can take

$$y'_j = (\psi^{n, j-mq}_{-,n+1})^{-1} \circ \cdots \circ (\psi^{n, \hat{i}^{n+m}_{max} - \hat{i}^n_{max} + j - mq}_{-,n+m})^{-1} \circ \psi^{n+m-1}_{+,n+m} \circ \ldots \circ \psi^n_{+,n+1}(y_j). \tag{4.6}$$

From Lemma 4.18, we conclude that all the negative bypass maps involved in (4.6) are all isomorphisms and

$$y'_j \in \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, j - mq) \in \mathcal{I}_+(-\widehat{Y}, \widehat{K}).$$

Finally, from Lemma 4.11, we know that $F_n(y'_j) = F_n(y_j)$. For

$$j \in [\hat{i}^n_{max} - 2g(S) - q - 1, \hat{i}^n_{max} - 2g(S)],$$

we can simply take $y'_j = y_j$. For $j \leqslant \hat{i}^n_{max} - 2g(S) - q - 1$, we can pick y'_j similarly, while switching the roles of $\psi^*_{+,*}$ and $\psi^*_{-,*}$ in (4.6). Hence we can take

$$y' = \sum_{j \in \mathbb{Z}} y'_j \in \mathcal{I}_+(-\widehat{Y}, \widehat{K})$$

and $F_n(y') = F_n(y) = x$. Hence F_n is surjective, and we obtain the desired result. □

In Definition 4.23, we use a large enough integer n to define $\mathcal{I}_+(-\widehat{Y}, \widehat{K})$. We can also use a small enough integer n to define a vector space similar to $\mathcal{I}_+(-\widehat{Y}, \widehat{K})$. Recall

$$\widehat{\Gamma}_n = \gamma_{(p_n, q_n)}$$

is defined as in Definition 4.4 and q_n is chosen to be always non-negative.

Definition 4.29. Suppose $n \in \mathbb{Z}$ satisfies $q_{n-1} - q = q_n \geqslant q + 2g(S)$. Define

$$\mathcal{I}_-(-\widehat{Y}, \widehat{K}, i) = \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, \hat{i}^n_{max} - 2g - i),$$

and

$$\mathcal{I}_-(-\widehat{Y}, \widehat{K}) = \bigoplus_{i=0}^{q-1} \mathcal{I}_-(-\widehat{Y}, \widehat{K}, i).$$

The arguments for $\mathcal{I}_-(-\widehat{Y}, \widehat{K})$ are similar to those for $\mathcal{I}_+(-\widehat{Y}, \widehat{K})$. We sketch them as follows.

Lemma 4.30. *Suppose $n \in \mathbb{Z}$ satisfies $q_{n-1} - q = q_n$. Then the map*

$$\psi^{n-1, i + \hat{i}^{n-1}_{max} - \hat{i}^n_{max}}_{+,n} : \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n-1}, S^\tau, i + \hat{i}^{n-1}_{max} - \hat{i}^n_{max}) \to \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, i)$$

is an isomorphism if $i \leqslant \hat{i}^n_{max} - 2g(S)$. Similarly, the map

$$\psi^{n-1, i - \hat{i}^n_{min} + \hat{i}^{n-1}_{min}}_{-,n} : \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n-1}, S^\tau, i - \hat{i}^n_{min} + \hat{i}^{n-1}_{min}) \to \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, i)$$

is an isomorphism if $i \geqslant \hat{i}^n_{min} + 2g(S)$.

Proof. The proof is similar to the proof of Lemma 3.20. □

Lemma 4.31. *For any $n \in \mathbb{Z}$ so that $q_{n-1} - q = q_n \geqslant 2g$, the map*

$$F_n : \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n) \to \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta)$$

defined as in (4.2) is the zero map.

Proof. If it is not, then let $j_{\max} \in \mathbb{Z}$ be the maximal index j so that there exists

$$x \in \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, j)$$

with $F_n(x) \neq 0$. Since $q_n \geqslant 2g$, by Lemma 4.14, we know that either

$$j_{max} \leqslant \hat{i}^n_{max} - 2g(S) \text{ or } j_{max} \geqslant \hat{i}^n_{min} + 2g(S).$$

Suppose, without loss of generality, that $j_{max} \geqslant \hat{i}^n_{min} + 2g(S)$ and

$$x \in \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, j_{max})$$

satisfying $F_n(x) \neq 0$. By Lemma 4.30, $\psi^{n-1, j_{max}+\hat{i}^{n-1}_{max}-\hat{i}^n_{max}}_{+,n}$ is an isomorphism, and we can take

$$y = \psi^{n-1, j_{max}+\hat{i}^{n-1}_{max}-\hat{i}^n_{max}}_{-,n} \circ (\psi^{n-1, j_{max}+\hat{i}^{n-1}_{max}-\hat{i}^n_{max}}_{+,n})^{-1}(x).$$

By Lemma 4.11 we know that

$$F_n(y) = F_n(x) \neq 0 \text{ and } y \in \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, j_{max} + q).$$

This is a contradiction. □

Lemma 4.32. *Suppose $n \in \mathbb{Z}$ satisfies $q_{n-1} - q = q_n \geqslant q + 2g(S)$, and suppose $i, j \in \mathbb{Z}$ satifying*

$$\hat{i}^n_{min} + 2g(S) \leqslant i, j \leqslant \hat{i}^n_{max} - 2g(S), \text{ and } i - j = q.$$

Then we have

$$\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, i) \cong \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, j).$$

Proof. The proof is similar to the proof of Lemma 4.22. □

Lemma 4.33. *Let $\mathcal{I}_-(-\widehat{Y}, \widehat{K})$ be defined as in Definition 4.29. Then we know that*

$$\dim_{\mathbb{C}} \mathcal{I}_-(-\widehat{Y}, \widehat{K}) = \dim_{\mathbb{C}} \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta).$$

Proof. The proof is similar to the proof of Lemma 4.27. □

Proposition 4.34. *Suppose $n \in \mathbb{Z}$ satisfies $q_{n-1} - q = q_n \geqslant q + 2g(S)$. Let Π_n be the projection*

$$\Pi_n : \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n) \to \mathcal{I}_-(-\widehat{Y}, \widehat{K}).$$

Then we have an isomorphism

$$\Pi_n \circ G_n : \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta) \xrightarrow{\cong} \mathcal{I}_-(-\widehat{Y}, \widehat{K}).$$

Proof. It suffices to show that $\Pi_n \circ G_n$ is injective. Suppose not, *i.e.*, there exists

$$x \neq 0 \in \underline{\mathrm{SHI}}(-\widehat{Y}(1), -\delta) \text{ with } \Pi_n \circ G_n(x) = 0.$$

Write

$$y = G_n(x) = \sum_{j \in \mathbb{Z}} y_j, \text{ where } y_j \in \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau, j).$$

From Lemma 4.11 and Lemma 4.31, we know that G_n is injective, and hence $y \neq 0$. From the assumption, we know that

$$y_j = 0 \text{ for } \hat{i}^n_{max} - 2g(S) \geqslant j \geqslant \hat{i}^n_{max} - 2g(S) - q + 1.$$

Also write

$$z = G_{n-1}(x) = \sum_{j \in \mathbb{Z}} z_j, \text{ where } z_j \in \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n-1}, S^\tau, j).$$

From Lemma 4.11, we know that

$$\psi^{n-1}_{-,n}(z) = y = \psi^{n-1}_{+,n}(z).$$

Suppose j_{min} is the minimal grading j so that

$$j > \hat{i}^n_{max} - 2g(S) \text{ and } y_j \neq 0.$$

Then we know that

$$y_{j_{min}-q} = 0 \text{ and } j_{min} - q \geqslant \hat{i}^n_{min} + 2g(s).$$

Hence by Lemma 4.30, we know that

$$\begin{aligned} y_{j_{min}} =& \psi^{n-1, j_{min}-\hat{i}^n_{min}+\hat{i}^{n-1}_{min}}_{-,n}(z_{j_{min}-\hat{i}^n_{min}+\hat{i}^{n-1}_{min}}) \\ =& \psi^{n-1, j_{min}-\hat{i}^n_{min}+\hat{i}^{n-1}_{min}}_{-,n} \circ (\psi^{n-1, j_{min}-\hat{i}^n_{min}+\hat{i}^{n-1}_{min}}_{+,n})^{-1}(y_{j_{min}-q}) \\ =& 0. \end{aligned}$$

This implies that $y_j = 0$ for all $j \geqslant \hat{i}^n_{max} - 2g(S) - q + 1$. Similarly we can prove that $y_j = 0$ for all $j < \hat{i}^n_{max} - 2g(S) - q + 1$, and $y = 0$, which contradicts to the injectivity of G_n. □

4.4. **Commutative diagrams for bypass maps.**

In this subsection, we show there are some commutative diagrams for bypass maps. With these commutative diagrams, we can prove $\mathcal{I}_+(-\widehat{Y}, \widehat{K})$ and $\mathcal{I}_-(-\widehat{Y}, \widehat{K})$ are isomorphic with respect to gradings and then finish the proof of Proposition 1.17.

Lemma 4.35 ([Li19, Corollary 2.20])**.** *For any surgery slope q/p, consider the bypass maps $\psi^*_{+,*}$ and $\psi^*_{-,*}$ in Proposition 4.10. For any integer $n \in \mathbb{Z}$, we have the following commutative diagram.*

$$\begin{tikzcd} \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n) \arrow[r, "\psi^n_{-,n+1}"] \arrow[d, "\psi^n_{+,n+1}"'] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}) \arrow[d, "\psi^{n+1}_{+,n+2}"] \\ \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}) \arrow[r, "\psi^{n+1}_{-,n+2}"] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+2}) \end{tikzcd} \tag{4.7}$$

Proof. In Subsection 2.3, we re-interpreted bypass maps by contact gluing maps. So the composition of bypass maps becomes the composition of contact gluing maps. To verify the commutative diagram, it suffices to verify that two components of contact structures coming from different bypasses are actually the same. Thus, it is free to change the basis of $H_1(T^2)$. It suffices to verify a special case $q/p = 1/0$ and $n = 0$. Then it follows from Honda [Hon00, Lemma 4.14] that the contact structures are the same.

Alternatively, to show the propoistion holds for $q/p = 1/0$ and $n = 0$, we can use the interpretation of bypass maps by contact handle decomposition. Explicitly, consider the proof of Proposition 4.10. We start with $\gamma = \gamma_{(2,-1)}$ and two bypass arcs α_+ and α_- as depicted in the left sub-figure of Figure 24. Let the suture γ' and the new bypass arc θ_+ be obtained by applying the bypass attachment twice along the arc α_+ (*c.f.* Figure 3). See the middle sub-figure of Figure 24. The suture γ' is a stabilization of $\gamma_{1,-1}$ and the bypass attachment along θ_+ gives the map $\psi^1_{+,2}$. The arc α_- becomes a bypass arc on γ'. Let the suture γ'' and the new bypass arc θ_- be obtained by applying the bypass attachment twice along the arc α_-. See the right sub-figure of Figure 24. The suture γ'' is a stabilization of $\gamma_{0,1}$ and the bypass attachment along θ_- gives the map $\psi^0_{-,1}$. Thus,

$$\psi^1_{+,2} \circ \psi^0_{-,1} = \psi_{\theta_+} \circ \psi_{\theta_-}.$$

By Lemma 2.33, these two bypass maps commute. Similar to the above discussion, we have

$$\psi^1_{-,2} \circ \psi^0_{+,1} = \psi_{\theta_-} \circ \psi_{\theta_+}.$$

Hence the lemma holds. □

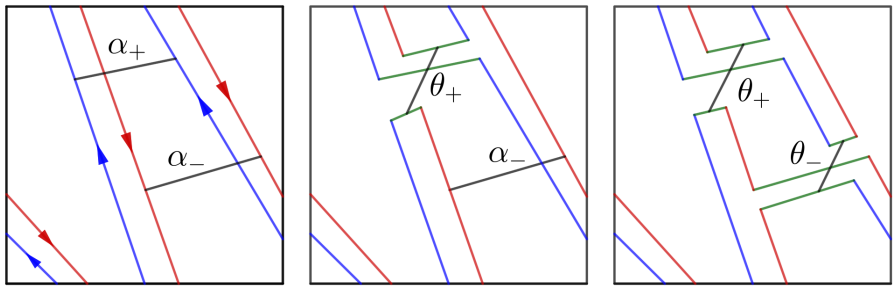


FIGURE 24. Bypass arcs on sutures.

Corollary 4.36. *For any surgery slope q/p, consider the bypass maps $\psi^*_{+,*}$ and $\psi^*_{-,*}$ in Proposition 4.10. For $i, j \in \mathbb{Z}$ with $i > j$, suppose*

$$\Psi^j_{\pm,i} = \psi^{i-1}_{\pm,i} \circ \psi^{i-2}_{\pm,i-1} \circ \cdots \circ \psi^j_{\pm,j+1}.$$

Then we have the following commutative diagram.

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_j) \arrow[r, "\psi^j_{+,j+1}"] \arrow[d, "\Psi^j_{-,i}"'] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{j+1}) \arrow[d, "\Psi^{j+1}_{-,i+1}"] \\
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_i) \arrow[r, "\psi^i_{+,i+1}"] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{i+1})
\end{tikzcd} \tag{4.8}$$

The similar commutative diagrams hold if we switch the roles of $\psi^*_{+,*}$ and $\psi^*_{-,*}$, $\Psi^*_{+,*}$ and $\Psi^*_{-,*}$, respectively.

Proof. Combining the commutative diagrams from Lemma 4.35 with $n = j, j+1, \ldots, i-2$, this lemma is straightforward. □

Lemma 4.37. *For any surgery slope q/p, consider the bypass maps $\psi^*_{+,*}$ and $\psi^*_{-,*}$ in Proposition 4.10. For any $n \in \mathbb{Z}$, we have two commutative diagrams*

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n) \arrow[rr, "\psi^n_{-,n+1}"] \arrow[dr, "\psi^n_{+,\mu}"'] & & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}) \arrow[dl, "\psi^{n+1}_{+,\mu}"] \\
& \underline{\mathrm{SHI}}(-\widehat{Y}(K), -\widehat{\Gamma}_\mu) &
\end{tikzcd} \tag{4.9}$$

and

$$\begin{tikzcd} \underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_n) \arrow[rr, "\psi^n_{-,n+1}"] & & \underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_{n+1}) \\ & \underline{\mathrm{SHI}}(-\widehat{Y}(K),-\widehat{\Gamma}_\mu) \arrow[ul, "\psi^\mu_{+,n}"] \arrow[ur, "\psi^\mu_{+,n+1}"'] & \end{tikzcd} \tag{4.10}$$

The similar commutative diagrams hold if we switch the roles of $\psi^*_{+,*}$ and $\psi^*_{-,*}$.

Remark 4.38. The bypass maps in Lemma 4.37 are from different bypass exact triangles.

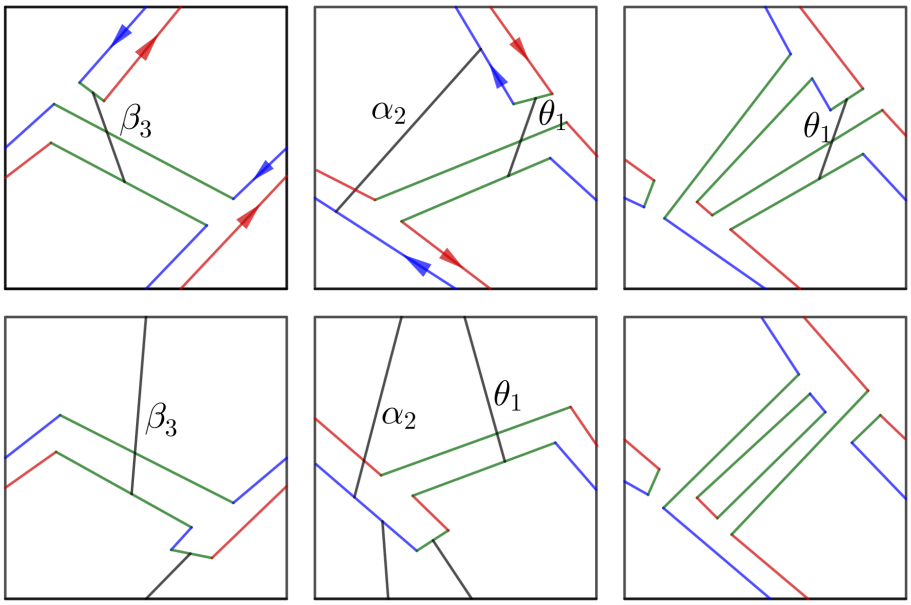


FIGURE 25. Top left, bypass β_3 from $\gamma_{(1,1)}$; top middle, bypasses α_2 and θ_1 from $\gamma_{(1,-1)}$; top right, after the bypass attachment along α_2; bottom left, top left subfigure after isotopy; bottom middle, top middle figure after isotopy; bottom right, the suture γ.

Proof of Lemma 4.37. We only give the proof of (4.9). The proof for (4.10) and other two commutative diagrams are similar. We use the same strategy as in the proof of Lemma 4.37. Hence it suffices to prove for $q/p = 1/0$ and $n = 0$. For $\psi^0_{+,\mu}$, we start with $\gamma_{(1,1)}$. For $\psi^0_{-,1} \circ \psi^1_{+,\mu}$, we start with $\gamma_{(1,-1)}$. See Figure 25, where

$$\psi_{\theta_1} = \psi^0_{-,1}, \psi_{\alpha_2} = \psi^1_{+,\mu}, \text{ and } \psi_{\beta_3} = \psi^0_{+,\mu}.$$

By Lemma 2.33, we have

$$\psi_{\theta_1} \circ \psi_{\alpha_2} = \psi_{\alpha_2} \circ \psi_{\theta_1}.$$

The bypass arcs α_2 and β_3 are isotopic. By Lemma 2.34, we have

$$\psi_{\alpha_2} = \psi_{\beta_3}.$$

After the bypass attachment along α_2, the bypass θ_1 is a trivial bypass. From the discussion in the proof of Lemma 3.21, the corresponding bypass map is an isomorphism. Explicitly, the third suture γ in the bypass triangle associated to θ_1 contains components that bounds disks. Hence the sutured manifold $(-Y(K), -\gamma)$ is not taut. By Theorem 2.13, $\underline{\mathrm{SHI}}(-Y(K), -\gamma)$ is trivial and the bypass triangle implies the bypass map ψ_{θ_1} is an isomorphism.

Thus, we conclude

$$\psi^0_{-,1} \circ \psi^1_{+,\mu} = \psi^0_{+,\mu}.$$

□

Remark 4.39. Indeed, Lemma 4.37 also follows from Honda's classification of tight contact structures on $T^2 \times I$ [Hon00].

Corollary 4.40. *For any surgery slope q/p, consider the bypass maps $\psi^*_{+,*}$ and $\psi^*_{-,*}$ in Proposition 4.10. For any $i, j \in \mathbb{Z}$, we have the following commutative diagrams related to $\psi^*_{+,*}$ and $\psi^*_{-,*}$, respectively.*

$$\begin{tikzcd} \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_\mu) \arrow[r, "\psi^\mu_{\pm,j}"] \arrow[d, "\psi^\mu_{\pm,i}"'] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_j) \arrow[d, "\psi^j_{\pm,\mu}"] \\ \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_i) \arrow[r, "\psi^i_{\pm,\mu}"] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_\mu) \end{tikzcd} \tag{4.11}$$

Proof. The commutative diagram related to $\psi^*_{+,*}$ follows from (4.9) and (4.9). Explicitly, for $i = j + 1$, both composition of maps are equal to

$$\psi^{j+1}_{+,\mu} \circ \psi^n_{-,j+1} \circ \psi^\mu_{+,j}.$$

The other commutative diagram follows from Lemma 4.37 similarly. □

Corollary 4.41. *For any surgery slope q/p, consider the bypass maps $\psi^*_{+,*}$ and $\psi^*_{-,*}$ in Proposition 4.10. For any $n \in \mathbb{Z}$, we have*

$$\psi^n_{+,\mu} \circ \psi^\mu_{-,n} = \psi^n_{-,\mu} \circ \psi^\mu_{+,n} = 0$$

and

$$\psi^\mu_{+,n} \circ \psi^n_{+,\mu} = \psi^\mu_{-,n} \circ \psi^n_{-,\mu} = 0.$$

Proof. By Lemma 4.37, we have

$$\psi^{n+1}_{+,\mu} \circ \psi^n_{-,n+1} \circ \psi^\mu_{-,n} = 0.$$

Other argument follow from Lemma 4.37 similarly. □

Corollary 4.42. *For any surgery slope q/p, consider the bypass maps $\psi^*_{+,*}$ and $\psi^*_{-,*}$ in Proposition 4.10. Consider $\Psi^j_{\pm,i}$ in Corollary 4.36. For any $i, j \in \mathbb{Z}$ with $i > j$, we have two commutative*

diagrams.

$$\begin{tikzcd} \underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_j) \arrow[rr, "\Psi^j_{-,i}"] \arrow[dr, "\psi^j_{+,\mu}"'] & & \underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_i) \arrow[dl, "\psi^i_{+,\mu}"] \\ & \underline{\mathrm{SHI}}(-\widehat{Y}(K),-\widehat{\Gamma}_\mu) & \end{tikzcd} \tag{4.12}$$

and

$$\begin{tikzcd} \underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_j) \arrow[rr, "\psi^j_{-,i}"] & & \underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_i) \\ & \underline{\mathrm{SHI}}(-\widehat{Y}(K),-\widehat{\Gamma}_\mu) \arrow[ul, "\psi^\mu_{+,j}"] \arrow[ur, "\psi^\mu_{+,i}"'] & \end{tikzcd} \tag{4.13}$$

The similar commutative diagrams hold if we switch the roles of $\psi^*_{+,*}$ and $\psi^*_{-,*}$, $\Psi^*_{+,*}$ and $\Psi^*_{-,*}$, respectively.

Proof. Combining the commutative diagrams from Lemma 4.35 with $n = j, j+1, \ldots, i-1$, this lemma is straightforward. □

Remark 4.43. The above commutative diagrams can be illustrated by the method described in Remark 4.17. The illustration of the special cases in the proofs are shown in Figure 26. Note that vector spaces are denoted by their sutures (we omit the minus signs) and all maps are bypass maps. They are grading preserving and commute with F_* and G_* by Proposition 4.16 and Lemma 4.11, respectively.

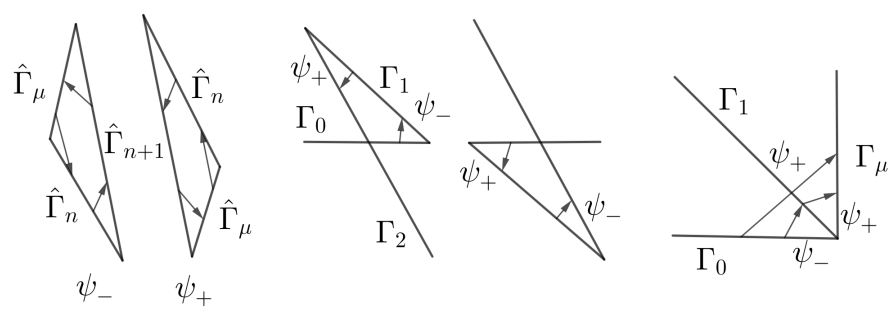


FIGURE 26. Left, bypass maps; middle, illustration of (4.7); right, illustration of (4.9).

With above commutative diagrams, the following proposition follows from diagram chasing.

Proposition 4.44. *For a surgery slope q/p with $q > 0$, consider $\mathcal{I}_+(-\widehat{Y},\widehat{K})$ and $\mathcal{I}_+(-\widehat{Y},\widehat{K})$ defined as in Subsection 4.3. We have a grading preserving isomorphism*

$$f_{K,\pm} : \mathcal{I}_+(-\widehat{Y},\widehat{K}) \xrightarrow{\cong} \mathcal{I}_-(-\widehat{Y},\widehat{K}),$$

up to a $\mathbb{Z}_q$ grading shift.

Proof. Suppose $n \in \mathbb{N}$ satisfies

$$q_n = -q_0 + nq > q + 2g(K), \text{ and } q_{-n} = q_0 + nq > q + 2g(K)$$

Note $q > -q_0 \geqslant 0$. In this case $\mathcal{I}_{\pm}(-\widehat{Y}, \widehat{K})$ are defined as direct summands of

$$\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{\pm n}),$$

respectively. Consider the following diagram

$$\begin{tikzcd}
A_n \arrow[dr, "\psi^{n}_{+,\mu}"] & & A_{n-1} \arrow[ll, "\psi^{n-1}_{+,n}"'] \arrow[dr, "\psi^{n-1}_{+,\mu}"] & & A_{n-2} \arrow[ll, "\psi^{n-2}_{+,n-1}"'] \arrow[dr, dashed, "\psi^{n-i+1}_{+,\mu}"] & & A_1 \arrow[ll, dashed, "\psi^{n-i}_{+,n-i+1}"'] \arrow[dr, "\psi^{1}_{+,\mu}"] & & A_0 \arrow[ll, "\psi^{0}_{+,1}"'] \\
& C_1 \arrow[ur, "\psi^{\mu}_{+,n-1}"] \arrow[dr, "\psi^{\mu}_{+,-1}"] \arrow[u, phantom, "\circlearrowleft"] & & C_2 \arrow[ur, "\psi^{\mu}_{+,n-2}"] \arrow[dr, "\psi^{\mu}_{+,-2}"] \arrow[u, phantom, "\circlearrowleft"] & & C_i \arrow[ur, dashed, "\psi^{\mu}_{+,n-i}"] \arrow[dr, dashed, "\psi^{\mu}_{+,-i}"] \arrow[u, phantom, "\circlearrowleft"] & & C_n \arrow[ur, "\psi^{\mu}_{+,0}"] \arrow[dr, "\psi^{\mu}_{+,-n}"] \arrow[u, phantom, "\circlearrowleft"] & \\
B_0 \arrow[ur, "\psi^{0}_{+,\mu}"] & & B_1 \arrow[ll, "\psi^{-1}_{+,0}"] \arrow[ur, "\psi^{-1}_{+,\mu}"] \arrow[uu, "\Psi^{-1}_{-,n-1}"] & & B_2 \arrow[ll, "\psi^{-2}_{+,-1}"] \arrow[ur, dashed, "\psi^{-i+1}_{+,\mu}"] \arrow[uu, "\Psi^{-2}_{-,n-2}"] & & B_{n-1} \arrow[ll, dashed, "\psi^{-i}_{+,-i+1}"] \arrow[ur, "\psi^{-n+1}_{+,\mu}"] \arrow[uu, "\Psi^{n-1}_{-,1}"] & & B_n \arrow[ll, "\psi^{-n}_{+,-n+1}"]
\end{tikzcd} \tag{4.14}$$

where

$$A_i = \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_i), B_i = \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{-i}), \text{ and } C_i = \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{\mu}).$$

The dashed arrows denote there are similar squares for C_i. By Corollary 4.42, the six triangles with circles are commutative. By Proposition 4.10, other eight triangles are bypass exact triangles. By Corollary 4.36, the squares involving $A_{n-i+1}, A_{n-i}, B_{i-1}$ and B_i are commutative for any integer i. By Proposition 4.16, all bypass maps are homogeneous, so this diagram can be decomposed into single gradings.

For $m \in \mathbb{Z}$, recall

$$\hat{i}^m_{max} = \lceil \frac{q_m - 1}{2} \rceil + g(S), \text{ and } \hat{i}^m_{min} = \lceil -\frac{q_m - 1}{2} \rceil - g(S).$$

For any $j \in \mathbb{Z}$, in diagram (4.14), we can replace A_i, B_i, C_i by

$$\begin{aligned}
A^j_i &= \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_i, S^\tau, \lceil \frac{1}{2}(-iq - q + q_0) \rceil + nq + j), \\
B^j_i &= \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{-i}, S^\tau, \lceil \frac{1}{2}(iq - q + q_0) \rceil + j), \\
\text{and } C^j_i &= \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{\mu}, S^\tau, (n-i)q + j), \text{ respectively.}
\end{aligned} \tag{4.15}$$

For a large enough integer n, there exist $j_0 \in \mathbb{Z}$ so that for any integer $j \in \{j_0, j_0+1, \dots, j_0+q-1\}$,

$$\lceil \frac{1}{2}(-q+q_0) \rceil + nq + j > \hat{i}^0_{max},$$
$$\lceil \frac{1}{2}(-q+q_0) \rceil + j < \hat{i}^0_{min},$$
$$\lceil \frac{1}{2}(-nq-q+q_0) \rceil + nq + j \in [\hat{i}^n_{min} + 2g(K), \hat{i}^n_{max} - 2g(K)],$$
$$\lceil \frac{1}{2}(nq-q+q_0) \rceil + j \in [\hat{i}^{-n}_{min} + 2g(K), \hat{i}^{-n}_{max} - 2g(K)].$$

Thus, we have

$$A^j_0 = B^j_0 = 0.$$

By Remark 4.24, there are grading preserving isomorphisms

$$f_A : \bigoplus_{j=j_0}^{j_0+q-1} A^j_n \xrightarrow{\cong} \mathcal{I}_+(-\hat{Y}, \hat{K})$$
$$f_B : \bigoplus_{j=j_0}^{j_0+q-1} B^j_n \xrightarrow{\cong} \mathcal{I}_-(-\hat{Y}, \hat{K})$$

up to $\mathbb{Z}_q$ grading shifts.

In the remain part of the proof, we find an isomorphism between A^j_n and B^j_n for $j \in \{j_0, j_0+1, \dots, j_0+q-1\}$ by a diagram chasing in diagram (4.14). For simplicity, we omit the superscript j. Note that all maps are linear maps between vector spaces over $\mathbb{C}$. Since $A_0 = B_0 = 0$, the maps $\psi^1_{+,\mu}$ and $\psi^\mu_{+,-1}$ are isomorphisms.

Step 1. We construct a map f from A_n to B_n.

For a nonzero element $a_n \in A_n$, consider $c_1 = \psi^n_{+,\mu}(a_n) \in C_1$. By exactness, either $c_1 \neq 0$, or there exists a nonzero element $a_{n-1} \in A_{n-1}$ so that $\psi^{n-1}_{+,n}(a_{n-1}) = a_n$.

If the first case happens, by the exact triangle, we know $\psi^\mu_{+,n-1}(c_1) = 0$. Since $\psi^\mu_{+,-1}$ is an isomorphism, $b_1 = \psi^\mu_{+,-1}(c_1) \in B_1$ is nonzero. From the commutative diagrams, we have

$$\Phi^{-1}_{-,n-1}(b_1) = \psi^\mu_{+,n-1}(c_1) = 0, \text{ and } \psi^{-1}_{+,\mu}(b_1) = \psi^{n-1}_{+,\mu} \circ \Phi^{-1}_{-,n-1}(b_1) = 0.$$

Hence there exists a nonzero element $b_2 \in B_2$ so that $\psi^{-2}_{+,-1}(b_2) = b_1$. Consider

$$a_{n-2} = \Psi^{-2}_{-,n-2}(b_2) \in A_{n-2}.$$

If $a_{n-2} = 0$, let $b'_2 = b_2$. If $a_{n-2} \neq 0$, then

$$\psi^{n-2}_{+,n-1}(a_{n-2}) = \Phi^{-1}_{-,n-1}(b_1) = \psi^\mu_{+,n-1}(c_1) = 0.$$

Then there exists a nonzero element $c_2 \in C_2$ so that $\psi^\mu_{+,n-2}(c_2) = a_{n-2}$. Let $b'_2 = b_2 - \psi^\mu_{+,-2}(c_2)$. In both cases for b'_2,

$$\psi^{-2}_{+,-1}(b'_2) = \psi^{-2}_{+,-1}(b_2) = b_1, \Psi^{-2}_{-,n-2}(b'_2) = 0,$$

and

$$\psi^{-2}_{+,\mu}(b'_2) = \psi^{n-2}_{+,\mu} \circ \Psi^{-2}_{-,n-2}(b'_2) = 0.$$

Hence there exists a nonzero element $b_3 \in B_3$ so that

$$\Psi^{-3}_{+,-1}(b_3) = \psi^{-2}_{+,-1} \circ \psi^{-3}_{+,-2}(b_3) = \psi^{-2}_{+,-1}(b'_2) = b_1.$$

By induction on i, there exists a nonzero element $b_n \in B_n$ so that $\Psi^{-n}_{+,-1}(b_n) = b_1$. Define $f(a_n) = b_n$.

If the second case happens, consider $c_2 = \psi^{n-1}_{+,\mu}(a_{n-1}) \in C_2$. If $c_2 = 0$, let $a'_{n-1} = a_{n-1}$. If $c_2 \neq 0$, consider $b_2 = \psi^{\mu}_{+,-2}(c_2) \in B_2$. If $b_2 \neq 0$, similar to the first case, there exists a nonzero element $b_n \in B_n$ so that $\Psi^{-n}_{+,-2}(b_n) = b_2$. Define $f(a_n) = b_n$. If $b_2 = 0$, then there exists a nonzero element $b_1 \in B_1$ so that $\psi^{-1}_{+,\mu}(b_1) = c_2$. Since $\psi^{\mu}_{+,-1}$ is an isomorphism, $\Psi^{-1}_{-,n-1}(b_1)$ is in the image of $\psi^{\mu}_{+,n-1}$ and then $\psi^{n-1}_{+,n} \circ \Psi^{-1}_{-,n-1}(b_1) = 0$. Let $a'_{n-1} = a_{n-1} - \Psi^{-1}_{-,n-1}(b_1)$. In both cases for a'_{n-1},

$$\psi^{n-1}_{+,n}(a'_{n-1}) = a_n, \text{ and } \psi^{n-1}_{+,\mu}(a'_{n-1}) = 0.$$

Hence there exists a nonzero element $a_{n-2} \in A_{n-2}$ so that $\psi^{n-2}_{+,n-1}(a_{n-2}) = a'_{n-1}$. Similarly, consider

$$c_3 = \psi^{n-2}_{+,\mu}(a_{n-2}) \in C_3 \text{ and } b_3 = \psi^{\mu}_{+,-3}(c_3).$$

If $c_3, b_3 \neq 0$, apply the argument in the first case. If $c_3 = 0$, let $a'_{n-2} = a_{n-2}$ and $a''_{n-1} = a'_{n-1}$. If $c_3 \neq 0, b_3 = 0$, there exists a nonzero element $b_2 \in B_2$ so that $\psi^{-2}_{+,\mu}(b_2) = c_3$. If

$$\psi^{n-2}_{+,n-1} \circ \Psi^{-2}_{-,n-2}(b_2) = \Psi^{-1}_{-,n-1} \circ \psi^{-2}_{+,-1}(b_2) \neq 0,$$

from the isomorphism $\psi^{\mu}_{+,-1}$, we know

$$\psi^{n-1}_{+,n} \circ \psi^{n-2}_{+,n-1} \circ \Psi^{-2}_{-,n-2}(b_2) = 0.$$

Let

$$a''_{n-1} = a'_{n-1} - \psi^{n-2}_{+,n-1} \circ \Psi^{-2}_{-,n-2}(b_2), \text{ and } a'_{n-2} = a_{n-2} - \psi^{n-2}_{+,n-1} \circ \Psi^{-2}_{-,n-2}(b_2).$$

In both cases for a'_{n-2} and a''_{n-1},

$$\psi^{n-1}_{+,n}(a''_{n-1}) = a_n, \psi^{n-2}_{+,n-1}(a'_{n-2}) = a''_{n-1}, \text{ and } \psi^{n-2}_{+,\mu}(a'_{n-2}) = 0.$$

Hence there exists a nonzero element $a_{n-3} \in A_{n-3}$ so that $\psi^{n-3}_{+,n-2}(a_{n-3}) = a'_{n-2}$. By induction on i, either there exists a nonzero element $a_1 \in A_1$ so that $\Psi^{1}_{+,n}(a_1) = a_n$, or there exist nonzero elements $b_n \in B_n, b_{k+1} \in B_{k+1}$, and $a_k \in A_{n-k}$ for some k, so that

$$\Psi^{-n}_{+,-k-1}(b_n) = b_{k+1} = \psi^{\mu}_{+,-k-1} \circ \psi^{k}_{+,\mu}(a_k), \text{ and } \Psi^{k}_{+,n}(a_k) = a_n.$$

Define $f(a_n) = \psi^{\mu}_{+,-n} \circ \psi^{1}_{+,\mu}(a_1)$ or $f(a_n) = b_n$ in either case, respectively.

Step 2. We construct a map g from B_n to A_n.

For a nonzero element $b_n \in B_n$, consider $b_{n-1} = \psi^{-n}_{+,-n+1}(b_n)$. By exactness, either $b_{n-1} \neq 0$, or there exists a nonzero element $c_n \in C_n$ so that $\psi^{\mu}_{+,-n}(c_n) = b_n$.

If the first case happens, From the commutative diagram, we have

$$\psi^{1}_{+,\mu} \circ \Psi^{n-1}_{-,1}(b_{n-1}) = \psi^{-n+1}_{+,\mu}(b_{n-1}) = 0$$

and then $\Psi^{n-1}_{-,1}(b_{n-1}) = 0$. Consider $b_1 = \Psi^{-n+1}_{+,-1}(b_{n-1})$. If $b_1 \neq 0$, then

$$\Psi^{-1}_{-,n-1}(b_1) = \Psi^{-1}_{-,n-1} \circ \Psi^{-n+1}_{+,-1}(b_{n-1}) = \Psi^{1}_{+,n-1}\Psi^{n-1}_{-,1}(b_{n-1}) = 0.$$

Consider $c_1 = (\psi^{\mu}_{+,-1})^{-1}(b_1)$. Then

$$\psi^{\mu}_{+,n-1}(c_1) = \Psi^{-1}_{-,n-1}(b_1) = 0.$$

Hence there exists a nonzero element $a_n \in A_n$ so that $\psi^{n}_{+,\mu} = c_1$. Define $g(b_n) = a_n$. If $b_1 = 0$, then there exists an integer k so that

$$b_k = \Psi^{-k}_{+,-1}(b_{n-1}) \neq 0, \text{ and } b_{k-1} = \Psi^{-k+1}_{+,-1}(b_{n-1}) = 0.$$

Hence there exists a nonzero element $c_k \in C_k$ so that $\psi^{\mu}_{+,k}(c_k) = b_k$. Then

$$\psi^{\mu}_{+,n-k+1}(c_k) = \Psi^{-k}_{-,n-k}(b_k) = \Psi^{1}_{+,n-k} \circ \Phi^{n-1}_{-,1}(b_{n-1}) = 0.$$

Hence there exists a nonzero element $a_{n-k+1} \in A_{n-k+1}$ so that $\psi^{\mu}_{+,k}(a_{n-k+1}) = c_k$. Define

$$g(b_n) = \Psi^{n-k+1}_{+,n}(a_{n-k+1}).$$

We claim that $\Psi^{n-k+1}_{+,n}(a_{n-k+1}) \neq 0$. Suppose there exists some integer $l \in [1, k-1]$ so that

$$a_{n-l+1} = \Psi^{n-k+1}_{+,n-l+1}(a_{n-k+1}) = 0, \text{ and } a_{n-l} = \Psi^{n-k+1}_{+,n-l}(a_{n-k+1}) \neq 0.$$

Then there exists a nonzero element in $c_l \in C_l$ so that $\psi^{\mu}_{+,n-l}(c_l) = a_{n-l}$. Consider $b_l = \psi^{\mu}_{+,-l}(c_l)$. Then

$$\psi^{-l}_{+,\mu}(b_l) = \psi^{n-l}_{+,\mu} \circ \Psi^{-l}_{-,n-l}(b_l) = \psi^{n-l}_{+,\mu}(a_{n-l}) = 0.$$

Hence there exists a nonzero element $b_{l-1} \in B_{l+1}$ so that $\psi^{l+1}_{+,l}(b_{l+1}) = b_l$. By induction we know there exists a nonzero element $b_{n-1} \in B_{n-1}$ so that $\psi^{n-1}_{+,l}(b_{n-1}) = b_l$. This is impossible because

$$\psi^{\mu}_{+,-n} \circ \psi^{-n+1}_{+,\mu}(b_{n-1}) = \psi^{\mu}_{+,-n}(c_n) = b_n \neq 0.$$

If the second case happens, Define

$$g(b_n) = \Psi^{1}_{+,n} \circ (\Psi^{1}_{+,\mu})^{-1}(c_n).$$

Similar argument to that in the first case shows $g(b_n) \neq 0$.

Step 3. The above maps f and g is not well-defined because we may have many preimages of the same element. We fix it by considering restrictions of maps.

Let β^{-i}_{-i+1} to be the restriction of $\psi^{-i}_{+,-i+1}$ on $\Psi^{-n}_{+,-i}(B_n)$. Let $\alpha^{n-1}_{n} = \psi^{n-1}_{+,n}$ and let A_{n-1} be decomposed as

$$A_{n-1} = \ker(\alpha^{n-1}_{n}) \oplus A'_{n-1}.$$

Then

$$A_n \cong \operatorname{coker}(\alpha^{\mathrm{n}-1}_{\mathrm{n}}) \oplus A'_{n-1}.$$

Let α^{n-2}_{n-1} be the restriction of $\psi^{n-2}_{+,n-1}$ on $(\psi^{n-2}_{+,n-1})^{-1}(A'_{n-1})$ and let

$$(\psi^{n-2}_{+,n-1})^{-1}(A'_{n-1}) = \ker(\alpha^{n-2}_{n-1}) \oplus A'_{n-2}.$$

Inductively define α^{n-i}_{n-i+1} and A'_{n-i}. Since $\psi^{-1}_{+,0} = \psi^{0}_{+,1} = 0$, we have the decomposition

$$A_n = \bigoplus_{i=1}^{n} \operatorname{coker}(\alpha^{n-i}_{n-i+1}), \text{ and } B_n = \bigoplus_{i=1}^{n} \ker(\beta^{-i}_{-i+1}). \tag{4.16}$$

The diagram chasing in Step 1 and Step 2 implies

$$\dim_{\mathbb{C}} \operatorname{coker}(\alpha^{n-i}_{n-i+1}) = \dim_{\mathbb{C}} \ker(\beta^{-i}_{-i+1}). \tag{4.17}$$

Indeed, for a fixed integer i, the diagram chasing related to elements in $\operatorname{coker}(\alpha^{n-i}_{n-i+1})$ and $\ker(\beta^{-i}_{-i+i})$ to will only go through C_i. After diagram chasing for a chosen element, we can quotient all spaces by the subspaces generated by involving elements. The diagram still holds and the diagram chasing works again.

For any i, we prove that the maps f and g are well-defined and both $f \circ g$ and $g \circ f$ are identity maps. The maps f and g on A_n and B_n, respectively, will be defined as sums of the maps on all direct summands.

For the well-definedness, we only prove for f. The proof for g is similar. Suppose there are nonzero elements $a_{n-i+1}, a'_{n-i+1} \in A_{n-i+1}$, $b_n, b'_n \in B_n$ and $b_i, b'_i \in B_i$ so that

$$\begin{aligned}\Psi^{n-i+1}_{+,n}(a_{n-i+1}) = a_n = \Psi^{n-i+1}_{+,n}(a'_{n-i+1}),\\ \psi^{\mu}_{+,-i} \circ \psi^{n-i+1}_{+,\mu}(a_{n-i+1}) = b_i = \Psi^{-n}_{+,-i}(b_n),\\ \psi^{\mu}_{+,-i} \circ \psi^{n-i+1}_{+,\mu}(a'_{n-i+1}) = b'_i = \Psi^{-n}_{+,-i}(b'_n).\end{aligned}$$

Then there exists an integer $k \in [0, i-1]$ so that

$$\Psi^{n-i+1}_{+,n-k+1}(a_{n-i+1} - a'_{n-i+1}) = 0, \text{ and } a''_{n-k} = \Psi^{n-i+1}_{+,n-k}(a_{n-k} - a'_{n-k}) \neq 0.$$

Then there exists a nonzero element $c''_k \in C_k$ so that

$$\psi^{\mu}_{+,n-k}(c''_k) = a_{n-k} - a'_{n-k}.$$

Consider $b''_k = \psi^{\mu}_{+,-2}(c''_k)$. Then $\psi^{-k}_{+,\mu}(b''_k) = 0$ since $a_{n-i+1} - a'_{n-i+1}$ is in the image of $\psi^{n-k-1}_{+,n-k}$. By induction there exists a nonzero element $b''_{i-1} \in B_{i-1}$ so that $\Psi^{-i+1}_{-,n-i+1}(b_{i-1}) = a_{n-i+1} - a'_{n-i+1}$. Then

$$\psi^{\mu}_{+,-i} \circ \psi^{n-i+1}_{+,\mu}(a_{n-i+1} - a'_{n-i+1}) = 0,$$

i.e., $b_i = b'_i$. Since $\Psi^{n}_{+,-i}$ is injective on $\ker(\beta^{-i}_{-i+1})$, we have $b_n = b'_n$.

On the direct summand about i, a_n and b_n are considered as images of a nonzero element $a_{n-i+1} \in A_{n-i+1}$. It is straightforward to check

$$f \circ g = g \circ f = \mathrm{id}.$$

□

Remark 4.45. In the above proof, the decomposition (4.16) is not canonical, *i.e.*, there are many choices of direct sum decompositions. As a result, the maps f and g are not canonical, neither. However, the equation (4.17) always holds and we only need the existence of the isomorphisms.

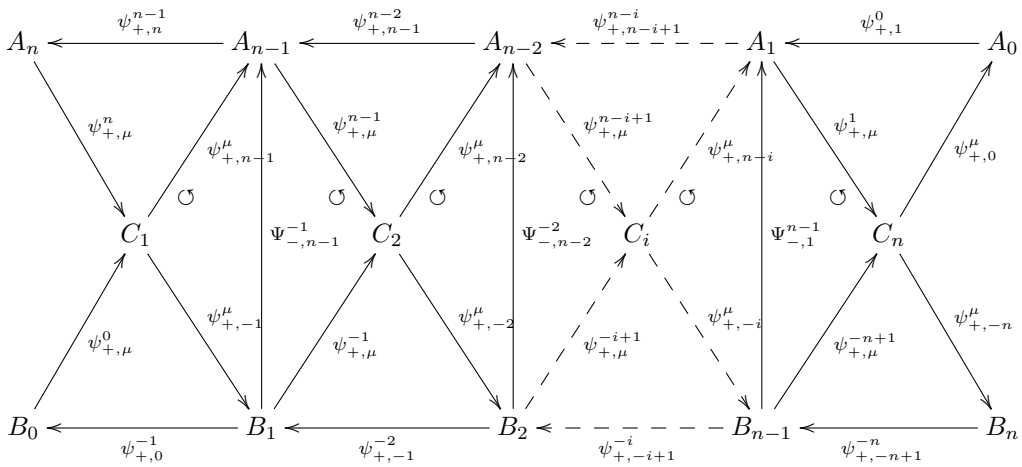
Proof of Proposition 1.17. Let Γ be the set of $-\widehat{\Gamma}_n$ for all n satisfying $q_n > q + 2g(K)$. The number n can be either positive or negative. Then term (1) is from Theorem 2.20. Term (2) is from Proposition 4.28 and Proposition 4.34. Term (3) is from Remark 4.24 and Proposition 4.44. Term (4) is from Lemma 4.11 and the fact that all isomorphisms in term (3) are from bypass maps or their inverse maps. □

Proof Theorem 1.10. We prove this theorem for $-\widehat{Y}$. Suppose $\mu \subset \partial\widehat{Y}(\widehat{K})$ is a simple closed curve so that $|\mu \cdot \lambda| = 1$ and suppose Y is the manifold obtained by Dehn filling along μ. Suppose further K is the dual knot in Y. Then K is a null-homologous knot in Y, and $(\widehat{Y}, \widehat{K})$ is obtained from Y by performing a q/p-surgery along (Y, K) with respect to the canonical framing in Subsection 4.1. The choice of μ is not important since it will only change the integer p. From Proposition 1.17, we can use any $\gamma \in \mathcal{G}$ and term (2) to define the decomposition. Explicitly we can choose either $\mathcal{I}_+(\widehat{Y}, \widehat{K})$ in Definition 4.23 or $\mathcal{I}_-(\widehat{Y}, \widehat{K})$ in Definition 4.29 to decompose $I^{\sharp}(\widehat{Y})$. □

4.5. **Spectral sequences and tau invariants.**

In this subsection, we derive more information from the diagram (4.14). We copy it here for the reader's convenience.

(4.14)



Here

$$A_i = \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_i), B_i = \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{-i}), \text{ and } C_i = \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_\mu),$$

or we can replace A_i, B_i, C_i by

$$\begin{aligned} A_i^j &= \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_i, S^\tau, \lceil \frac{1}{2}(-iq - q + q_0)\rceil + nq + j), \\ B_i^j &= \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{-i}, S^\tau, \lceil \frac{1}{2}(iq - q + q_0)\rceil + j), \\ \text{and } C_i^j &= \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_\mu, S^\tau, (n-i)q + j), \text{ respectively,} \end{aligned} \tag{4.15}$$

for some shift number $j \in \mathbb{Z}$.

Definition 4.46. Suppose K is a null-homologous knot in Y and $(\widehat{Y}, \widehat{K})$ is obtained from a q/p surgery with respect to the canonical framing as in Subsection 4.1. Following the idea in [Li19, Section 5], we define $\underline{\mathrm{KHI}}^-(\widehat{Y}, \widehat{K})$ to be the direct limit of the following direct system

$$\cdots \to \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau)[j_n^-] \xrightarrow{\psi^n_{-,n+1}} \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau)[j_{n+1}^-] \to \cdots,$$

where

$$j_m^- = g(K) + \lceil \frac{q-1}{2} \rceil - \hat{i}^m_{max}.$$

There is a $\mathbb{Z}$ grading on $\underline{\mathrm{KHI}}^-(-\widehat{Y}, \widehat{K})$ induced by S^τ. By Proposition 4.16 and Lemma 4.35, there is a commutative diagram

$$\begin{tikzcd} \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n, S^\tau)[j_n^-] \arrow[r, "\psi^n_{-,n+1}"] \arrow[d, "\psi^n_{+,n+1}"] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau)[j_{n+1}^-] \arrow[d, "\psi^{n+1}_{+,n+2}"] \\ \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1}, S^\tau)[j_{n+1}^-] \arrow[r, "\psi^{n+1}_{-,n+2}"] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+2}, S^\tau)[j_{n+2}^-] \end{tikzcd}$$

Hence the maps $\{\psi^n_{+,n+1}\}$ induces a map U_- of degree is $-q$ on $\underline{\mathrm{KHI}}^-(-\widehat{Y},\widehat{K})$. Similarly, we can define $\underline{\mathrm{KHI}}^+(-\widehat{Y},\widehat{K})$ to be the inverse limit of the following system

$$\cdots \to \underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_{-n-1},S^\tau)[j^+_n] \xrightarrow{\psi^n_{-,n+1}} \underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}_{-n},S^\tau)[j^+_{n+1}] \to \cdots$$

where

$$j^+_m = -g(K) + \lceil -\frac{q-1}{2} \rceil - \hat{i}^n_{min}.$$

Also, there is a $\mathbb{Z}$ grading and a map U_+ on $\underline{\mathrm{KHI}}^+(-\widehat{Y},\widehat{K})$.

If the underlying space is clear, we will denote U_+ and U_- by U.

Proposition 4.47. *Consider $\underline{\mathrm{KHI}}^-(-\widehat{Y},\widehat{K})$ and $\underline{\mathrm{KHI}}^+(-\widehat{Y},\widehat{K})$ in Definition 4.46. For any grading i apart from the top $2g(K)$ gradings of $\underline{\mathrm{KHI}}^-(-\widehat{Y},\widehat{K})$ and the bottom $2g(K)$ gradings of $\underline{\mathrm{KHI}}^+(-\widehat{Y},\widehat{K})$, the direct sums of the vector spaces*

$$\bigoplus_{m=0}^{q-1} \underline{\mathrm{KHI}}^-(-\widehat{Y},\widehat{K},i-m), \text{ and } \bigoplus_{m=0}^{q-1} \underline{\mathrm{KHI}}^+(-\widehat{Y},\widehat{K},i+m)$$

are isomorphic to

$$\underline{\mathrm{SHI}}(-\widehat{Y}(1),-\delta) \cong I^\sharp(\widehat{Y})$$

and the maps U_- and U_+ are an isomorphisms.

Proof. This follows from Lemma 4.18, Lemma 4.22, Lemma 4.30 and Lemma 4.32. □

Then the diagram (4.14) induces the following proposition directly.

Proposition 4.48. *Consider $\underline{\mathrm{KHI}}^-(-\widehat{Y},\widehat{K})$ and $\underline{\mathrm{KHI}}^+(-\widehat{Y},\widehat{K})$ defined as in Definition 4.46. There is a diagram*

(4.18)

$$\begin{tikzcd}
\underline{\mathrm{KHI}}^-(-\widehat{Y},\widehat{K}) \arrow[dr, "\mathrm{P}"] \arrow[rr, leftarrow, "U_-"] \arrow[dr, phantom, shift right=4ex, "\circlearrowleft"] & & \underline{\mathrm{KHI}}^-(-\widehat{Y},\widehat{K}) \\
& KHI(-\widehat{Y},\widehat{K}) \arrow[ur, "\partial_1"] \arrow[dr, "\mathrm{I}"] & \\
\underline{\mathrm{KHI}}^+(-\widehat{Y},\widehat{K}) \arrow[uu, "\partial_3"] \arrow[ur, "\partial_2"] \arrow[rr, leftarrow, "U_+"'] \arrow[ur, phantom, shift left=4ex, "\circlearrowleft"] & & \underline{\mathrm{KHI}}^+(-\widehat{Y},\widehat{K}) \arrow[uu, "\partial_3"']
\end{tikzcd}$$

where the maps $\mathrm{I}, \mathrm{P}, \partial_1, \partial_2$ and ∂_3 are induced by $\psi^\mu_{+,*}, \psi^*_{+,\mu}, \psi^\mu_{+,*}, \psi^*_{+,\mu}$ and $\Psi^*_{-,*}$, respectively. The top triangle and the bottom triangle are exact triangles. Two triangle with circles and the square are commutative.

Proof. Since the maps in the diagram (4.18) are homogeneous, it suffices to prove the proposition for finite many gradings (*c.f.* Proposition 4.47 and [OS08, Lemma 2.5]).

For a fixed integer $i_0 \in \mathbb{N}$, we consider

$$\underline{\mathrm{KHI}}^-_{fin} := \bigoplus_{m\in\mathbb{N}} \underline{\mathrm{KHI}}^-(\widehat{Y},\widehat{K},-i_0+mq), \text{ and } \underline{\mathrm{KHI}}^+_{fin} := \bigoplus_{m\in\mathbb{N}} \underline{\mathrm{KHI}}^+(\widehat{Y},\widehat{K},i_0-mq).$$

From Lemma 4.18 and Lemma 4.30, we can choose a large enough integer n and choose the shift number j properly in (4.15) for the diagram (4.14) to recover the diagram (4.18) for $\underline{\mathrm{KHI}}^-_{fin}, \underline{\mathrm{KHI}}^+_{fin}$ and the corresponding direct summands of gradings of KHI. □

Remark 4.49. The maps ∂_1, ∂_2 and ∂_3 in the diagram (4.18) are counterparts of boundaries maps from the following short exact sequences (*c.f.* [OS04a]).

$$0 \to^b CFK^- \to CFK^- \to CFK^0 \to 0 \tag{4.19}$$

$$0 \to CFK^0 \to CFK^+ \to CFK^+ \to 0 \tag{4.20}$$

$$0 \to^b CFK^- \to CFK^\infty \to CFK^+ \to 0 \tag{4.21}$$

Lemma 4.50. *Consider $(-\widehat{Y}, \widehat{K})$ in Definition 4.46. For $l \in \{0, \dots, q-1\}$, define*

$$KHI_l(-\widehat{Y}, \widehat{K}) := \bigoplus_{m \in \mathbb{Z}} KHI(-\widehat{Y}, \widehat{K}, l + mq).$$

For any fixed l, there exist a spectral sequence $\{E_r, d_r\}$ with $E_1 = KHI_l(-\widehat{Y}, \widehat{K})$.

Proof. As in the proof of Proposition 4.48, we can choose a large enough integer n and choose the shift number j properly in (4.15) for the diagram (4.14) to recover the diagram (4.18) for finite gradings. For any fixed l, there is a choice of (n, j) so that

$$\bigoplus_{i=0}^{n} C_i^j = KHI_l(-\widehat{Y}, \widehat{K}).$$

Note in this case $A_i^j = B_i^j = C_i^j = C_{n-i}^j = 0$ for any small integer $i\mathbb{N}$. In the following, we omit the superscript j.

We focus on the bottom exact triangles about C_i and B_i. Let d_1 be the composition $\psi^{-i}_{+,\mu} \circ \psi^{\mu}_{+,-i}$, or equivalently $\partial_2 \circ \mathrm{I}$ in the diagram (4.18). The fact $d_1 \circ d_1 = 0$ follows from the exactness at C_i. Let E_2 be the homology of (E_1, d_1). We define d_2 on E_2 by diagram chasing as follows.

For a nonzero element $c_i \in \ker(d_1) \subset C_i$, suppose $b_i = \psi^{\mu}_{+,-i}(c_i) \in B_i$. By construction, $\psi^{-i}_{+,\mu}(b_i) = 0$, so there exists an element $b_{i+1} \in B_{i+1}$ so that $\psi^{-i-1}_{+,-i}(b_{i+1}) = b_i$. Define

$$d_2(c_i) = \psi^{-i-1}_{+,\mu}(b_{i+1}).$$

If there is another element $b'_{i+1} \subset B_{i+1}$ so that $\psi^{-i-1}_{+,-i}(b'_{i+1}) = b_i$. Then

$$\psi^{-i-1}_{+,-i}(b_{i+1} - b'_{i+1}) = 0.$$

Hence there is an element $c_{i+1} \in C_{i+1}$ so that $\psi^{\mu}_{+,-i-1}(c_{i+1}) = b_{i+1} - b'_{i+1}$. Then

$$\psi^{-i-1}_{+,\mu}(b_{i+1}) - \psi^{-i-1}_{+,\mu}(b'_{i+1}) = d_1(c_{i+1}).$$

Thus, the map d_2 is well-defined on E_2. The fact $d_2 \circ d_2 = 0$ is also from the exactness.

Let E_3 be the homology of (E_2, d_2). Consider $\ker(d_2)$ as a subspace of $\bigoplus_i^n C_i$. Then we can define d_3 similarly: for elements $c_i \in \ker(d_2)$, choose an element $b_{i+2} \in B_{i+2}$ so that

$$\Psi^{-i-2}_{+,-i}(b_{i+2}) = \psi^{\mu}_{+,-i}(c_i).$$

Then define

$$d_3(c_i) = \psi^{-i-2}_{+,\mu}(b_{i+2}).$$

By similar diagram chasing, we know d_3 is well-defined on E_3. For $r \geqslant 3$, define (E_r, d_r) similarly. □

Remark 4.51. When fixing a basis for B_i and C_i, or equivalently fixing all direct sum decompositions, we can lift all d_r to C_i. The exactness implies $d_r \circ d_s = 0$ for any $r, s \in \mathbb{Z}_{\geqslant 1}$. Thus,

$$(\bigoplus_{i=1}^{n} C_i, d = \sum_r d_r)$$

is a filtered chain complex. The map d_r on C_i is roughly the composition

$$\psi_{+,\mu}^{-i-r+1} \circ (\Psi_{+,-i-r+1}^{-i})^{-1} \circ \psi_{+,-i}^{\mu}.$$

Here $(\Psi_{+,-i-r+1}^{-i})^{-1}$ is the map from B_i to B'_{i+r-1}, where

$$B_{i+r-1} = B'_{i+r-1} \oplus \ker(\Psi_{+,-i-r+1}^{-i}).$$

The restriction of $\mathrm{im}(d)$ on B_i is well-defined. Indeed, it is the direct sum of images of

$$(\Psi_{+,-i}^{-i+r-1})^{-1} \circ \psi_{+,-i+r-1}^{\mu}$$

for all $r \in \mathbb{Z}_{\geqslant 1}$.

Lemma 4.52. *Consider the spectral sequence $\{E_r, d_r\}$ with $E_1 = KHI_l(-\widehat{Y}, \widehat{K})$ for some l in Lemma 4.50. This spectral sequence stabilizes at some finite page, whose homology is isomorphic to $\mathcal{I}_-(\widehat{Y}, \widehat{K}, N - l)$ for some fixed integer N.*

Proof. First, consider the choice of (n, j) and the corresponding diagram (4.18) in the proof of Lemma 4.50. By definition of $\mathcal{I}_-(\widehat{Y}, \widehat{K})$, there exists a fixed integer N so that

$$\mathcal{I}_-(\widehat{Y}, \widehat{K}, N - l) \cong B_n \cong B_{n-1} \cong \cdots \cong B_{n-i}$$

for any small integer $i \in \mathbb{N}$. Then it is straightforward to check the spectral sequence stabilizes at some finite page. Suppose $H(C_i)$ is the homology of this page at C_i.

Second, fix a basis for C_i, consider d_r on $\bigoplus_{i=1}^{n} C_i$ and define $d = \sum_r d_r$. We prove

$$\mathrm{im}(d)|_{C_s} = \mathrm{im}(\psi_{+,-s+1}^{\mu}) \tag{4.22}$$

by induction on s. For $s = 2$, we have $B_0 = B_1 = C_1 = C_2 = 0$, so both images are zeros. Assume the argument has been proved for $s = i$, we prove it for $s = i + 1$. By assumption and Remark 4.51, we have $\mathrm{im}(d)|_{B_{i-1}} = B_{i-1}$.

From the exactness at B_i, we have

$$\mathrm{im}(\psi_{+,-i}^{\mu}) = \ker(\psi_{+,-i+1}^{-i}) \cong B_i / B'_i,$$

where $B'_i \cong \mathrm{im}(\psi_{+,-i+1}^{-i})$ by the map $\psi_{+,-i+1}^{-i}$. From the fact $\mathrm{im}(d)|_{B_{i-1}} = B_{i-1}$, we have

$$\mathrm{im}(d)|_{B_i} = B'_i \oplus \mathrm{im}(\psi_{+,-i}^{\mu}) = B_i.$$

Then

$$\mathrm{im}(d)|_{C_{i+1}} = \psi_{+,-i}^{\mu}(\mathrm{im}(d)|_{B_i}) = \mathrm{im}(\psi_{+,-i}^{\mu}).$$

Finally, consider the decomposition of B_n in (4.16):

$$B_n = \bigoplus_{i=1}^{n} \ker(\beta_{-i+1}^{-i}),$$

where β_{-i+1}^{-i} is the restriction of $\psi_{+,-i+1}^{i}$ on $\Psi_{+,-i}^{-n}(B_n)$. The diagram chasing in the proof of Proposition 4.44 and the fact (4.22) implies

$$\ker(\beta_{-i+1}^{-i}) \cong H(C_i).$$

Hence we conclude

$$\bigoplus_{i=1}^{n} H(C_i) \cong B_n \cong \mathcal{I}_-(\widehat{Y}, \widehat{K}, N-l).$$

□

Proof of Proposition 1.11. This follows directly from the proof of Theorem 1.10, Lemma 4.50 and Lemma 4.52. Since the spectral sequence comes from $\underline{\mathrm{KHI}}^+$, it is independent of the choice of (n, j) satiesfying the condition in the proof of Lemma 4.50. □

Remark 4.53. We can also construct a spectral sequence by A_i and C_i in the diagram (4.18). The commutative diagrams in the diagram (4.18) imply these spectral sequences are indeed the same one. By reversing the order of B_i and replacing $\psi^*_{+,*}$ by $\psi^*_{-,*}$, we will obtain another spectral sequence. This spectral sequence stabilizes at some finite page whose homology is isomorphic to the one contructed by $\psi^*_{+,*}$ in Lemma 4.50. However, the spectral sequences coming from $\psi^*_{+,*}$ and $\psi^*_{-,*}$ are not the same one and this difference might be useful to construct differentials d_z and d_w on the filtered chain complex $(\bigoplus_{i=1}^n C_i, d)$, and moreover a mapping cone formula similar to the work of Ozsváth and Szabó [OS08, OS11].

Then we discuss about tau invariants. Suppose $Y = S^3$.

Definition 4.54 ([Li19])**.** $\tau_I(K) = \max\{i \in \mathbb{Z} | \exists x \in \underline{\mathrm{KHI}}^-(-S^3, K, i), U_-^k(x) \neq 0, \forall k \in \mathbb{N}\}$.

Proof of Proposition 1.18. From [Li18], There is a canonical duality

$$\underline{\mathrm{SHI}}(-Y(\bar{K}), -\widehat{\Gamma}_n) \cong \underline{\mathrm{SHI}}(Y(\bar{K}), -\widehat{\Gamma}_{-n}) \cong \mathrm{Hom}_{\mathbb{C}}(\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{-n}), \mathbb{C}).$$

The bypass maps

$$\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_n) \xrightarrow{\psi^n_{\pm,n+1}} (\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{n+1})$$

become

$$\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{-n-1}) \xrightarrow{\psi^{-n-1}_{\pm,-n}} \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_{-n})$$

since the cobordisms inducing the bypass maps have reverse orientations. Considering the gradings, we have

$$\underline{\mathrm{KHI}}^-(-Y, \bar{K}, i) \cong \mathrm{Hom}_{\mathbb{C}}(\underline{\mathrm{KHI}}^+(-Y, K, -i), \mathbb{C}),$$

Applying to $Y = S^3$, we have

$$\tau_I(\bar{K}) = -\min\{i \in \mathbb{Z} | \exists x \in \underline{\mathrm{KHI}}^+(-S^3, K, i), U_+^{-k}(x) \neq \varnothing, \forall k \in \mathbb{N}\}.$$

From Proposition 4.47, apart from the top gradings of $\underline{\mathrm{KHI}}^-(-S^3, K)$ and the bottom gradings of $\underline{\mathrm{KHI}}^+(-S^3, K)$, the maps U_- and U_+ are isomorphisms. Note $I^\sharp(-S^3) \cong \mathbb{C}$. The condition $k \in \mathbb{N}$ in the definition of τ_I can be replaced by $k \in [1, 2g]$. Then this proposition is induced by the diagram (4.14) as follows.

Let $n = 2g(K) + 1$. Choose j in the proof of Proposition 4.44 so that

$$A_n^j \cong \underline{\mathrm{KHI}}^-(-S^3, K, -g(K)) \cong \mathbb{C}.$$

Then

$$A_{n-i}^j \cong \underline{\mathrm{KHI}}^-(-S^3, K, i - g(K))$$

for any integer $i \in [0, n]$. In particular $A_0^j \cong \underline{\mathrm{KHI}}^-(-S^3, K, g(K) + 1) = 0$. Moreover, the map $\psi^*_{+,*}$ becomes the map U_- under the isomorphims. Then it is straightforward to check

$$\tau_I(K) = \max\{i \in \mathbb{Z} | \mathrm{coker}(\alpha_i^{i-1}) \neq 0\},$$

where α_i^{i-1} is defined as in the proof of Proposition 4.44.

Under this grading shift j, we have

$$B_i^j \cong \underline{\mathrm{KHI}}^+(-S^3, K, i - \frac{n+1}{2})$$

for any integer $i \in [0, n]$. In particular

$$B_n^j \cong \underline{\mathrm{KHI}}^+(-S^3, K, -g(K)) \cong \mathbb{C}, \text{ and } B_0^j \cong \underline{\mathrm{KHI}}^+(-S^3, K, -g(K)-1) = 0.$$

Similarly, the map $\psi^*_{+,*}$ becomes U_+ under the isomorphims. Then it is straightforward to check

$$\tau_I(\bar{K}) = \max\{i \in \mathbb{Z} | \ker(\beta_{i-n}^{i-n-1}) \neq 0\}.$$

From the equation (4.17), we know that

$$\dim_{\mathbb{C}} \operatorname{coker}(\alpha_{n-k+1}^{n-k}) = \dim_{\mathbb{C}} \ker(\beta_{-k+1}^{-k}).$$

Substituting $k = n - i + 1$, we conclude that $\tau_I(\bar{K}) = -\tau_I(K)$. □

4.6. **The stabilization of Integral surgeries.**

Throughout this subsection, suppose $K \subset Y$ is a null-homologous knot and S is a minimal genus Seifert surface of K. The stabilization S^τ of S is chosen according to Definition 4.12. Since we might work with different surgery slopes, we will use the notation $\gamma_{(p,q)}$ to denote the suture on the knot complement. The maximal and minimal gradings of $\underline{\mathrm{SHI}}(-Y(K), -\gamma_{(p,q)}, S^\tau)$ are described explicitly according to Lemma 4.14 and we write them as

$$i^q_{max} = \lceil \frac{q-1}{2} \rceil + g(S), \text{ and } i^q_{min} = \lceil -\frac{q-1}{2} \rceil - g(S).$$

Note they are independent of p.

Since we will deal with different surgeries in the current subsection, we will write out the surgery slope explicitly: for the 3-manifold obtained by the q/p-surgery, we write $\widehat{Y}_{q/p}$. For the special class of sutures, we write $\widehat{\Gamma}_n(q/p)$ instead of $\widehat{\Gamma}_n$. For bypass maps, we write $\psi^{n+1}_{\pm,n}(q/p)$. However, if $q/p = 1/0$, we still omit it from the notation, then in this case, we simply write $\widehat{\Gamma}_n$ as Γ_n.

In this subsection, we deal with large integral surgeries. In this case, we know that $\Gamma_n = \gamma_{(-1,n)}$. Hence we have the following by Lemma 4.22 and Proposition 4.28. (Note $(p,q) = (0,1)$.)

Lemma 4.55. *For any $n > 2g(S)$ and $i \in \mathbb{Z}$ so that*

$$\lceil -\frac{n-1}{2} \rceil + g(S) \leqslant i \leqslant \lceil \frac{n-1}{2} \rceil - g(S),$$

we have

$$\underline{\mathrm{SHI}}(-Y(K), -\Gamma_n, S^\tau, i) \cong \underline{\mathrm{SHI}}(-Y(1), -\delta).$$

Remark 4.56. A more direct explanation of Lemma 4.55 is that, apart from the top $2g(S)$ many and bottom $2g(S)$ gradings, the vector spaces in all gradings are isomorphic to $\underline{\mathrm{SHI}}(-Y(1), -\delta)$.

Next, suppose we perform a $(-n)$-surgery. We can take

$$\hat{\lambda} = (0, -1) = -\mu \text{ and } \hat{\mu} = (-1, n) = \lambda - n\mu.$$

Then we compute

$$\widehat{\Gamma}_\mu(-n) = \Gamma_n,\ \widehat{\Gamma}_0(-n) = \Gamma_\mu,\ \widehat{\Gamma}_1(-n) = \gamma_{(-1,n-1)} = \Gamma_{n-1},\ \widehat{\Gamma}_2(-n) = \gamma_{(-2,2n-1)}, \tag{4.23}$$

and also

$$\widehat{\Gamma}_{-1}(-n) = \gamma_{(-1,n+1)} = \Gamma_{n+1},\ \widehat{\Gamma}_{-2}(-n) = \gamma_{(-2,2n+1)}. \tag{4.24}$$

Observe that

$$\widehat{\Gamma}_{-2}(-n) = \widehat{\Gamma}_2(-n-1). \tag{4.25}$$

Proposition 4.57 (Proposition 1.14)**.** *Suppose integer $n \geqslant 2g(S)+1$, then*

$$\mathcal{I}_+(-\widehat{Y}_{-n}, \widehat{K}, i) \cong \underline{\mathrm{SHI}}(-Y(1), -\delta)$$

for any integer $i \in [0, n-2g(S)-1]$ and $i = n-1$.

Proof. When $0 \leqslant i \leqslant n-1-2g(S)$, we know from equality (4.23), Definition 4.23, Lemma 4.18, and Lemma 4.55 that

$$\begin{aligned}\mathcal{I}_+(-\widehat{Y}_{-n}, \widehat{K}, i) =& \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_2(-n), S^\tau, i^{2n-1}_{max} - 2g(S) - i)\\ \cong& \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_\mu(-n), S^\tau, i^{n}_{max} - 2g(S) - i)\\ =& \underline{\mathrm{SHI}}(-Y(K), -\Gamma_n, S^\tau, i^{n}_{max} - 2g(S) - i)\\ \cong& \underline{\mathrm{SHI}}(-Y(1), -\delta).\end{aligned}$$

For $i = n-1$, we know similarly that

$$\begin{aligned}\mathcal{I}_+(-\widehat{Y}_{-n}, \widehat{K}, i) =& \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_2(-n), S^\tau, i^{2n-1}_{max} - 2g(S) - n + 1)\\ \cong& \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}_\mu(-n), S^\tau, i^{n}_{max} - 2g(S))\\ =& \underline{\mathrm{SHI}}(-Y(K), -\Gamma_n, S^\tau, i^{n}_{max} - 2g(S))\\ \cong& \underline{\mathrm{SHI}}(-Y(1), -\delta).\end{aligned}$$

□

Proposition 4.58 (Proposition 1.15)**.** *Suppose integer $n \geqslant 2g(S)+1$, then for any integer $i \in [0, n-1]$,*

$$\mathcal{I}_+(-\widehat{Y}_{-n-1}, \widehat{K}, i+1) \cong \mathcal{I}_+(-\widehat{Y}_{-n}, \widehat{K}, i).$$

Proof. For integer $i \in [0, n-1-2g(S)]$ and $i = n-1$, we know from Proposition 4.57 that

$$\mathcal{I}_+(-\widehat{Y}_{-n-1}, \widehat{K}, i+1) \cong \underline{\mathrm{SHI}}(-Y(1), -\delta) \cong \mathcal{I}_+(-\widehat{Y}_{-n}, \widehat{K}, i)$$

For the rest $(2g(S)-1)$ gradings, we use the following commutative diagram.

$$\begin{tikzcd}\underline{\mathrm{SHI}}(-Y(K), -\Gamma_{n-1}) \arrow[r, "\psi^{n-1}_{+,n}"] & \underline{\mathrm{SHI}}(-Y(K), -\Gamma_n)\\ \underline{\mathrm{SHI}}(-Y(K), -\Gamma_n) \arrow[u, "\psi^{\mu}_{-,1}(-n)"] \arrow[r, "\psi^{n}_{-,n+1}"] & \underline{\mathrm{SHI}}(-Y(K), -\Gamma_{n+1}) \arrow[u, "\psi^{\mu}_{-,1}(-n-1)"']\end{tikzcd} \tag{4.26}$$

This is directly from facts (4.23) and (4.24), and Corollary 4.40 by taking $i = 1$ and $j = -1$. Note

$$\psi^{n-1}_{+,n} = \psi^{1}_{-,\mu}(-n), \psi^{n}_{-,n+1} = \psi^{\mu}_{-,-1}(-n),\ \text{and}\ \psi^{\mu}_{-,1}(-n-1) = \psi^{-1}_{-,\mu}(-n).$$

From Proposition 4.16 and Definition 4.23, we obtain a graded commutative diagram

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\Gamma_{n-1}, S^\tau, i^{n-1}_{min} + n - 2 - i) \arrow[r, "\psi^{n-1}_{+,n}"] & \underline{\mathrm{SHI}}(-Y(K), -\Gamma_n, S^\tau, i^{n}_{min} + n - 2 - i) \\
\underline{\mathrm{SHI}}(-Y(K), -\Gamma_n, S^\tau, i^{n}_{max} + n - 2g - 1 - i) \arrow[u, "\psi^{\mu}_{-,1}(-n)"] \arrow[r, "\psi^{n}_{-,n+1}"] & \underline{\mathrm{SHI}}(-Y(K), -\Gamma_{n+1}, S^\tau, i^{n+1}_{max} + n - 2g - 1 - i) \arrow[u, "\psi^{\mu}_{-,1}(-n-1)"']
\end{tikzcd}$$

For any fixed integer $i \in [n - 2g(S), n - 1]$, we have a graded exact triangle

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\Gamma_{n-1}, S^\tau, i^{n-1}_{min} + n - 2 - i) \arrow[r] & \mathcal{I}_+(-\widehat{Y}_{-n}, \widehat{K}, i) \arrow[dl] \\
\underline{\mathrm{SHI}}(-Y(K), -\Gamma_n, S^\tau, i^{n}_{max} + n - 2g - 1 - i) \arrow[u, "\psi^{\mu}_{-,1}(-n)"] &
\end{tikzcd}$$

Hence we know that

$$\dim_{\mathbb{C}} \mathcal{I}_+(-\widehat{Y}_{-n}, \widehat{K}, i) = \dim_{\mathbb{C}} \ker(\psi^{\mu}_{-,1}(-n)) + \dim_{\mathbb{C}} \operatorname{coker}(\psi^{\mu}_{-,1}(-n)).$$

Similarly, we know that

$$\dim_{\mathbb{C}} \mathcal{I}_+(-\widehat{Y}_{-n-1}, \widehat{K}, i+1) = \dim_{\mathbb{C}} \ker(\psi^{\mu}_{-,1}(-n-1)) + \dim_{\mathbb{C}} \operatorname{coker}(\psi^{\mu}_{-,1}(-n-1))$$

Since the maps $\psi^{n-1}_{+,n}$ and $\psi^{n}_{-,n+1}$ are both isomorphisms in the corresponding gradings by Lemma 4.18, we conclude that

$$\dim_{\mathbb{C}} \mathcal{I}_+(-\widehat{Y}_{-n-1}, \widehat{K}, i+1) = \dim_{\mathbb{C}} \mathcal{I}_+(-\widehat{Y}_{-n}, \widehat{K}, i).$$

□

4.7. **The splitting of exact triangles.**

In this subsection, we study the surgery exact triangle about framed instanton Floer homology and prove that it splits under certain conditions. Suppose Y is a closed 3-manifold and $K \subset Y$ is a null-homologous knot. Let S be a minimal genus Seifert surface of K. Suppose we have three pairs of integers (p_1, q_1), (p_2, q_2) and (p_3, q_3) that satisfy the following conditions.

(1) For $i \in \{1, 2, 3\}$, $q_i > 0$, $p_i \leqslant 0$.
(2) We have $q_3 = q_1 + q_2 > 2g(S)$ and $p_3 = p_1 + p_2$.

For $i \in \{1, 2, 3\}$, write $\widehat{Y}_i$ be the 3-manifold obtained from Y by the (q_i/p_i)-surgery along K. From Theorem 2.4, we have an exact triangle

$$\begin{tikzcd}
I^\sharp(-\widehat{Y}_{q_1/p_1}) \arrow[rr, "f_1"] & & I^\sharp(-\widehat{Y}_{q_2/p_2}) \arrow[dl, "f_2"] \\
& I^\sharp(-\widehat{Y}_{q_3/p_3}) \arrow[ul, "f_3"] &
\end{tikzcd} \tag{4.27}$$

Then we have the following proposition.

Proposition 4.59 (Theorem 1.19)**.** *The map f_1 in the exact triangle (4.27) is zero, or equivalently, we have*

$$\dim_{\mathbb{C}} I^\sharp(-\widehat{Y}_{q_3/p_3}) = \dim_{\mathbb{C}} I^\sharp(-\widehat{Y}_{q_1/p_1}) + \dim_{\mathbb{C}} I^\sharp(-\widehat{Y}_{q_2/p_2}).$$

Proof. Define

$$(p_0, q_0) = (p_2, q_2) - (p_1, q_1), \mu^1 = (-p_1, -q_1), \lambda^1 = (-p_0, -q_0),$$
$$\mu^2 = (-p_2, -q_2),\ \lambda^2 = (-p_0, -q_0), \mu^3 = (-p_3, -q_3),\ \text{and } \lambda^3 = (-p_2, -q_2).$$

As in Definition 4.4, for $n, j \in \mathbb{Z}$, we define $\widehat{\Gamma}^j_n$ to be the suture on $Y(K)$ consisting of two curves of class $\pm(\lambda^j - n\mu^j)$. It is straightforward to compute

$$\widehat{\Gamma}^1_{-2} = \gamma_{(p_3,q_3)},\ \widehat{\Gamma}^1_{-3} = \gamma_{(p_1+p_3,q_1+q_3)}, \widehat{\Gamma}^2_2 = \gamma_{(p_3,q_3)},\ \widehat{\Gamma}^2_3 = \gamma_{(p_2+p_3,q_2+q_3)}, \tag{4.28}$$

$$\widehat{\Gamma}^3_2 = \gamma_{(p_1+p_3,q_1+q_3)},\ \widehat{\Gamma}^3_3 = \gamma_{(p_1+2p_3,q_1+2q_3)},\ \widehat{\Gamma}^3_{-1} = \gamma_{(p_2+p_3,q_2+q_3)},\ \text{and } \widehat{\Gamma}^3_{-2} = \gamma_{(p_2+2p_3,q_2+2q_3)}. \tag{4.29}$$

We sketch the idea of the proof as follows:

(1) The sutures $\widehat{\Gamma}^1_{-3} = \widehat{\Gamma}^3_2$ encode information of $I^\sharp(\widehat{Y}_{q_1/p_1})$.
(2) The sutures $\widehat{\Gamma}^2_3 = \widehat{\Gamma}^3_{-1}$ encode information of $I^\sharp(\widehat{Y}_{q_2/p_2})$.
(3) The sutures $\widehat{\Gamma}^3_3$ and $\widehat{\Gamma}^3_{-2}$ encode information of $I^\sharp(\widehat{Y}_{q_3/p_3})$.
(4) The sutures $\widehat{\Gamma}^3_{-2}$ can be related to $\widehat{\Gamma}^3_3$ in a way similar to that of the proof of Proposition 4.58.
(5) Finally, comparing the information coming from the above four terms, we conclude the proof of the proposition.

We apply Proposition 4.16 and perform the computation as in the proof of Lemma 4.27:

size	$-\widehat{\Gamma}^3_\mu$	$\xrightarrow{\psi^\mu_{+,2}(q_3/p_3)}$	$-\widehat{\Gamma}^3_2$	$\xrightarrow{\psi^2_{+,3}(q_3/p_3)}$	$-\widehat{\Gamma}^3_3$	$\xrightarrow{\psi^3_{+,\mu}(q_3/p_3)}$	$-\widehat{\Gamma}^3_\mu$
$2g(S)$	E				E		E
$q_3 - 2g(S)$	F				F		F
$2g(S)$	G		A		I		G
q_1			B		B		
$q_3 - 2g(S)$			C		C		
$2g(S)$			D		D		

and

size	$-\widehat{\Gamma}^3_\mu$	$\xrightarrow{\psi^\mu_{-,2}(q_3/p_3)}$	$-\widehat{\Gamma}^3_2$	$\xrightarrow{\psi^2_{-,3}(q_3/p_3)}$	$-\widehat{\Gamma}^3_3$	$\xrightarrow{\psi^3_{-,\mu}(q_3/p_3)}$	$-\widehat{\Gamma}^3_\mu$
$2g(S)$			A		A		
q_1			B		B		
$q_3 - 2g(S)$			C		C		
$2g(S)$	E		D		H		E
$q_3 - 2g(S)$	F				F		F
$2g(S)$	G				G		G

Comparing two expressions of $\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_3)$, we conclude that

$$A = E,\ \begin{pmatrix} B \\ C \\ H \end{pmatrix} = \begin{pmatrix} F \\ I \\ B \end{pmatrix},\ F = C,\ \text{and } D = G. \tag{4.30}$$

Similarly, we compute

size	$-\widehat{\Gamma}^3_{-1}$ $\xrightarrow{\psi^{-1}_{+,\mu}(q_3/p_3)}$	$-\widehat{\Gamma}^3_{\mu}$ $\xrightarrow{\psi^{\mu}_{+,-2}(q_3/p_3)}$	$-\widehat{\Gamma}^3_{-2}$ $\xrightarrow{\psi^{-2}_{+,-1}(q_3/p_3)}$	$-\widehat{\Gamma}^3_{-1}$
$2g(S)$	J		J	J
q_2	K		K	K
$q_3-2g(S)$	L		L	L
$2g(S)$	M	E	N	M
$q_3-2g(S)$		F	F	
$2g(S)$		G	G	

and

size	$-\widehat{\Gamma}^3_{-1}$ $\xrightarrow{\psi^{-1}_{-,\mu}(q_3/p_3)}$	$-\widehat{\Gamma}^3_{\mu}$ $\xrightarrow{\psi^{\mu}_{-,-2}(q_3/p_3)}$	$-\widehat{\Gamma}^3_{-2}$ $\xrightarrow{\psi^{-2}_{-,-1}(q_3/p_3)}$	$-\widehat{\Gamma}^3_{-1}$
$2g(S)$		E	E	
$q_3-2g(S)$		F	F	
$2g(S)$	J	G	O	J
q_2	K		K	K
$q_3-2g(S)$	L		L	L
$2g(S)$	M		M	M

Comparing two expressions of $\underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}^3_{-2})$, we conclude that

$$J=E,\ \begin{pmatrix} K\\ L\\ N\end{pmatrix}=\begin{pmatrix} F\\ O\\ K\end{pmatrix},\ F=L,\ \text{and } G=M. \tag{4.31}$$

Now we argue in two cases.

Case 1. If either $q_1\geqslant 2g(S)$ or $q_2\geqslant 2g(S)$.

We only prove the case when $q_2\geqslant 2g(S)$. The other is similar. When $q_2\geqslant 2g(S)$, from Proposition 4.28 and Proposition 4.34, we know that the block

$$F\subset \underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}^3_{\mu})=\underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}^1_{-2})$$

contains all information of $I^{\sharp}(-\widehat{Y}_{q_1/p_1})$. Similarly, the block

$$\begin{pmatrix} K\\ L\end{pmatrix}\subset \underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}^3_{-1})=\underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}^2_{3})$$

contains all information of $I^{\sharp}(-\widehat{Y}_{q_2/p_2})$, and

$$I^{\sharp}(-\widehat{Y}_{q_3/p_3})\cong\begin{pmatrix} F\\ O\end{pmatrix}\subset \underline{\mathrm{SHI}}(-Y(K),-\widehat{\Gamma}^3_{-2}).$$

Since $q_2\geqslant 2g(S)$, we can split L as

$$L=\begin{pmatrix} L_1\\ L_2\end{pmatrix},$$

where L_1 is of size $q_3 - q_2$ and L_2 is of size $q_2 - 2g(S)$. Then from (4.31), we know that

$$I^\sharp(-\widehat{Y}_{q_3/p_3}) \cong \begin{pmatrix} F \\ O \end{pmatrix} = \begin{pmatrix} K \\ L_1 \end{pmatrix}.$$

Now, the top q_1 gradings of the block

$$\begin{pmatrix} F \\ O \end{pmatrix} = \begin{pmatrix} K \\ L_1 \end{pmatrix}$$

is contained in F and hence is isomorphic to $I^\sharp(-\widehat{Y}_{q_1/p_1})$. Likewise, the rest $q_3 - q_1 = q_2$ gradings of the block

$$\begin{pmatrix} F \\ O \end{pmatrix} = \begin{pmatrix} K \\ L_1 \end{pmatrix}$$

is isomorphic to $I^\sharp(-\widehat{Y}_{q_2/p_2})$, and thus we conclude that

$$\dim_{\mathbb{C}} I^\sharp(-\widehat{Y}_{q_3/p_3}) = \dim_{\mathbb{C}} I^\sharp(-\widehat{Y}_{q_1/p_1}) + \dim_{\mathbb{C}} I^\sharp(-\widehat{Y}_{q_2/p_2}).$$

Case 2. We have $0 < q_1, q_2 < 2g(S)$.

As in case 1, we know that the block

$$\begin{pmatrix} B \\ C \end{pmatrix} \subset \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_2) = \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^1_{-3})$$

contains all information of $I^\sharp(-\widehat{Y}_{q_1/p_1})$, the block

$$\begin{pmatrix} K \\ L \end{pmatrix} \subset \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_{-1}) = \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^2_3)$$

contains all information of $I^\sharp(-\widehat{Y}_{q_2/p_2})$, and

$$I^\sharp(-\widehat{Y}_{q_3/p_3}) \cong \begin{pmatrix} H \\ F \end{pmatrix} \subset \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_3).$$

Since $q_2 < 2g(S)$, we can split O as

$$O = \begin{pmatrix} O_1 \\ O_2 \end{pmatrix},$$

where O_1 is of size q_2 and O_2 is of size $2g(S) - q_2$. Thus from (4.31), we know that

$$\begin{pmatrix} F \\ O_1 \end{pmatrix} = \begin{pmatrix} K \\ L \end{pmatrix}.$$

In particular, we conclude that

$$I^\sharp(-\widehat{Y}_{q_2/p_2}) \cong O_1.$$

Also, since we split I as

$$I = \begin{pmatrix} I_1 \\ I_2 \end{pmatrix},$$

where I_1 is of size q_1 and I_2 is of size $2g(S) - q_1$. We make the following claim.

Claim. The blocks H and O are isomorphic by a grading preserving isomorphism as $\mathbb{Z}_{2g(S)}$ graded spaces.

Before proving the Claim, we first use it to conclude the proof of the proposition. From (4.30), we know that

$$I^\sharp(-\widehat{Y}_{q_3/p_3}) \cong \begin{pmatrix} H \\ F \end{pmatrix} = \begin{pmatrix} O \\ F \end{pmatrix} = \begin{pmatrix} O_1 \\ O_2 \\ F \end{pmatrix} = \begin{pmatrix} I_2 \\ B \\ C \end{pmatrix}.$$

Finally, we observe that the top q_2 gradings of the block

$$\begin{pmatrix} O_1 \\ O_2 \\ F \end{pmatrix} = \begin{pmatrix} I_2 \\ B \\ C \end{pmatrix}$$

is

$$O_1 \cong I^\sharp(-\widehat{Y}_{q_2/p_2}),$$

The rest $q_2 = q_3 - q_1$ gradings is contained in the block

$$\begin{pmatrix} B \\ C \end{pmatrix}.$$

Hence it is isomorphic to $I^\sharp(-\widehat{Y}_{q_1/p_1})$. Thus, we conclude that

$$\dim_{\mathbb{C}} I^\sharp(-\widehat{Y}_{q_3/p_3}) = \dim_{\mathbb{C}} I^\sharp(-\widehat{Y}_{q_1/p_1}) + \dim_{\mathbb{C}} I^\sharp(-\widehat{Y}_{q_2/p_2}).$$

It remains to prove Claim. First, note from the computations to derive (4.30), we have a graded exact triangle

$$\begin{tikzcd} E \arrow[rr] & & D \arrow[dl] \\ & H \arrow[ul] & \end{tikzcd}$$

which is covered by

$$\begin{tikzcd} \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_\mu) \arrow[rr, "\psi^\mu_{-,2}(q_3/p_3)"] & & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_2) \arrow[dl, "\psi^2_{-,3}(q_3/p_3)"] \\ & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_3) \arrow[ul, "\psi^3_{-,\mu}(q_3/p_3)"] & \end{tikzcd}$$

Similarly from the computation to derive (4.31), we have a graded exact triangle

$$\begin{tikzcd} J \arrow[rr] & & G \arrow[dl] \\ & O \arrow[ul] & \end{tikzcd}$$

which is covered by

$$\begin{tikzcd} \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_{-1}) \arrow[rr, "\psi^{-1}_{-,\mu}(q_3/p_3)"] & & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_\mu) \arrow[dl, "\psi^\mu_{-,-2}(q_3/p_3)"] \\ & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_{-2}) \arrow[ul, "\psi^{-2}_{-,-1}(q_3/p_3)"] & \end{tikzcd}$$

From (4.28) and (4.29), we know that

$$\widehat{\Gamma}^3_\mu = \widehat{\Gamma}^2_2, \text{ and } \widehat{\Gamma}^3_{-1} = \widehat{\Gamma}^2_3.$$

Thus there is a map

$$\psi^2_{-,3}(q_2/p_2) : \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_\mu) \to \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_{-1}),$$

which restricts to a grading-preserving isomorphism by Proposition 4.16 and Lemma 4.18:

$$\psi^2_{-,3}(q_2/p_2) : E \xrightarrow{\cong} J.$$

Similarly, from (4.28) and (4.29), we know that

$$\widehat{\Gamma}^3_2 = \widehat{\Gamma}^1_{-2}, \text{ and } \widehat{\Gamma}^3_\mu = \widehat{\Gamma}^1_{-1}.$$

Thus, there is a map

$$\psi^{-2}_{-,-1}(q_1/p_1) : \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_2) \to \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_\mu),$$

which restricts to a grading-preserving isomorphism by Proposition 4.16 and Lemma 4.30:

$$\psi^{-2}_{-,-1}(q_1/p_1) : D \xrightarrow{\cong} G.$$

To conclude that $H \cong O$, it suffices to show that the following diagram commutes:

$$\begin{tikzcd}
E \arrow[r, "\psi^{\mu}_{-,2}(q_3/p_3)"] \arrow[d, "\psi^2_{-,3}(q_2/p_2)"'] & D \arrow[d, "\psi^{-2}_{-,-1}(q_1/p_1)"] \\
J \arrow[r, "\psi^{-1}_{-,\mu}(q_3/p_3)"] & G
\end{tikzcd}$$

which is covered by

$$\begin{tikzcd}
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_\mu) \arrow[r, "\psi^{\mu}_{-,2}(q_3/p_3)"] \arrow[d, "\psi^2_{-,3}(q_2/p_2)"'] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_2) \arrow[d, "\psi^{-2}_{-,-1}(q_1/p_1)"] \\
\underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_{-1}) \arrow[r, "\psi^{-1}_{-,\mu}(q_3/p_3)"] & \underline{\mathrm{SHI}}(-Y(K), -\widehat{\Gamma}^3_\mu)
\end{tikzcd} \tag{4.32}$$

This is directly from Corollary 4.40 by taking $\hat{\mu} = \mu^3, i = 2,$ and $j = -1$. □

5. Some remarks for future directions

First, we would like to make comments on the proof of Proposition 3.14. In the proof, we take great effort to pick the arc ζ, which is disjoint from the surface S_T and intersects the suture γ once. This ensures the bypass maps shift the gradings, as stated in Lemma 3.18. In the case of a rational homology sphere Y compatible with a Heegaard diagram, we are dealing with a vertical tangle $T = \alpha_1 \cup ... \cup \alpha_m$ in a balanced sutured manifold $(Y(1), \delta)$. Since the suture δ is connected, it is easy to pick an ζ_i for each component α_i of T so that each ζ_i intersects δ once and they are pair-wise disjoint. The cost to pay is that we may no longer be possible to make them disjoint

with the surfaces S_i associated to α_i. This might make a difference on the proof of Lemma 3.18. However, the authors do not find any essential gaps to conclude Proposition 3.14.

Since we choose ζ_i simultaneously and disjointly, we can deal with all different components of T at the same time. Thus, the following result is expected.

Conjecture 5.1. Suppose Y is a rational homology sphere. Then there exists a balanced sutured handlebody (H,γ) and surfaces $S_1,...,S_m$ properly embedded in H. If we consider the $\mathbb{Z}^m$ grading on $\underline{\mathrm{SHI}}(-H,-\gamma)$ induced by $S_1,...,S_m$ simultaneously, then there is a finite subset $V\subset\mathbb{Z}^m$ so that

$$I^\sharp(-Y)=\bigoplus_{\vec{v}\in V}\underline{\mathrm{SHI}}(-H,-\gamma,\vec{v}).$$

Here $\underline{\mathrm{SHI}}(-H,-\gamma,\vec{v})$ denote the sutured instanton Floer homology at the multi-grading $\vec{v}\in\mathbb{Z}^m$. Moreover, the data (H,γ), m, S_1,..., S_m, and $\vec{v}\subset\mathbb{Z}^m$ all depend only on the homeomorphism type of Y.

Second, there is a possible way to improve the estimate of $I^\sharp$ from the Heegaard diagrams. In Example 3.25, we conclude that

$$\dim_{\mathbb{C}} I^\sharp(-Y)\leqslant 10.$$

However, the equality holds only if the map ψ_4^3 in (3.9) is zero. Hence we could possibly improve the bound if we can understand the map ψ_4^3 better. A possible way is to study contact elements. It might be possible to show that the vector space $\underline{\mathrm{SHI}}(-H,-\gamma_3)$, or at least a subspace of it, is generated by the contact elements of tight contact structures on (H,γ_3). So if we could understand how the bypass attachment associated to ψ_4^3 changes the contact structures, we might be able to understand more about the map ψ_4^3, and hence obtain a more refined bound on $\dim_{\mathbb{C}} I^\sharp(-Y)$.

Third, the proof of Theorem 1.8 can be applied to computations of KHI for more examples. For example, the induction based on anti-waves can be used to give an alternate proof of Theorem 1.4 for (1,1)-L-space knots, namely (1,1)-knots that are also Heegaard Floer L-space knots, whose (1,1)-diagrams have explicit description by the work of Greene, Lewallen, and Vafaee [GLV18]. Sketchily, replace the anti-wave bypass and the bypass exact triangle in case 1 of the proof of Theorem 3.33 by an anti-wave and the surgery exact triangle from Theorem 2.18. Considering the orientations of rainbows, it can be checked that the other two knots in the surgery exact triangle are both (1,1)-L-space knots. Thus, we can prove an inequality about KHI of (1,1)-L-space knots by induction without using the inequality about $\underline{\mathrm{SHI}}$. The remainder of the proof is the same.

The above strategy may work for Floer simple knots admitting doubly-pointed Heegaard diagrams with the higher genus. However, the essential problem is the existence of the anti-wave bypass. For example, Floer simple knots in strong Heegaard diagrams introduced by Greene and Levine [GL16] are generalizations of simple knots, while the existence of the anti-wave bypass is a nontrivial problem mentioned in [GL16, Section 7].

Fourth, as mentioned in Remark 1.13, it is not known if the decomposition of $I^\sharp(\widehat{Y})$ is independent of the choice of $\widehat{K}$. Explicitly, we have the following conjecture.

Conjecture 5.2. Suppose $(\widehat{Y},\widehat{K})$ satisfies the hypothesis of Theorem 1.10. Suppose further that another knot $\widehat{K}'\subset\widehat{Y}$ satisfies the similar conditions to those of $\widehat{K}$, and

$$[\widehat{K}]=[\widehat{K}']\in H_1(\widehat{Y}).$$

Then there exists a grading preserving isomorphism

$$\mathcal{I}_{\pm}(\widehat{Y},\widehat{K}) \cong \mathcal{I}_{+}(\widehat{Y},\widehat{K}')$$

up to a $\mathbb{Z}_q$ grading shift, where $\mathcal{I}_+$ is defined as in Definition 4.23.

The condition in Theorem 1.10 that the boundary of the Seifert surface S of $\widehat{K}$ is connected can be removed by modifying the hypothesis and the statement as follows.

(1) Suppose the order of $[\widehat{K}] \in H_1(\widehat{Y})$ is a, *i.e.*, a is the minimal positive integer so that $a[K] = 0$. Suppose the number of the boundary components of S is b, and λ is a simple closed curve on $\partial\widehat{Y}(\widehat{K})$ so that $\partial S = b\lambda$.
(2) Choose another simple closed curve μ on $\partial\widehat{Y}(\widehat{K})$ so that $\mu\cdot\lambda = -1$. Suppose the meridian of $\widehat{K}$ has homology class $q\mu + p\lambda$. It can be shown that q is independent of the choice of μ. Indeed, we know that $q = a/b$.
(3) Let definitions of i^y_{max} and i^y_{min} in Definition 4.15 be replaced by

$$i^y_{max} = \lceil \frac{1}{2}(yb - \chi(S)) \rceil, \text{ and } i^y_{min} = \lceil -\frac{1}{2}(yb - \chi(S)) \rceil.$$

(4) All proofs about Theorem 1.10 and Proposition 1.11 apply without change and we will obtain a decomposition associated to $\widehat{K}$

$$I^{\sharp}(\widehat{Y}) \cong \bigoplus_{i=0}^{a-1} I^{\sharp}(\widehat{Y}, i).$$

Note in the original statement of Theorem 1.10, the integer q is indeed the order of $[\widehat{K}]$.

Moreover, the decomposition in Theorem 1.10 can be generalized to a rationally null-homologous link, *i.e.*, the union of rationally null-homologous knots, to recover more decompositions as a candidate for the counterpart of the torsion spin^c decompositions in monopole or Heegaard Floer homology. However, the statement would be much more complicated. We leave the discussion to the future.

Fifth, though we have had a decomposition of $I^{\sharp}(-\widehat{Y})$, we do not know if it works well with the cobordism maps. For example, let $K \subset S^3$ be a knot and let $S^3_{-n}(K)$ be obtained from S^3 by a $(-n)$-surgery along K. Then there is a natural cobordism W from S^3 to $S^3_{-n}(K)$, which induces a cobordism map

$$I^{\sharp}(W_{-n}) : I^{\sharp}(-S^3_{-n}(K)) \to I^{\sharp}(-S^3).$$

Baldwin and Sivek [BS19, Section 7] proved that the cobordism map decomposes in basic classes:

$$I^{\sharp}(W_{-n}) = I^{\sharp}(W_{-n}, t_i),$$

where $t_i : H_2(W_{-n}) \to \mathbb{Z}$ maps the homology class of the surface obtained by capping off of the Seifert surface of K to $(2i)$. We have the following conjecture, basically saying that the decomposition of the cobordism map $I^{\sharp}(W_{-n})$ is compatible with the decomposition of $I^{\sharp}(-S^3_{-n}(K))$.

Conjecture 5.3. There exists an integer N so that for any integer $i \in [0, n-1]$, under the identifications $I^{\sharp}(-S^3_{-n}(K)) \cong \mathcal{I}_+(-S^3_{-n}(K), \widehat{K})$ and $I^{\sharp}(-S^3) \cong \mathcal{I}_+(-S^3_{1/0}(K), K)$, we have

$$I^{\sharp}(W_{-n}, t_i) = I^{\sharp}(W_{-n})|_{\mathcal{I}_+(-S^3_{-n}(K), \widehat{K}, N-i)}$$

and this cobordism map can be recovered by bypass maps.

If the above conjecture holds, based on the bypass maps

$$\psi^*_{\pm,*} : \mathcal{I}_+(-S^3_{-n}(K), \widehat{K}, i) \to \mathcal{I}_+(-S^3_{1/0}(K), K, 0),$$

it is possible to construct a mapping cone formula similar to the work of Ozsváth and Szabó [OS08, OS11] (*c.f.* Remark 4.53).

Sixth, from Remark 4.49, we can construct the infinity version of instanton homology $\underline{\mathrm{KHI}}^\infty$ as the mapping cone of

$$\underline{\mathrm{KHI}}^+(-\widehat{Y}, \widehat{K}) \xrightarrow{\partial_3} \underline{\mathrm{KHI}}^-(-\widehat{Y}, \widehat{K})$$

in the diagram (4.18). Also, we can define the reduce version $\underline{\mathrm{KHI}}_{\mathrm{red}}$ as the image of ∂_3. Alternatively, we may construct $\underline{\mathrm{KHI}}^\infty$ and $\underline{\mathrm{KHI}}_{\mathrm{red}}$ from the spectral sequence in Proposition 1.11 and the related filtered chain complex in Remark 4.51. We may induce some properties from the diagram (4.18) and indicate they are counterparts of HFK^∞ and HFK_{red}, respectively. However, it is still unknown how to construct a chain complex from $(-\widehat{Y}, \widehat{K})$ directly, rather than from its closures, and recover different versions of instanton knot Floer homology.

Seventh, consider $\underline{\mathrm{KHI}}^-$ in Definition 4.46, KHI in Definition 2.16, and $\mathcal{I}_+(-S^3_{-n}(K), \widehat{K})$ in Definition 4.23. Based on Corollary 4.41, Proposition 4.16, and [OSS15, Lemma A.3.10], it is evident that we have the following surgery-type formula.

Conjecture 5.4. For a knot $K \subset S^3$ and $n \in \mathbb{Z}$ so that $n > 2g(K)$, there exists an $N \in \mathbb{Z}$ so that there exist an exact triangle

$$\begin{tikzcd}
KHI(-S^3, K, i) \arrow[rr] & & \underline{\mathrm{KHI}}^-(-S^3, K, i) \oplus \underline{\mathrm{KHI}}^-(-S^3, K, -i) \arrow[dl] \\
& \mathcal{I}_+(-S^3_{-n}(K), \widehat{K}, N-i) \arrow[ul] &
\end{tikzcd}$$

We write the above as a conjecture because one also has to check that the proof of [OSS15, Lemma A.3.10] is compatible with the proof of the bypass exact triangle for $\underline{\mathrm{SHI}}$ and the construction of the gradings, which requires some extra work.

Finally, we would like to present an example that the decomposition of $I^\sharp(\widehat{Y})$ from Propostion 1.17 is capable of performing some computations.

Example 5.5. Suppose $K \subset S^3$ is a knot. We adopt the notation Γ_n in Definition 4.4 to denote sutures consisting of two curves of slope $-n$ and Γ_μ to denote two meridians. Pick a minimal genus Seifert surface S of K. If we know that

$$\underline{\mathrm{SHI}}(-S^3(K), -\Gamma_\mu, S^\tau, i) \cong \begin{cases} \mathbb{C} & -2 \leqslant i \leqslant 2 \\ 0 & else \end{cases} \tag{5.1}$$

then we can show that K admits an instanton L-space surgery. Note such a knot do exist: the $T_{(2,5)}$ torus knot, though it is obvious that $T_{2,5}$ admits a lens space surgery.

From (4.25), Proposition 4.16 and Proposition 4.34, we know that

$$I^\sharp(-S^4_{-4}(K)) \cong \mathcal{I}_-(-S^3_{-4}(K), \widehat{K}) \subset \underline{\mathrm{SHI}}(-S^3(K), -\gamma_{(-2,9)}).$$

By definition, for $i \in \{0, 1, 2, 3\}$, we have

$$\mathcal{I}_-(-S^3_{-4}(K), \widehat{K}, i) \cong \underline{\mathrm{SHI}}(-S^3(K), -\gamma_{(-2,9)}, S^\tau, 2-i). \tag{5.2}$$

Hence our goal is to compute $\underline{\mathrm{SHI}}(-S^3(K), -\gamma_{(-2,9)})$. From Proposition 4.8, there is a bypass exact triangle relating the sutures $\gamma_{(-2,9)}, \Gamma_4$ and Γ_5. Hence we will first compute $\underline{\mathrm{SHI}}(-S^3(K), -\Gamma_4)$ and $\underline{\mathrm{SHI}}(-S^3(K), -\Gamma_5)$ as follows.

First, from (5.1), we know that $g(K) = 2$ and K is fibred by Kronheimer and Mrowka [KM10b, KM10a]. Then, the main result of Baldwin and Sivek [BS18] can be restated in terms of $\underline{\mathrm{KHI}}^-$ as follows.

Theorem 5.6 (Baldwin and Sivek [BS18])**.** *Suppose K is a non-trivial fibred knot which is not right veering, then*

$$\underline{\mathrm{KHI}}^-(-S^3, K, g(K)) \cong \mathbb{C} \subset \ker(U).$$

A priori, we do not know if K is right veering or not. However, it is a basic observation that either K or the mirror $\bar{K}$ is not right veering. Note a surgery of slope r along $\bar{K}$ can be thought of as a surgery of slope $-r$ along K in the mirror manifold. Without loss of generality, we can assume that K is not right veering. Thus,

$$\underline{\mathrm{KHI}}^-(-S^3, \bar{K}, g(K)) \cong \mathbb{C} \subset \ker(U). \tag{5.3}$$

Substituting $n = 4$, we can apply Proposition 4.16 and compute as follows. Note this time we consider gradings instead of blocks. The number at each grading now denotes the dimension of the corresponding vector space. A letter implies we do not know the precise value.

grading	$-\Gamma_\mu$	$\xrightarrow{\psi^\mu_{+,4}}$	$-\Gamma_4$	$\xrightarrow{\psi^4_{+,5}}$	$-\Gamma_5$	$\xrightarrow{\psi^5_{+,\mu}}$	$-\Gamma_\mu$
4	1				1		1
3	1		1		a		1
2	1		a		b		1
1	1		b		c		1
0	1		c		1		1
-1			c		c		
-2			b		b		
-3			a		a		
-4			1		1		

The structures of $\underline{\mathrm{SHI}}(-S^3(\bar{K}), -\Gamma_4)$ and $\underline{\mathrm{SHI}}(-S^3(\bar{K}), -\Gamma_5)$ can be found in [GLW19, Proposition 3.3]. According to Proposition 4.16, the vector space for Γ_4 after the grading shift in the above computation is

$$\underline{\mathrm{SHI}}(-S^3(\bar{K}), -\Gamma_4, S^\tau)[-1].$$

From [Li19, Section 5], we know for $i \geqslant -1$ that

$$\underline{\mathrm{KHI}}^-(-S^3, \bar{K}, i) \cong \underline{\mathrm{SHI}}(-S^3(\bar{K}), -\Gamma_4, S^\tau, i+1)[-1].$$

Moreover, the U $(= U_-)$ map on these gradings are also identified with $\psi^4_{+,5}$. From (5.3), we know that

$$\psi^4_{+,5}|_{\underline{\mathrm{SHI}}(-S^3(\bar{K}), -\Gamma_4, S^\tau, 3)[-1]} = 0.$$

Hence $a = 0$, which implies $b = 1$ by the exactness. Although we do not know the value of c, it would not be a problem. The computation for $\underline{\mathrm{SHI}}(-S^3(K), -\gamma_{(-2,9)})$ is presented as follows.

grading	$-\Gamma_5$ $\xrightarrow{\psi^5_{+,4}}$	$-\Gamma_4$ $\xrightarrow{\psi^5_{+,9/2}}$	$-\gamma_{(-2,9)}$ $\xrightarrow{\psi^5_{+,\mu}}$	$-\Gamma_5$
6	1		1	1
5	0		0	0
4	1		1	1
3	c		c	c
2	1		1	1
1	c	1	d	c
0	1	0	1	1
-1	0	1	1	0
-2	1	c	e	1
-3		c	c	
-4		1	1	
-5		0	0	
-6		1	1	

In the above computation, it is clear that for $i \in \{-1, 0, 2\}$,

$$\underline{\mathrm{SHI}}(-S^3(K), -\gamma_{(-2,9),S^\tau}, i) \cong \mathbb{C}.$$

We can also compute using $\psi^*_{-,*}$ instead of $\psi^*_{+,*}$, which will show that

$$\underline{\mathrm{SHI}}(-S^3(K), -\gamma_{(-2,9),S^\tau}, 1) \cong \mathbb{C}.$$

Hence we conclude from (5.2) that $I^\sharp(-S^3_{-4}(K)) \cong \mathbb{C}^4$, *i.e.*, $S^3_{-4}(K)$ is an instanton L-space.

Remark 5.7. Using τ_I defined as in Definition 4.54, it is possible to show $c = 0$.

Remark 5.8. The same technique can also be used to prove the following statement. Suppose $K \subset S^3$ is a knot so that for any $i \in \mathbb{Z}$, we have

$$KHI(S^3, K, i) \cong KHI(S^3, T_{(4,5)}, i),$$

then K admits an L-space surgery. Here $T_{(4,5)}$ is the $(4, 5)$-torus knot.

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